



Energy modelling to support subnational sustainable planning in developing countries:

The case of Kakamega County in Kenya

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Abstract

Kenya is at the forefront of a socioeconomic transformation, aiming to turn into an industrialized middle income country by 2030. Kenya Vision 2030 has identified energy as a key foundation and one of the infrastructural “enablers” upon which the economic, social and political pillars of this long-term development strategy will be built.

Predicting the future of energy systems however, involves risks due to various uncertainties. Therefore, systematic energy planning at national and sub-national/County level is highly recommended through the adoption of more realistic assumptions on the future evolution and profile of demand and robust pre-feasibility of prospective projects including the integration of renewable energy sources, which the country is endowed with.

This thesis provides a comprehensive analysis of the energy sector for Kakamega County in Western Kenya. The current energy demand level was estimated for six selected sectors of the County namely Residential, Industrial, Transportation, Commercial, Public and Agricultural. Additionally, the renewable energy resources potential was assessed at local level using GIS and other available data. LEAP software was used in order to model and project the energy demand and supply based on three 15-year scenarios till 2030, developed to support the economic, social and environmental sustainability of the County.

This study intended to create a framework aiming to facilitate sub-national energy planning in developing countries and it is expected that the findings will be complementary to already existing energy planning models but also the base for future research towards energy poverty elimination.

Keywords: Rural electrification, Energy Access, Energy modelling, LEAP, Kakamega, Kenya.

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Alexandros Korkovelos

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NOMENCLATURE

Here all the notations – abbreviations used in this thesis are described.

Abbreviations

NGO	Non-governmental organization
SE4ALL	Sustainable Energy for all
LEAP	Long range Energy Alternative Planning system
GDP	Gross Domestic Product
USD	United States Dollars
Ksh	Kenyan Shilling
MoEP	Kenyan Ministry of Energy & Petroleum
LPG	Liquefied Petroleum Gas
LNG	Liquefied Natural gas
KPC	Kenyan Pipeline Company Ltd.
KPLC	Kenyan Power & Lighting Company
RAE	Rural Electrification Authority
PV	Photovoltaic
KETRACO	Kenyan Electricity Transmission Company Ltd.
REP	Rural Electrification Programme
CBE	Cross Border electrification Scheme
ERC	Energy Regulatory Commission
LCPDP	Load Forecasting Power Development Plan
FiT	Feed in Tariff
IMF	International Monetary Fund
SWH	Solar Heating System
HPS	High Pressure Sodium
UN	United Nations
FAO	Food & Agriculture Organization of the UN
GIS	Geographic Information System
GHI	Global Horizontal Insolation
DNI	Direct Normal Insolation
ESMAP	Energy Sector Management Assistance Programme
PUoE	Productive Use of Energy
VAT	Value-added tax
WHO	World Health Organization
DSM	Demand Side Management
CAPEX	Capital Expenditure
OPEX	Operating Expenditure

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INTRODUCTION

Energy holds a privileged place in the construction of durable human societies (United Nations 2002). It is a vital commodity highly interconnected with socioeconomic development and well-being. It is therefore a crucial prerequisite for the achievement of continuous growth both on global and regional level.

However, economic development, improvement of living standards and population growth are leading to an exponential increase of energy consumption, especially in developing countries which are currently at the forefront of a tremendous transformation. While this transformation should not be contestable, the associated production and consumption of energy is encompassing several geopolitical and environmental issues, such as land loss, climate change, resources depletion and environmental degradation (Urban 2009) (Emmanuel Bergasse 2013).

It is crucial that the path followed by the industrialized countries over the past decades should be avoided and instead a more sustainable path should be envisioned. A sustainable energy transition, can create new possibilities for developing countries in terms of growth while at the same time safeguarding natural resources, environment and human health. Technological leap-frogging is assumed to play an important role in this growth as it should give access to modern technologies and innovation, enabling this sustainable transformation to happen (Urban 2009).

Renewable energy technologies are of significant importance in this attempt. Many developing countries have seen them as an opportunity towards their prospective rapid growth, while their integration is estimated to expand as a wider range of cost effective technologies will be employed. Renewables potentially improve energy security by reducing the reliance on imported fuels and help to diversify the power mix. Their decentralized manner, enables them to be deployed faster than centralized power plants while providing local employment for deployment and maintenance. Furthermore, they can substantially affect the energy access to remote communities (IEA 2014).

Globally, there are a number of initiatives trying to enhance this sustainable transition by building cooperation between governments, private sector, communities, non-governmental organizations (NGO's) and other involved stakeholders in the energy sector. This thesis is elaborated under the Sustainable Energy for All (SE4ALL) framework aiming at the following goals by 2030 (The World Bank 2013).

- Ensuring universal access to modern energy services
- Double the global rate of improvements in energy efficiency
- Double the share of renewable energy in the global energy mix

Energy models are a useful tool in order to analyze, plan and manage the energy transition imposed by the aforementioned goals. They can highlight the most environmentally friendly, institutionally sound, social acceptable and cost-effective solutions of the best mix of energy supply and demand options, for a defined area. The development of long term projections and scenarios even though subjected to several

uncertainties, is a necessary step in order to support long-term regional sustainable development (Atom Mirakyan 2015).

Predicting the future of energy systems in developing countries involves risks. Technological leap-frogging, ambiguity towards different trajectories of development, significant regional and income-related differences between consumption patterns are issues which can critically affect the credibility of the models (Urban 2009). There is therefore an evident need to properly adapt energy models towards a more decentralized and flexible environment, better representing the energy future of developing countries.

Thesis scope and objectives

The scope of this thesis is to develop a model in LEAP software that will be able to perform a comprehensive energy analysis on a sub national level in a developing country. Kakamega County in western Kenya was selected as a suitable candidate for this analysis.

The objectives of the thesis are to:

- Conduct a base line survey in order to understand and record the current energy situation.
- Create a benchmark of energy uses and greenhouse gas emissions.
- Assess the renewable energy resources potential in the county.
- Model the energy demand & supply including projection till 2030.
- Create scenarios suggesting relevant energy interventions at a regional level.

The overall objective of the thesis, is to develop an energy model able to analyze and optimize energy systems at a sub national level, facilitating the decentralized and renewable energy generation and supporting a multi-sectorial sustainable development.

In chapter one, an introduction in the Kenyan content is presented, highlighting the current socioeconomic environment and the framework under which the energy sector is operating. Chapter two, focuses into the subject area of Kakamega. A description of the County's characteristics is followed by a comprehensive bottom-up approach regarding energy consumption data collection for the county. The current energy generation mechanisms and the renewables potential are also discussed. In chapter three, LEAP software is introduced, analyzing the energy modelling procedure and the main factors affecting its implementation in the Kakamega case study. The construction and purpose of various scenarios is also elaborated. Finally, chapter four discusses on the results of the optimization process, specifically commenting on the possible energy related interventions in the involved sectors.

CHAPTER 1. GENERAL ENVIRONMENT AND ENERGY SECTOR IN KENYA

This chapter illustrates the general environment in Kenya in terms of socioeconomic conditions. Special focus is given in the structure of the of the energy sector, whose understanding is fundamental for the successful development of any energy plan in the country.

1.1 Background

Kenya, with a population of 44.08 million¹ (2015), is a lower-middle income country and the biggest economy within East Africa community with the gross domestic product (GDP) at 54.993 billion USD in 2013 (International Monetary Fund n.d.). That is, the GDP per capita for the same year was 1,245.5 USD and an annual GDP growth of 5.7%. However, the past ten year average GDP growth was calculated as 5.2% which is a more representative index and will be used in this report (The World Bank n.d.) (International Monetary Fund n.d.).

Although the economy of the country has experienced various shocks in the last ten years (e.g. global economic recession, political instability, drought), it has sustained its capacity and seems ready to take off leveraging its natural resources, its significant geopolitical location, the expanding skilled youthful population and its dynamic private sector (The World Bank n.d.) (Barkan 2011).

Energy is a critical factor towards the socioeconomic development of the country. As at June 2014, almost 68% of the population did not have access to electricity, with the percentage being higher in the rural areas (MoEP 2015). Frequent electricity service interruptions are also evident and significantly affect economic activities, imposing higher costs due to production loss and/or self-generation cost. Therefore, higher levels of electricity access and service quality are necessary in order to enhance the national energy mix and bust the social and economic development.

In 2011 it was estimated that the primary energy use per capita in Kenya was 480 kg of oil equivalent (The World Bank n.d.). That implies an equivalent of 19,536 ktoe for the whole country in 2011². The socioeconomic stability and the population increase is expected to have a direct impact on the energy use in Kenya. In 2015 the primary energy use in the country is estimated to be 21,376 ktoe. Biomass accounts for the 68% of the total energy consumption in the country (55% of which derives from farmland as woody biomass and crop residue while 45% from forest), petroleum for 22%, electricity for 9% and coal for roughly 1% (MoEP 2015).

¹ In 2015 the population is estimated to be 44,076,000 (International Monetary Fund n.d.)

² In 2011 the population of Kenya was 40,700,000 (International Monetary Fund n.d.)

1.2 The energy sector

Petroleum

According to the Ministry of Energy and petroleum (MoEP) of Kenya in 2013, the petroleum products imported were 3,996 thousand tons. Almost 61.3% of the petroleum consumption accounted for retail pump outlets and road transport, 14.24% for industrial and commercial use and 6.37% for power generation. A refinery is operating in Mombasa producing roughly 12,400 tons LPG per year (2013) covering almost 14% of the country's demand (MoEP 2015). The refinery also produces premium and regular motor spirit, automotive gas oil, dual purpose kerosene, fuel oil, grease and bitumen. It is projected that the demand for petroleum products will show an annual increase of about 3.1% till 2030 (MoEP 2015). Petroleum exploration processes are in progress in the sedimentary basins both on and off shore.

The major transport of petroleum products is being performed by the existing pipeline system (operated by Kenya Pipeline Company Ltd. (KPC)) and road transportation, while roughly 1% was transported by rail. The pipe line is connecting Mombasa – Nairobi – Nakuru – Eldoret and Kisumu while an extension is planned from Nakuru through Nanyuki to Isiolo as well as to Taveta and Konza. LAPSSET project is also under development interconnecting Lamu port with Ethiopia and South Sudan (for crude oil transportation). Interconnections with Kampala (Uganda), Kigali (Rwanda) and Goma (DRC) are also planned (MoEP 2015).

Coal

As in December 2014, all the coal used in Kenya (~208,900 tons) was imported and mainly used for industrial purposes (cement manufacturers). Coal consumption is expected to be increased due to the discovery of coal deposits in Mui basin in Kitui County and its exploitation plans for power generation purposes (MoEP 2015).

Electricity generation

Even though the electricity sector accounts for 9% of the total energy consumption in Kenya, it significantly affects the economy. The total generating capacity by December 2014 was 2,173 MW with annual energy generation of 8,840 GWh between 2013/2014. Renewable energy sources account for almost 68% of electricity generation while the rest 32% is deriving from fossil fuels. Table 1 provides a detailed list of the main generation sources and their capacity.

In the period 2013/2014, 85 GWh were imported (mainly from Uganda and to a smaller extend from Ethiopia and Tanzania) while 39 GWh were exported (37 GWh to Uganda, 2 GWh to Tanzania) (MoEP 2015). As at December 2014, the peak demand was 1,512 MW showing an increase of almost 33% over the last five years (MoEP 2015).

Table 1. Sources of electric power generation in Kenya (MoEP 2015).

Sources of Electric Power Generation		Installed Capacity (December 2014)		Annual Generation (FY 2013/14)	
		(MW)	Percentage	(GWHrs)	Percentage
Renewable Energy	Hydro	821	37.8	3,945	44.6
	Geothermal	593.5	27.3	2,008	22.7
	Wind	25	1.2	18	0.2
	Cogeneration	38	1.7	57	0.6
	Imports	-	-	85	1.0
	Total	1477.5	68.0	6,112	69.1
Fossil Fuels	MSD	579.5	26.7	2,533	28.6
	Gas Turbines	60	2.8	41	0.5
	HSD (Isolated Stations)	25.8	1.2	61	0.7
	Emergency Power Plant	30	1.4	94	1.1
	Total	695.3	32.0	2,729	30.9
Installed Capacity and Units Generated		2173MW		8,840GWHrs	

Hydropower

Hydropower is a significant source of electric power generation in Kenya. Large hydroelectric power plants (above 10 MW) account for the majority of the production while small scale hydro projects have an accumulated capacity of 25 MW (15 MW public – 10 MW private investments). According to the Ministry of Energy and Petroleum there is a high potential of both large and small hydroelectric power in the country reaching 6,000 MW of which 1,449 MW are of high economic significance.

The major drainage basins are Lake Victoria, Rift Valley, Athi River, Tana River, Ewaso Ng'iro North River. However, competing interests over the land and water use, environmental degradation, lack of adequate hydrological data and vandalism are some of the challenges needed to overcome in order to harvest the full potential of hydropower in Kenya (MoEP 2015).

Geothermal

Geothermal power is the second biggest source of electricity in Kenya with a current installed and operating capacity of 593.5 MW. According to the Ministry of Energy and Petroleum there is high potential of geothermal power (~ 10,000 MWe) in the region of Rift Valley, due to significant volcanic activity in the area (Figure 1). In October 2014, the biggest single turbine geothermal plant in the world (Olkaria IV) was inaugurated in Naivasha County with a capacity of 140 MW (GDC n.d.). Further exploration and development activities are taking place in indicative sites, aiming to step up and leverage the geothermal potential.

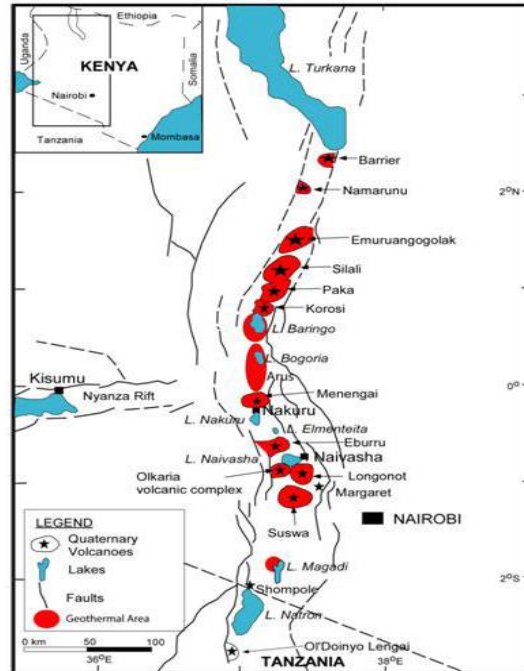


Figure 1. Geothermal sites in Kenya (GDC n.d.).

Wind

By November 2014, wind power installed capacity was 25 MW in Kenya. The wind power potential is estimated at 346 W/m² with several locations however to experience wind speeds above the threshold of 6 m/s and indicate the large scale exploitation of wind power (Figure 2).

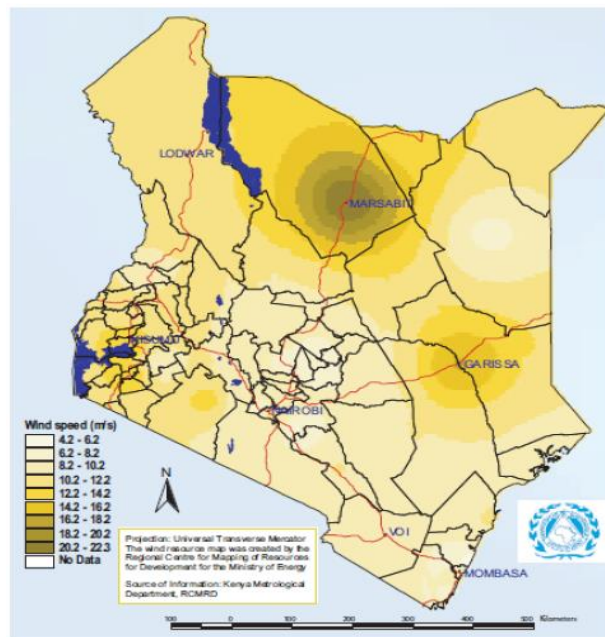


Figure 2. Maximum wind speed map for Kenya (Sklivaniotis, Integrated Techno-Economic Comparative and Socio-Economic Impact Study for Increasing Energy Access in Rural Kenya 2014).

The 300 MW Lake Turkana project is expected to be commissioned by 2017 while other feasibility studies for additional 650 MW are in progress for Kinangop, Marsabit, Isiolo/Meru and Ngong locations (MoEP 2015). Some pilot small wind power projects also exist so far in Kenya as part of hybrid systems developed in areas where the grid extension is not currently possible (APPENDIX A).

Solar

Due to its geographic location Kenya's potential for solar energy exploitation is high. The average solar insolation ranges between 4-6 kWh/m²/day all year round, with moderate to high temperatures. Figure 3 schematically represents the solar potential for Kenyan territory. According to the annual report of Kenya Power & Lighting Company (KPLC) by June 2014 there were 0.7 MW of solar power installed and all of them under the collaboration with the Rural Electrification Authority (RAE) (KPLC 2014).

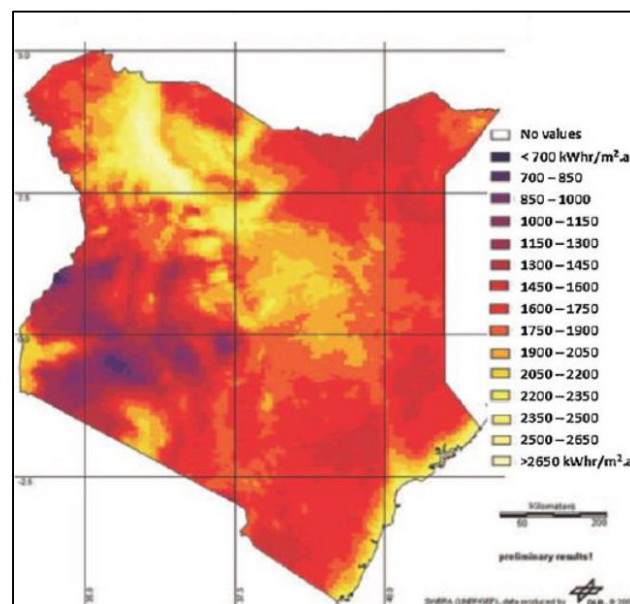


Figure 3. Solar insolation for Kenya (Sklivaniotis, Integrated Techno-Economic Comparative and Socio-Economic Impact Study for Increasing Energy Access in Rural Kenya 2014).

It is evident that the on grid solar power has been stagnant in Kenya over the past few year while the off-grid, small PV applications (12-50 Wp) show a significant increase in popularity and high market penetration mainly due to private sector activity (MoEP 2015) (Sklivaniotis, Integrated Techno-Economic Comparative and Socio-Economic Impact Study for Increasing Energy Access in Rural Kenya 2014). As at December 2014 there were 6 MW of solar home and small commercial PV systems (MoEP). Regardless the national grid extension plan, it is estimated that the long term potential for solar home systems (SHS) is about 30 MWp while the potential for other off-grid PV applications amounts to roughly 6 MWp (Ondraczek, The sun rises in the east (of Africa): A comparison of the development and status of solar energy markets in Kenya and Tanzania 2013) (APPENDIX A).

Solar water heating (SWH) is used at households, hospitals, hotels. There is an estimated demand of 800,000 units by 2020 (equivalent to 300,000 toe) implying an impressive 20% growth per year. This

demand derives mainly from domestic, institutional and small commercial applications, spurred by Energy Regulations 2012 (MoEP 2015).

Other

Biomass is the major primary energy source in Kenya. Sustainable wood fuel supply management is crucial in order to meet the growing demand and halt the environmental degradation in the country. A strategy for introduction of biofuel blending (ethanol – gasoline) was developed in 2010, with facilities constructed in Kisumu, Eldoret and Nakuru. Main source of feed stock is expected to be sugarcane and sweet sorghum for ethanol and jatropha, castor, coconut, croton and cotton seed for biodiesel. Further exploration is being performed towards this direction (MoEP 2015).

Biogas projects have also been initiated both by public and private initiative. According to the Ministry of Energy and Petroleum, there is a 19,852 kW potential capacity from the floriculture industry (especially from sisal production) (MoEP 2015).

A pre-feasibility study by Ministry of Energy and petroleum in 2007 indicated potential of 120 MWe from sugar companies (through cogeneration) in Kenya under minor investments, and 200 MWe under modest investments. Currently, Mumias Sugar company in Kakamega County has an installed capacity of 38 MW running on sugarcane bagasse, from which 26 MW are connected to the national grid. It is estimated that the generation capacity by sugar companies will reach 60 MW by 2016 (MoEP 2015).

In October 2014, the Ministry of Energy and Petroleum granted the approval for a 100 MW electric power plant running on wave energy. Nuclear power is on an initial development stage in Kenya, however it is aspired that it will be the main contributor of electricity generation by 2030.

Electricity Transmission Network

As at June 2014, the transmission network in Kenya involved 1,434 km of 220 kV, 2,513 km of 132 kV, 132 kV double circuit with Uganda, 9 generation stations with capacity of 1,846 MVA and 45 transmission substations with capacity of 3,181 MVA.

The transmission capacity is severely constrained especially during peak hours (partly due to inadequate reactive power and transmission constraints especially in Western regions and Nairobi) (MoEP 2015). Power outages are a daily phenomenon across the country leading to reduced power reliability. In order to temporarily cope with this problem an online, location based power outage notification service (Power Alert) was introduced in 2014. However, additional transmission capacity development is necessary for the electrification goals to be achieved in the future.

To this direction, the transmission entity KETRACO (established in 2008) has undertaken several projects in order to reach geographical locations without access to electricity, enhance the evacuation capacity for current and planned generating plants and facilitate regional connection with neighbor countries (Kenya Electricity Transmission Company Ltd. KETRACO n.d.). It is expected the 5,000 km of new transmission lines will be developed short term and 16,000 km by 2030 (Figure 4) (MoEP 2015). The on-going and planned transmission projects are presented in APPENDIX B.



Figure 4. Transmission network planned extension in Kenya (MoEP 2015).

Electricity Distribution Network

Kenya Power & Lighting Company (KPLC) is the main licensee of the distribution network in Kenya with Rural Electrification Authority (REA) undertaking responsibility for grid extension at low and medium voltages in areas that is it consider unprofitable.

REA usually cooperates with each County authorities in order to enhance energy access through electricity and gas reticulation and regulation. As at June 2014, there were 1,212 km of 66 kV lines, 20,788 km of 33 kV lines, 30,860 km of 11 kV lines and low voltage lines in Kenya with 3,311 MVA primary distribution substations and 6,317 MVA total transformer capacity. Through the Rural Electrification Programme (REP) and the Cross Border Electrification scheme (CBE) it is expected that a big part of the population will be electrified in the next few years (MoEP 2015). As at 2014, 430,000

households were connected to grid electricity while 100% electrification of market centers and primary schools is due to be completed by 2016 (The World Bank n.d.).

Key stakeholders involved in the Kenyan Energy sector

The Kenyan energy sector is vertically integrated with the Ministry of Energy and Petroleum (MoEP) being the responsible body for energy policy while administrating a system of performance contracts with the public sector entities:

Kenyan Pipeline Company Ltd. (KPC), Kenyan Petroleum Refinery Ltd. (KPRL) National Oil Company of Kenya (NOCK), Kenya Electricity Generation Company (KenGen), Geothermal Development Corporation (GDC), Kenya Nuclear Electricity Board (KNEB), Kenya Transmission Company (KETRACO), Kenya Power and Lighting Company (KPLC) and Rural Electrification Authority (REA).

In addition, a semi-autonomous regulatory agency, the Energy Regulatory Commission (ERC) formulates, enforces and reviews regulations, codes and standards for the energy sector and reviews and adjusts electric power tariffs and tariff structures (The World Bank 2014).

The power sector has been significantly reformed since 1997 with the sector's governance anchored by the Energy policy 2004 and the Energy Act 2006. The generation, transmission and distribution entities are separated and operate based on commercial principles and transparent financial relationships between the involved stakeholders.

There is a strong private sector presence in the sector, with the independent power producers (IPPs) comprising 23% of the electricity supply in the country (483 MW) while about 1.2 billion USD additional investments have reached financial closed and new power generation plants are under development (The World Bank 2014).

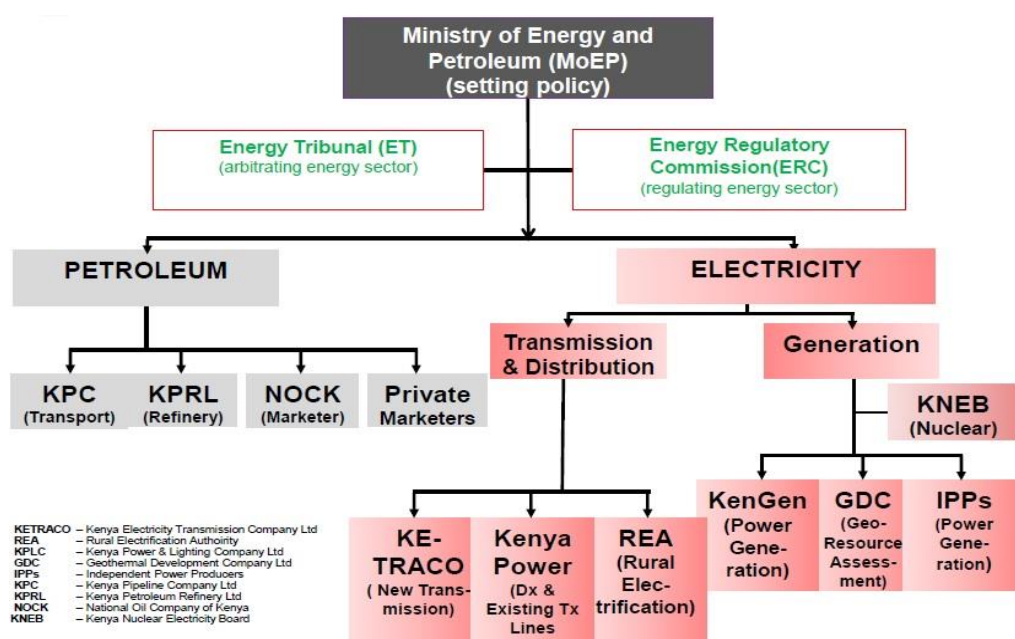


Figure 5. Energy sector structure in Kenya (Ministry of Energy and Petroleum MoEP n.d.).

Acts and policies that set the current framework in the Kenyan Energy Sector

- Constitution of Kenya 2010
- Sessional paper No.4 (2004)
- Energy Act No.12 (2006)

Assigned responsibility for development of national energy plans to ERC. In 2009 established the 20 year rolling Least Cost Power Development Plan (LCPDP) aiming at Load forecasting and Generation, Transmission & Distribution planning. The last update was published in 2013 with the next one expected in 2015.

- Rural Electrification Master Plan
- Millennium Development Goals
- Feed-in Tariff (FiT) policy
- Kenya National Climate Change Response Strategy
- Kenia Vision 2030

Development plan started in 2012 aiming to transform Kenya into an industrialized middle income country by 2030. Part of the VISION 2030 plan is to dramatically change the energy profile of the country by providing universal and reliable electricity services in an affordable yet cost effective way.

- Government 5000+ project

Power generation development plan started in October 2013 aiming in the addition of 5,000 MW of generating capacity by April 2017. The plan is projected by that time to have reduced the generation cost from 11.30 USD cents to 7.41 USD cents per kWh and the electricity tariff from 14.14 USD cents to 9 USD cents per kWh for commercial/industrial users and from 19.78 USD cents to 10.43 USD cents per kWh for domestic consumers (Table 2) (MoEP 2015).

- Geothermal Resources Act No.12 (1982)
- Petroleum (Exploration and Production) Act Cap. 308
- Environmental Management and Co-ordination Act, No. 8
- Occupational Safety and Health Act of 2007
- Water Act of 2002
- Merchant Shipping Act & Kenya Maritime Authority Act of 2006

Table 2. Development activities of 5000+ plan (MoEP 2015).

TECHNOLOGY	CUMULATIVE INSTALLED CAPACITY (MW)							
	Number of Months from start of the Project							
	0	6	12	18	24	30	36	40
HYDRO	770	794	794	794	794	794	794	794
THERMAL	622	709	782	782	782	432	432	432
GEO THERMAL	241	331	507	697	747	952	1102	1887
WIND	5	5	5	25	85	385	635	635
COAL	0	0	0	0	0	960	960	1920
LNG	0	0	0	0	700	1050	1050	1050
CO-GENERATION	26	26	26	44	44	44	44	44
RETIRED PLANTS		90				350		
CUMMULATIVE TOTAL	1,664	1,775	2,114	2,342	3,152	4,617	5,017	6,762
Generation Tariff (US\$/kWh)	11.3	10.14	9.93	8.74	8.07	7.38	7.58	7.41
Industrial/Commercial Tariff (US\$/kWh)	14.14	12.77	12.49	11.03	10.08	9.03	9.32	9
Domestic Tariff Progression (US\$/kWh)	19.78	18.3	17.73	15.85	13.46	11.14	11.19	10.43

1.3 Current challenges

Kenya is at the forefront of a socioeconomic transformation, aiming to turn into an industrialized middle income country by 2030. Kenya Vision 2030 has identified energy as a key foundation and one of the infrastructural “enablers” upon which the economic, social and political pillars of this long-term development strategy will be built.

The three overarching objectives of the government are to increase electricity access, to provide efficient and reliable electricity services and to secure adequate electricity supply at least cost. While the energy sector has been significantly reformed over the last decade, the enabling environment to realize these objectives is not yet fully in place. It is necessary that new policies and institutional arrangements should be implemented in order to secure the optimal planning and put in place a robust framework for competitive procurement of new capacity (The World Bank 2014).

In particular, governance in the public sector entities should be strengthened for greater accountability and improved efficiency. The investment needs for electrification should be met by the government and not by the public entities (like KPLC) in line with the international practice of successful electrification programs. In addition, private sector participation should be mobilized through better procurement processes and competitive bidding (The World Bank 2014).

Substantial investments are required in the transmission and distribution network in order to ensure the achievement of good standards of service quality. However, decentralized electrification approaches should be thoroughly taken into consideration through mini-grids and/or off grid systems in areas where connection to the national system is not envisaged in the short and medium term (The World Bank 2014).

For that to be achieved, systematic energy planning (at national and sub-national/County level) is highly recommended through the adoption of more realistic assumptions on the future evolution and profile of demand and robust pre-feasibility of prospective projects including the integration of renewable energy sources, which the country is endowed with. It was under this framework that this thesis was elaborated.

CHAPTER 2. KAKAMEGA COUNTY CASE STUDY

In this chapter Kakamega County in Western Kenya is introduced. A brief introduction on the socioeconomic characteristics of the County is followed by a comprehensive analysis of the energy demand in various sectors and an assessment of the available energy resources.

2.1 Demographics and socioeconomic background

Geographic Location

Kakamega County is located in the western part of Kenya, about 40 km northeast of the Lake Victoria and 40 km from the borders with Uganda. It has a surface area of 3,020 square kilometers and an average elevation of 1,535 meters. The County borders with Bungoma to the north, Trans Nzoia to the north east, Uasin Gishu and Nandi Counties to the East, Vihiga to the South, Siaya to the south west and Busia to the west. The County has 9 constituencies namely Butere, Mumias, Matungu, Khwisero, Shinyalu, Lurambi, Ikolomani, Lugari and Malava (CRA 2011), (SoftKenya n.d.).

Demographics

Kakamega is the second most populous county in Kenya after Nairobi with as estimated population of 1,644,328 (2013 census) and a density of 544 ppl/km², higher in the central and south parts of the county (KNBS - SID 2013). The population growth is estimated at 2.5% per year. The urban population accounts for 15.2% of the inhabitants while growing at a rate of 4.1% per year (CRA 2011), (Unicef n.d.). The biggest towns in the region are:

- Mumias: 99,987
- Kakamega (Capital): 91,768
- Butere: 12,780
- Lumakanda: 10,580
- Malava: 4,070

In Kakamega there are registered 355,679 households with the annual growth rate being 2.54%. The average household size ranges between 5.5 people in rural areas and 4 people in urban areas. The average value is 4.7 people per household that matches with the distribution of the estimated population to the registered households. Household access to electricity is 5.6% with its usage mainly involve lighting and small electric appliances. The monthly income of a household is ranging from 3,000 to 150,000 Ksh with the average value being near 15,000 Ksh per month. 76.1% of the households have access to improved water infrastructure, 54.1% has access to access to good/fair roads while only 4.9% of the households have access to paved roads (KNBS 2014), (CRA 2011), (Ngetich 2013), (Eston Ngugi 2013).

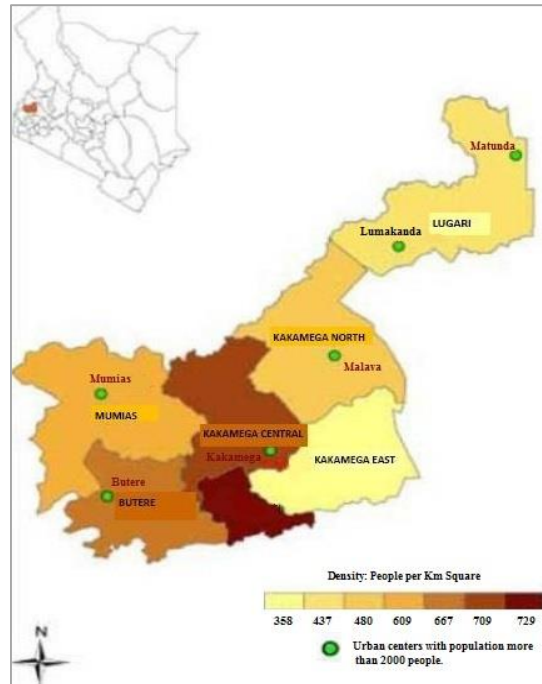


Figure 6. Population density in Kakamega (CRA 2011).

Economy

Kakamega had the 5th higher local revenue among the Kenyan counties as in 2013/2014 with 3,500 Million Ksh. An equitable share of 6515.51³ Million Ksh and a conditional grant of 840.7⁴ Million Ksh set the total revenue of the County at 10,856 Million Ksh for 2013/2014 (KNBS 2014).

Estimation of Kakamega's GDP

According to IMF in 2013 the estimated GDP per capita in Kenya was 1245.5 USD. In absence of a detailed county-scale estimation for the GDP, an estimation of the total GDP of the Kakamega can therefore be made using the population of the county. Hence, it was estimated that the local GDP was 2.048 billion USD (190 billion Ksh) at the base year, set in 2013.

Main economic activities

The main economic activities are (SoftKenya n.d.):

- Large and small scale sugar cane farming.
- Sugar industry.
- Small scale mixed farming of millet, tea, maize, sugar beans, sunflower & dairy products.
- Commercial center: 5 Commercial Banks & 3 Microcredits.
- Commercial businesses: Supermarkets, grocery shops, open markets.
- Transportation business: Boda-Boda (35 registered societies, 1450 boda-boda cyclists).

³ 3.1% of the 210 Billion Ksh distributed from Kenya Government according to the County Allocation of Revenue Act 2013.

⁴ 4.2 % of the 20 Billion Ksh of conditional grant

- Informal economy: Street vendors, hawkers.

2.2 Energy demand

A bottom-up approach was followed, assuming and estimating disaggregated energy demands for selected activities of each sector. The following paragraphs contain detailed description of these data.

2.2.1 Residential sector (households)

The residential sector holds a great percentage of the energy consumption in Kakamega. The following paragraphs provide an analysis of the main activities in a typical household in Kakamega under an energy consumption and fuel utilization approach.

Cooking

The main energy source for cooking in Kakamega is solid biomass (firewood and charcoal) as it accounts for 95.7% of the energy fuels used. The constituencies of Lurambi and Shieywe near Kakamega town show the highest charcoal adoption at 32% and 54% respectively. On the other hand, there are areas where firewood is predominant reaching adoption rates above 95%. Table 3, shows in more detail the allocation of fuels for cooking purposes in Kakamega County (Eston Ngugi 2013).

It is estimated that the average firewood consumption in Kenya is around 5-10 kg/household/day (KERECA 2015). Another indication suggests 1.45 kg of firewood/person/day (Geoffrey Ndegwa 2011) while a study conducted by Boulkaid in 2015 for Nyeri County shows similar intensities of approximately 1.3 kg of firewood/person/day (Boulkaid 2015). With the average household size being 4.7 persons, that induces a daily average firewood consumption between 6 - 6.8 kg per household. The upper limit of 6.8 kg of firewood/household/day will be used now on in this paper. Consequently, the monthly household consumption of firewood is 204 kg and the annual consumption 2,448 kg. Firewood has a heating value ranging from 8 MJ/kg (if freshly harvested, moisture above 50%) to 16 MJ/kg (if it is air dried, moisture around 15%) (Steve Sepp 2014). However, 84.1% of it is self-collected (especially in rural areas), which implies no monetary cost but very low heating value (Dalberg 2013). When purchased, firewood price is about 76 Ksh per head log or approximately 9 Ksh/kg (Wambua, Household energy consumption and dependency on common pool forest resources: The case of Kakamega forest, Western Kenya 2011).

The charcoal consumption is higher in regions with increased income. Charcoal has higher heating value than firewood, therefore lower quantities are necessary in order to cover the cooking needs. It is estimated that an average household consumes about 750 kg of charcoal per year (Dalberg 2013) (Wambua, Household energy consumption and dependency on common pool forest resources: The case of Kakamega forest, Western Kenya 2011). That value is translated into 62.5 kg of charcoal per month and roughly 2 kg of charcoal per day. The heating value of charcoal ranges between 27-33 MJ/kg according to the fixed carbon and the carbonization process (Steve Sepp 2014). The price of charcoal is estimated at 14 Ksh/kg (average value between 9.8 – 18.3 Ksh/kg) (Dalberg 2013) (Wambua, Household energy

consumption and dependency on common pool forest resources: The case of Kakamega forest, Western Kenya 2011).

Table 3. Percentage of fuel used for cooking purposes in Kakamega County (Eston Ngugi 2013).

Fuel	Percentage (%)
Firewood	86.8
Charcoal	8.9
Kerosene/Paraffin	2.3
LPG	0.8
Electricity	0.5
Biogas	0.4
Other	0.3

Both firewood and charcoal consumption depend on the cook stove utilized and its efficiency. The three stone stove (open fire), traditional stove and improved stove are commonly found in Kakamega. Table 4, presents a more detailed description of the stove technologies and their respective efficiencies.

Table 4. Cooking stoves technologies and efficiencies (Amarasekara 1994), (Boulkaid 2015).

Fuel	Stove	Efficiency (%)	Mean eff. (%)
Firewood	Three stone fire	5 to 10	7.5
	Traditional stoves	8 to 14	11
	Improved stoves	20 to 30	25
Charcoal	Traditional stoves	8 to 14	11
	Improved stoves	20 to 30	25
Kerosene	Wick type	25 to 35	30
	Pressure type	35 to 55	45
LPG	LPG stove	40 to 60	50
Electricity	Hot plates	55 to 75	65
	Rice cooker	85	85
	Pressure cooker	90	90
	Electrical Kettle	80 to 90	85
Biogas	Biogas stoves	45	45

Kerosene is used for cooking to a relatively low percentage even though it is a more efficient solution. The higher cost and the availability are mainly to be blamed. It is estimated that the average consumption of kerosene for cooking is about 200 liters per household per year (Dalberg 2013). That is translated to 16.7 liters per month or 0.54 liters per day. The lower heating value of Kerosene is 43 MJ/kg (The

Engineering Toolbox 2015) and its price is following a declining path since 2012, set at 54.35 Ksh/liter as in March 2015 (Energy Regulatory Commission (ERC) 2015).

LPG has a very low adoption rate in Kakamega of about 0.8% (Eston Ngugi 2013). The high price (especially in comparison with the cheaper charcoal) and the limited storage, distribution and retail capacity in the County are the basic barriers for that. It is estimated that on average households consume 128.6 liters of LPG per year for cooking (Dalberg 2013). That implies an average value of 10.7 liters per household per month or 0.35 liters per day. The specific heat of LPG (including propane/butane) is 46.4 MJ/kg (The Engineering Toolbox 2015) while its price at 256.2 Ksh/liter as in April 2013 (Dalberg 2013).

According to the data described above, a comparative analysis was performed in order to estimate the average daily energy required by a household for cooking purposes in Kakamega. Therefore, depending on fuel and cooking technology (stove) it was estimated that a household in Kakamega requires about 7.8 MJ per day for cooking purposes. That approximation was used in order to determine the electricity consumed for cooking by electrified households in the County.

According to the statistics, about 1,778 out of 19,918 electrified households use electricity for cooking in Kakamega. In order to cover the estimated 7.8 MJ of necessary energy per household for daily cooking and with an average efficiency of 0.8 for electric cooking devices, 2.7 kWh are needed. That means that the average electricity consumption is about 986 kWh per year per household or 82 kWh per month for cooking purposes, while the electricity price 2.5 KES/kWh (under fixed tariff of 120 KES and charge 2.5 KES/kWh for the first domestic users tier) (ERC 2013).

Finally, Biogas is used by 0.4% of the households for cooking purposes. The calorific value of biogas is estimated at 20 MJ/kg and the biogas stove efficiency 45%, so the daily average consumption of biogas is around 0.865 m³. That means 26.8 m³ per month or 321.8 m³ per year for an average household. Assuming a yield of 0.035 m³ of biogas per kg of slurry (3.5% efficiency), that is translated to 24.7 kg of slurry per day. If 50% of the slurry is water that implies that manure should be supplied at 12.4 kg per day per household. It is estimated that cows produce 4.5 kg of dung/head per day from which 60% is recoverable. That accounts to 4-5 cows per household as a fuel feed for biogas cooking (author's estimation).

Lighting

The main source of lighting for households in Kakamega is kerosene, which accounts 93.4% of the lighting fuels and is utilized in multiple devices with predominant the tin lamps and lanterns (Eston Ngugi 2013). It is estimated that 5.5% of households use electricity for lighting, percentage that matches with the total electrification rate in the County (5.6%). That means that the households that have access to electricity, mainly use it for lighting purposes. In Lurambi constituency the percentage of electricity usage for lighting reaches 21% while in Mumias North the percentage is around 50%. Table 5, presents the lighting sources allocation according to the household preference in the County (Eston Ngugi 2013).

In a survey conducted in 2014 in western Kenya, it was found that the average monthly expenditure on kerosene for lighting was around 891.4 Ksh per household (Sklivaniotis, Integrated Techno-Economic Comparative and Socio-Economic Impact Study for Increasing Energy Access in Rural Kenya 2014). That accounts for 5.9% of the average household income (15,000 Ksh/month) (Denise Law, Kopernik 2013). According to the price of kerosene in Kakamega at that time (May-June 2014) which was 84.68 KES per liter (Energy Regulatory Commission (ERC) 2015), it was estimated that the monthly consumption of Kerosene is about 10.5 liters. That is translated to 126 liters of kerosene per year or 0.34 liter per day for an average household in Kakamega.

In order to give a practical dimension to these numbers, 0.34 liters of kerosene can provide 340 lumen-hours per day (if an average efficacy of kerosene lamps is assumed to be 1000 lumen-hours per liter) (Mills 2003). With average 4.5 hours of lighting necessity per day that accounts to 75.6 lumens available. That is a very low number in terms of usefulness.

On the other hand, for the electrified households, the lighting needs slightly change. It is assumed that an average electrified household in Kakamega has 3-6 lamps (Plas 1994). The assumption involves 25W tungsten incandescent light bulbs providing light for 4-5 hours per day. That implies a daily electricity consumption of about 0.56 kWh for lighting purposes of a household. The average monthly and annual consumption are respectively estimated at 17.4 kWh and 209 kWh. It should be mentioned at this point that under this scenario the luminous flux available is roughly thirteen times higher (1000 lumen) than in kerosene fueled appliances which is an appropriate light source for everyday activities.

Table 5. Percentage of lighting sources for households in Kakamega County (Eston Ngugi 2013).

Source	Percentage (%)
Tin lamp	63.9
Lantern	28.2
Electricity	5.5
Gas lamp	0.8
Solar	0.7
Pressure lamp	0.5
Fuel wood	0.4

Other sources of lighting such as solar power, dry cell (torches), firewood and candles are also evident in Kenya but no concrete data were found for Kakamega County. As a matter of fact, solar powered lighting is a solution for 1.4% of Kenyan households. The same percentage accounts for dry cell devices. It is interesting to notice that in lack of electricity access, households turn mainly to kerosene, dry cell and firewood in order to cover their lighting needs. Biogas is also an option but at a very small extent (Lighting Africa 2012) (Jann Lay 2012).

Water heating

It is estimated that the water usage for bathing and clothes washing is about 13.5 liters/day/household (IIED 2001). Assuming that the necessary temperature of the water should be 45 °C that would imply an energy requirement of 1,129 kJ/day/household⁵ or 0.31 kWh/day/household. Electric heaters show an overall efficiency of 0.96 so that would imply a total energy requirement of 117 kWh/year/household. Additionally, the efficiency of solar water heating systems (SWH) is estimated at 0.55 while the firewood burners for water heating have an efficiency of 0.25. Therefore, the primary energy requirement for the two technologies is about 749 MJ/year/household and 1,648 MJ/year/households respectively. That is, around 165 kg of firewood/year/household for water heating purposes.

Other Uses

The estimated electricity consumed by a household on an annual basis in Kenya is 844 kWh in urban and 544 kWh in rural areas (Sklivaniotis, Integrated Techno-Economic Comparative and Socio-Economic Impact Study for Increasing Energy Access in Rural Kenya 2014). The electrification rate in Kakamega is quite low (5.6%) and even more, electricity is available mainly in the biggest towns (Kakamega – Mumias). Therefore, the energy profile of electrified households is better described by the average 694 kWh per year. That is further enhanced by a survey of 2012, indicating that the mean electricity expenditure of a household in Western Kenya is roughly 300 Ksh per month (Jann Lay 2012). According to the Energy Regulatory Commission of Kenya (ERC) tariff policy, an electricity consumption of 57.8 kWh/month would cost 265 Ksh (under fixed tariff of 120 Ksh and charge 2.5 Ksh/kWh for the first domestic users tier) (ERC 2013).

In addition, as described in a previous paragraph, electricity is mainly used for lighting and minor additional services like radio, TV etc. Cooking is still based on traditional sources of energy (firewood, charcoal). It was also estimated that the average electricity consumption of a household for lighting is about 17.4 kWh per month and for water heating 9.75 kWh per month. With a simple calculation it can be approximated that “other” uses of electricity account for 31 kWh per month or 368 kWh per year for a household. That is also translated to roughly 1 kWh per day for appliances except lighting. TV (188 W), radio (10 W) or fridge (1,200 Wh/day) could be possible uses.

Conclusion

According to the assumptions made, the residential sector shows quite high energy consumption levels. The high consumption mainly derives from the use of inefficient technologies (consequently high requirements of primary energy inputs) in the daily activities (mainly cooking and lighting). Disaggregated data are available in Table 12 at the end of this chapter.

⁵ Water temperature set at 45 °C while ambient temperature at 25 °C. Also the specific heat capacity of water was set at 4.18 kJ/kg.K.

2.2.2 Industrial sector

Sugar production is the main industrial activity in the Kakamega. Three companies are active in this field; Mumias Sugar Company Limited, West Kenya Sugar Company and Butali Sugar Mills. It is estimated that sugar factories consume 28 kWh of electricity per ton of sugarcane crushed and 570 kg of steam per ton of sugar produced (C. Mbohwa 2013) (Francis X. Johnson 2012). The steam used in sugar processing is typically at pressure between 1.2-2 bars and temperature of 200-300 °C, having a specific enthalpy around 2,972 MJ/ton of steam (Mbohwa 2009) (Francis X. Johnson 2012). Under this assumption, the heat requirements in sugar processing can be estimated at about 1.694 GJ/ton of sugar produced. It should be mentioned at this point that the majority of sugar factories cover their energy needs through cogeneration, using bagasse as fuel. The electricity production highly depends on the power house specifications. It is estimated that for small – medium plants it ranges between 24 kWh (Mukhongo 2012) and 30 kWh per sugarcane crushed (C. Mbohwa 2013). A more analytical energy balance for each factory in Kakamega is presented in the following paragraphs.

Mumias Sugar Factory Limited (MSFL)

Mumias sugar factory has the highest capacity for sugarcane crushing in Kenya and accounted to the 36.1% of the county's total sugar production in 2012. More specifically, 1,964,063 tons of sugarcane were crushed in the factory producing 181,372 tons of sugar and 743,791 tons of bagasse⁶ (KETS 2013). According to that data, the electricity consumption of the factory is about 55 GWh/year while the heat consumption about 307,224 GJ per year. It is estimated that the energy intensity is 2.786 GJ/ton of sugar produced with the energy mix involving 60.8% if form of heat and 39.2% as electricity. It should be mentioned at this point that the factory has an installed capacity of 38 MW (35 MW effective capacity) 26 MW of which are connected to the national grid. The generation process runs 7200 hours per year under a low capacity factor of 0.3 (MoEP 2015) (CDM 2006).

West Kenya Sugar Company (WKSC)

As in 2012, WKSC crushed 539,329 tons of sugarcane producing 49,565 tons of sugar and 204,244 tons of bagasse (KETS 2013). Following the same assumptions as in the previous case that would imply an electricity consumption of 15.1 GWh/year and a heat consumption of 83,963 GJ/year. It is estimated that the energy intensity is 2.791 GJ/ton of sugar produced with the energy mix involving 60.7% if form of heat and 39.3% as electricity.

Butali Sugar Mills

Finally, according to KETS report, Butali Sugar Mills crushed 415,546 tons of sugarcane and produced 42,671 tons of sugar and 157,367 tons of bagasse in 2012 (KETS 2013). Following the same process as in the previous cases it is estimated that the electricity consumption for the plant is about 11.6 GWh/year while the heat consumption about 72,284 GJ/year. It is estimated that the energy intensity is 2.676 GJ/ton of sugar produced with the energy mix involving 63.3% if form of heat and 36.7% as electricity.

⁶ Sugarcane to bagasse ratio was found to be 37.87% (Mukhongo 2012).

Conclusion

Therefore, the total industrial electricity consumption in Kakamega is 81.7 GWh/year while the heat requirements for industrial processes is estimated at 463,471 GJ per year.

2.2.3 Transportation sector

The number of vehicles is estimated according to the national data on registered vehicles and the population (Kakamega/Kenya – 3.7%) ratio. Table 6, gives an initial estimation of the number of vehicles in Kakamega while Table 7 provides the number of new registered cars in the county (KNBS 2014) (KNBS 2014).

Table 6. Estimated number of registered vehicles in Kakamega (KNBS 2014).

Motor cars	26263
Utilities, parcel vans, pick ups	9331
Lorries, Vans & heavy Vans	4350
Busses, Mini Busses	3539
Motor & Auto cycles	27314
Trailers	1470
Other motor vehicles	2176
Total	74443

Table 7. Estimated number of new registered (2013) vehicles in Kakamega (KNBS 2014).

Motor cars	2405
Utilities, parcel vans, pick ups	363
Lorries, Vans & heavy Vans	354
Busses, Mini Busses	85
Motor & Auto cycles	4742
Trailers	147
Other motor vehicles	124
Total	8220

In order to calculate the energy consumption in the transportation sector, a number of parameters were introduced and assumed. Therefore, by taking into account the average annual distance covered, the fuel economy for each segment and the load factors, the final energy intensities were estimated as shown in Table 8 by using the following equation:

$$\text{Energy Intensity} \left(\frac{\text{liter}}{\text{pass_km}} \right) = \frac{1}{\text{Fuel economy} \left(\frac{\text{veh_km}}{\text{liter}} \right) \times \text{Load factor} (\text{pass_km/veh_km})}$$

Two categories were suggested, namely passenger transportation and freight transportation.

The first category involves all the means of transport that are used in passenger mobility (motor cycles, cars, busses) and are highly correlated with the population and income level. An average value of 1,482 passenger-km/person/year was estimated based on the following equation:

$$Pass_{km_{person_{year}}} = \frac{1}{population} \times \sum_{i=1}^3 [NoV_i \times mileage_i \times Load\ factor_i]$$

Where,

NoV: Number of Vehicles

Mileage: Distance covered (in km) in one year

Load factor: Pass_km/Veh_km

And **i=1** stands for motor cars, **i=2** stands for busses and mini busses and **i=3** stands for motor cycles.

It should be mentioned at this point that walking and cycling were also introduced as means of transport in this category as high portion of the population in Kakamega is based on them. These two activities do not contribute to the energy consumption in actual terms, but they should be taken into consideration in the policy formation as they highly affect the future of the sector.

The second category includes freight transport in Kakamega. Following the same principle as before, the energy intensities were calculated for all the vehicles involved (utilities, parcel vans, pick-ups, lorries, trailers and other) on the base of GDP contribution. The average value was estimated 1.27 ton-km/USD and used in the model to calculate the total energy consumption.

$$Ton_{km_{year}} = \frac{1}{County's\ GDP} \times \sum_{k=1}^4 [NoV_k \times mileage_k \times Load\ factor_k]$$

Where, **k=1** stands for utilities, parcel vans and pick-ups, **k=2** stands for lorries, vans and heavy vans, **k=3** stands for trailers and **k=4** stands for mixed other vehicles.

It was observed that motorcycles and conventional cars are the most common vehicles in the roads of Kakamega. The average growth rate of registered vehicles was calculated for the years 2004-2013 to 12.5% per year with motorcycles having the highest rate of 36.3% (U.S Department of Energy n.d.), (Oak Ridge National Laboratory n.d.), (United Nations Environment Programme n.d.), (UNEP 2006).

Conclusion

The energy consumption was estimated as 745.9 GWh for gasoline and 898.5 GWh for diesel, accounting for 1,644 GWh/year for the total transportation sector in Kakamega. It should be mentioned at this point that measuring the long run fuel demand in transportation sector is a quite challenging task. This thesis adopted a basic approach in order to have some initial preliminary results. It is suggested that several factor should be added (fuel price, income, taxation, road investment policies etc.) in order to provide higher accuracy on the projections. Here is some suggested literature (Olof Johansson 1997).

Table 8. Energy Intensities in transportation sector in Kakamega (UNEP n.d.), (Kimmo Erkkilä 2005), (US Department of Energy n.d.), (US Department of Transportation n.d.) (OAK Ridge National Laboratory n.d.).

Transportation	Number of Vehicles	Percentage	Growth rate (%)	Vehicle-km per year	Fuel economy (liter per vehicle-km)	Load factor (Pass-km/veh-km)	Energy Intensity (liter/pass-km)
Motor cars	26263	35%	9.75	17082		2.5	
Gasoline (75%)	19697.25				0.072		0.0288
Diesel (25%)	6565.75				0.091		0.0364
Busses, Mini Busses (Matatu)	3539	5%	6.97	22000		15	
Diesel	3539				0.280		0.0187
Motor & Auto cycles	27314	37%	36.26	3574		1.5	
Gasoline	27314				0.054		0.0360
Walking							
Cycling							
						(ton-km/veh-km)	(liter/ton-km)
Utilities, parcel vans, pick ups	9331	13%	3.85	21306		2.5	
Gasoline (75%)	6998.25				0.123		0.0493
Diesel (25%)	2332.75				0.156		0.0624
Lorries, Vans & heavy Vans	4350	6%	7.02	21671		7.5	
Gasoline (25%)	1087.5				0.126		0.0167
Diesel (75%)	3262.5				0.157		0.0209
Trailers	1470	2%	16.65	50000		15	
Diesel	1470				0.375		0.0250
Other motor vehicles	2176	3%	6.99	16093		8.5	
Gasoline (50%)	1088				0.094		0.0110
Diesel (50%)	1088				0.212		0.0249
Total	74443	100%	12.5				

2.2.4 Commercial sector

The commercial sector in Kakamega mainly comprises of small and medium scale enterprises. These enterprises are characterized by low production cost, accessibility to raw materials, proximity to the markets and generally low level of mechanization which entails low productivity. Examples of such enterprises found across the county are engaged in activities like pottery, crude sugar production, brick-making, liquor production, quarrying and masonry, carpentry, traditional medicine production, charcoal production, basketry and weaving, baking, bicycle repair, flour-grinding, and shoe-making and repair (Makokha 2014). The division is made regarding the energy consumption into these two categories respectively.

Small scale enterprises

In this category the energy demand is respectively low and mainly regards low load electrical applications. Lighting and professional equipment few hours per day compose the consumption profile, while the total energy demand is up to 10 kWh/day. It was assumed that there are 1,000 businesses of this kind (according to data retrieved from Open Street Map for the region) in the county accounting for 3,650 MWh/year.

Medium/large scale enterprises

This category contains commercial activities with higher energy consumption between 10 kWh/day and 100 kWh/day. Such activities involve super markets, open markets, shopping malls, banks, hotels etc.

Hotel – Guest houses - Restaurants

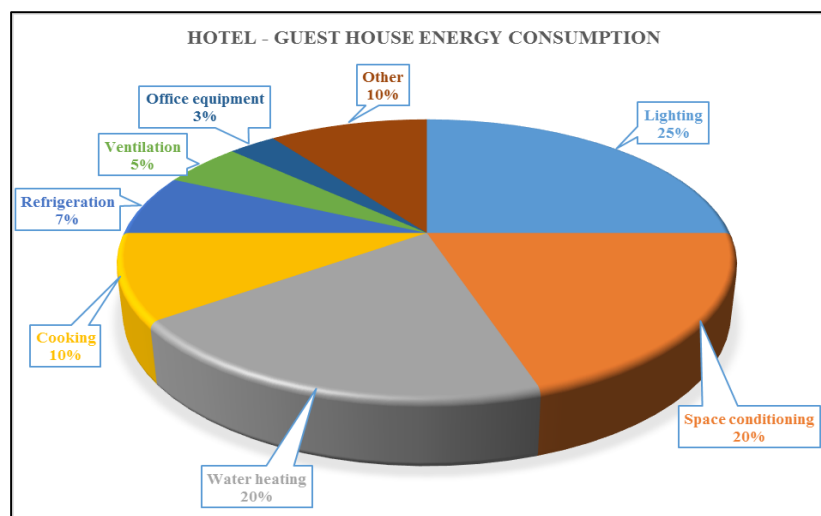
The energy consumption in hotels and guest houses in Kakamega follows the profile presented in Graph 1. It is estimated that there are 174 hotels – guest houses in the area with a capacity of 1,340 beds and an occupancy rate of 30% (KNBS 2014). It is assumed that the average electricity consumption is 30 kWh/day while heat required for water heating⁷ is about 30 MJ/day and for cooking⁸ 10 MJ/day. That would imply a total consumption of 1,905 MWh/year of electricity and 2,540 GJ/year of heat.

Super markets, Open markets & Shopping malls

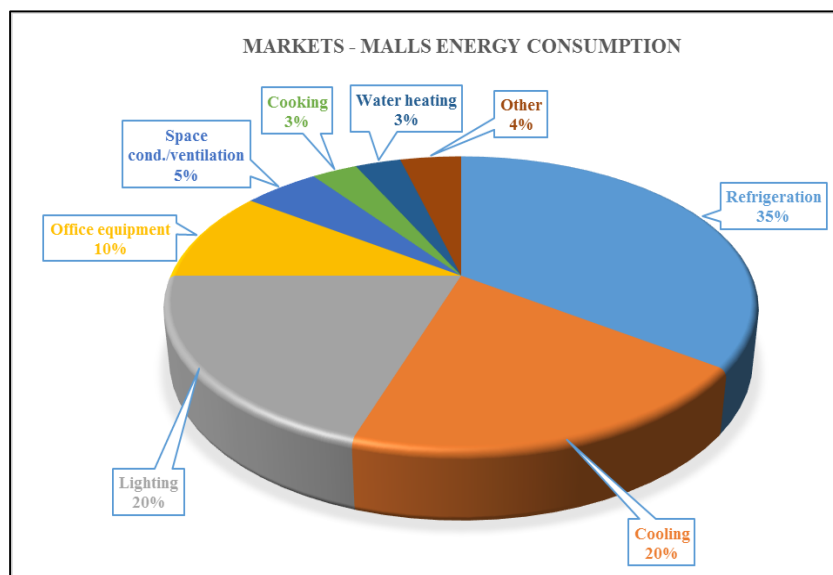
Electrification of markets is one of the direct plans in Kakamega (governor speech 2015). The energy demand is mainly derived from electricity requirements for refrigeration, cooling and lighting. Heat is necessary for cooking/baking purposes. The energy profile of markets in Kakamega is assumed as shown in Graph 2. It is estimated that the electricity consumption is 50 kWh/day while 25 MJ/day are necessary for heating purposes. It is assumed that there are 300 markets of this kind in the county. That implies an electricity consumption of 5,475 MWh/year and heat consumption of 2,738 GJ/year.

⁷ It was assumed that the average water consumption for a 10-minute shower is 35 liters and that hotels have 5 guests per day on average. Water temperature set at 60 °C while ambient temperature at 25 °C. Also the specific heat capacity of water was set at 4.18 kJ/kg.K. Efficiency of the boiler at 0.85.

⁸ Average energy required for cooking assumed at 1.7 MJ/day. Also 5 guest/day was assumed for each hotel.



Graph 1. Estimation of the energy consumption profile in hotels and guest houses in Kakamega.



Graph 2. Estimation of the energy consumption profile in big open markets in Kakamega.

Banks and other businesses

This category includes mainly service oriented commercial activities, whose energy demand derives mainly from electricity requirements for office equipment and lighting. It is assumed that 25 kWh/day cover these needs with 20% devoted to lighting and 80% to equipment (computers, monitors etc.). With 250 businesses of this type active in Kakamega, that implies an energy consumption of about 2,281 MWh/year.

Conclusion

In total, the commercial sector in Kakamega shows an electricity demand of 13,356 MWh/year and a heat demand of 5,278 GJ/year.

2.2.5 Community sector & Public services

This section involves the energy demand of activities addressing community needs. More specifically, the energy requirements are estimated for Hospitals and health clinics, Schools, Community buildings & Public offices and Street lighting.

Hospitals & health clinics

It is found that in Kakamega there are 119 health facilities from which 17 are hospitals, 32 clinics the rest health centers and dispensaries (Kakamega County n.d.) (Google maps 2015) (Open Street Map). According to USAID, health facilities can be categorized into three groups according to their energy requirements. In this report it is assumed that hospitals in Kakamega fall into the third category having a daily energy requirement of 25 kWh while clinics, health centers and dispensaries have an average energy requirement of 10 kWh per day (combination of category 1 and 2) (USAID n.d.). That is, the energy consumption for hospitals and health clinics in Kakamega is estimated at 1,445 kWh per day or 527.4 MWh per year. Further explanation of the categories can be found in APPENDIX D.

Educational buildings

Primary & secondary education

Recent data shows that 24% of the population in Kakamega are children between the ages of 6-13 and 11% children between 13-17 years old (Republic of Kenya 2006). The attendance to primary education is very high in the region and in this report is assumed 100%. On the other hand attendance to secondary education is estimated at 55% (UNESCO-IBE 2011). Under these assumptions, the primary education students in Kakamega in 2013 are estimated 394,639 while the secondary education students 99,482.

With 300-500 students registered in each school, that implies a rough estimation of 1000 primary schools and 300 secondary schools in the County. In this report the latest data obtained on the number of schools will be used. Therefore the number of primary schools is 938 while the secondary schools are 276 (Kakamega County n.d.).

The estimated 1.515 kWh/day (Table 9) can be an indicative value of average energy consumption in schools in Kakamega. Under this assumption the total estimated energy demand is estimated at 1,840 kWh/day or 671.6 MWh/year.

Higher education

In terms of higher education, Kakamega is the home of Masinde Muliro University of Science and Technology (MMUST). According to Dimitriadis, the energy consumption of a university building averages at 0.12 kWh/day/m² (Dimitriadis 2011). Assuming that MMUST area is 5,000 m² that would imply an energy consumption of 600 kWh/day or 219 MWh/year.

Therefore, the total energy demand of educational buildings in Kakamega is assumed 1,418 MWh/year.

Table 9. Energy consumption of various appliances utilized in schools (NREL 2000).

Appliances	Power (Watts)	Duration (h/day)	Energy/day (Wh)
Lights (fluorescent)	5-30	2-12	122.5
Overhead fan	40	4-12	320
TV (19" color)	60	1-4	150
VCR/DVD	30	1-4	75
AM/FM Radio	15	1-12	97.5
Computer (desktop+screen)	150W	2-8	750
Electric consumption	-	-	1515

Street Lighting

Kakamega has approximately 400 km of main roads including (A1, B2, C33, C40 and C41) and 200 km additional road connections within the county (Google maps 2015). It is assumed that 25% of these roads are lit so that 150 km are covered by street lighting. With one light pole per 30 meters that means that 5,000 lamps light up the streets in the county every night. Assuming that 70W - HPS (High Pressure Sodium) lamps are used, operating 4,000 hours/year, that implies an annual energy consumption of 1,400 MWh/year.

Community buildings

In this category are included all the energy demands of activities that serve community needs and are not in the previous categories, namely public offices, police stations, military buildings, sport facilities, libraries etc. It is assumed that there are approximately 100 building serving such purposes in Kakamega base on data retrieved from open street map. The energy requirements of such buildings is set at 50 kwh/day with 15% devoted to lighting and 75% to electric appliances (computer, cooling, special equipment etc.). That would imply an electrical demand of 1,825 Mwh/year.

Conclusions

The total energy consumption in the Community sector is 4.6 GWh/year, which account all for electricity. It should be mentioned at this point that additional fuels (charcoal, briquettes, oil products etc.) were not included in this sector. That is one of the limitations of this study and should be revised in future work.

2.2.6 Agriculture, Aquaculture & Livestock

Agriculture activities contribute by 10%⁹ to Kakamega's GDP, while accounting for 0.54% of the County's energy consumption. In the absence of validated data for Kakamega, this value was estimated based on the national energy shares between sectors in Kenya (J.K. Kiplagata 2011).

About 90% (Lübker 2013) of the population in Kakamega depend directly or indirectly from activities related to agriculture such as farming, agro-processing, post-harvest and storage facilities, marketing, retail and infrastructure. All these activities are part of the value chain and show an increasing demand for modern energy services and related equipment. Energy needs in agricultural activities include direct and indirect inputs. The first term refers to energy requirements for land preparation, cultivation, irrigation, harvesting, post-harvest processing, food production, storage and transport of agricultural quantities. The latter, refers to needs in the form of sequestered energy in fertilizers, herbicides, pesticides, and insecticides.

The aggregated energy consumption of agricultural activities in Kakamega, is estimated at 31 GWh per year. The form of energy required highly depends on the activity and it can vary between electrical, mechanical (motive power) and thermal energy. It is assumed that 50% of the energy demand derives from motive power, 30% by electrical energy and 20% by thermal energy requirements in the sector. Table 10, presents examples of activities and their respective energy requirements.

Table 10. Farming activities and respective energy requirements

Activity	Modern Energy form
Land preparation/tilling	Motive power
Seeding	Motive power
Harvesting	Motive power
Drip-feed/sprinkler irrigation	Electrical
Grain milling	Mechanical/electrical
Oil pressing	Mechanical/electrical
Drying (fruits, vegetables, coffee, tea, meat, fish, spices)	Thermal
Smoking (fish, meat, cheese)	Thermal
Food & drink cooling (milk chilling/pasteurisation)	Electrical
Ice-making (fish storage)	Electrical
Water heating	Thermal
Sawmilling	Mechanical/electrical
Improved warehousing	Electrical
Water circulation and purification (fish hatcheries and fish farms)	Mechanical/electrical
Lighting	Lighting

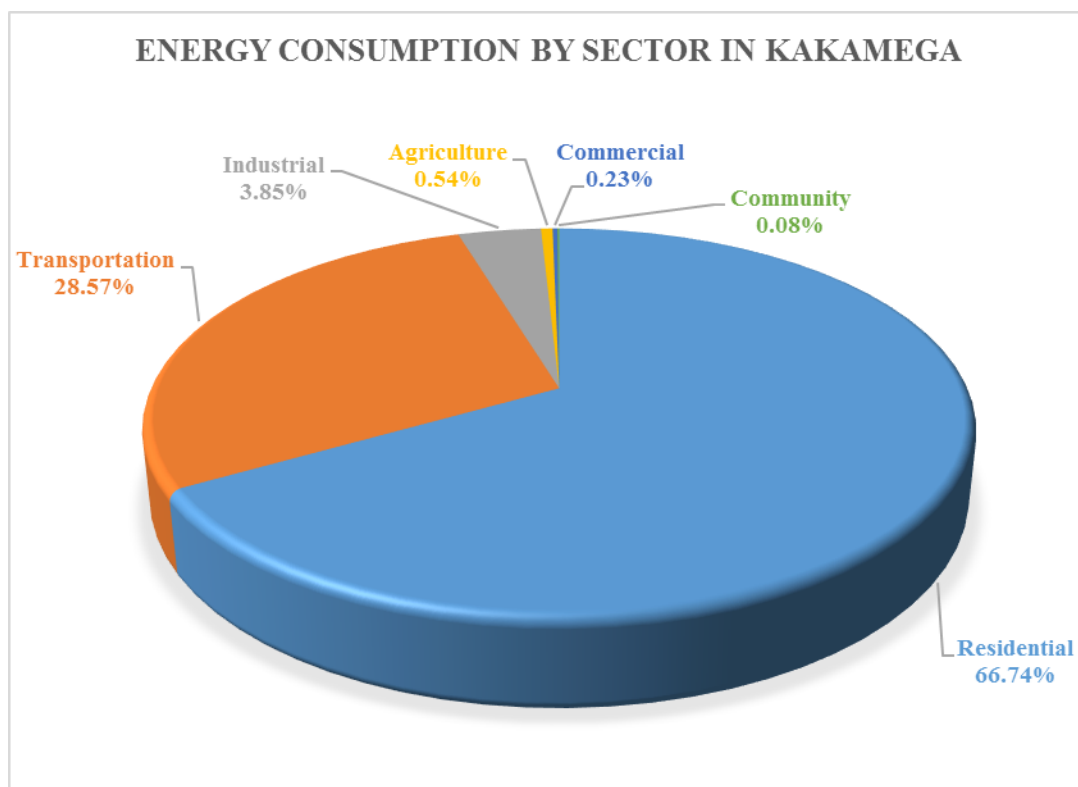
⁹These are assumptions made by the author due to the absence of validated data of these values in Kakamega. The assumptions reflect regional trends in Eastern African counties. It should be noticed that the LEAP model was developed on these values.

2.2.7 Estimated energy demand profile in Kakamega

According to the disaggregated analysis presented above, the following results were estimated for the sectorial energy consumption in Kakamega.

Table 11. Aggregated energy consumption estimations per sector in Kakamega.

Sector	Units		Percentage (%)
	(ktoe)	(GWh)	
Residential	330.2	3841.0	66.74
Transportation	141.4	1644.0	28.57
Industrial	19.0	221.3	3.85
Agriculture	2.7	31.0	0.54
Commercial	1.1	13.0	0.23
Community	0.4	4.6	0.08
Total	494.8	5755	100.00



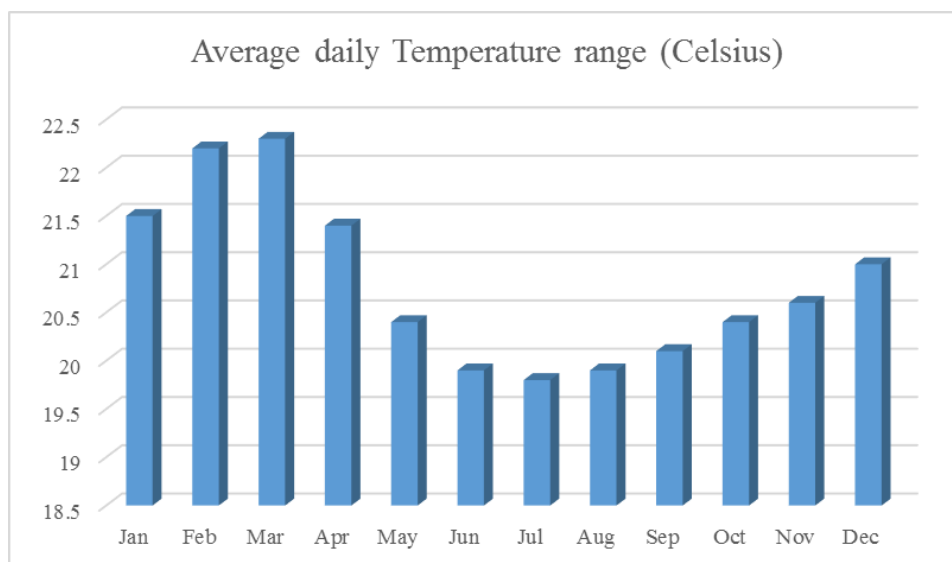
Graph 3. Energy consumption by sector in Kakamega.

Table 12. Estimated fuel consumption in GWh per sector in Kakamega in 2014.

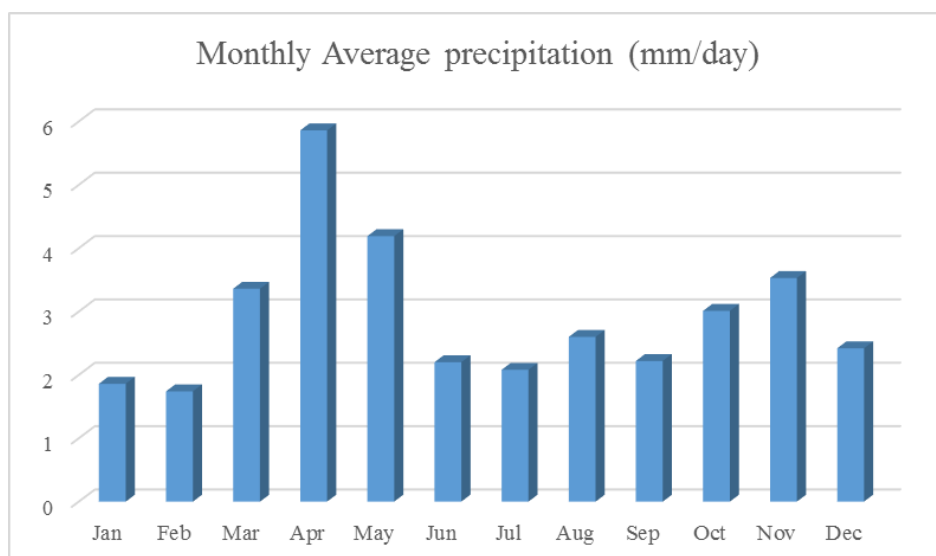
Sectors	Electricity (grid)	Gasoline	Kerosene	Diesel	LPG	Wood	Charcoal	Biogas	Bagasse	Electricity (solar)	Heat	Total
Residential	13.5	-	434.8	-	3.3	3,153.7	230.7	4.2	-	1.0	-	3,841.3
Electrified	13.5	-	2.1	-	0.3	84.2	51.9	0.2	-	0.3	-	152.4
Unelectrified	-	-	432.7	-	3.0	3,069.6	178.8	4.0	-	0.7	-	3,688.8
Industry	-	-	-	-	-	-	-	-	221.3	-	-	221.3
Mumias Sugar Company Limited	-	-	-	-	-	-	-	-	147.4	-	-	147.4
West Kenya Sugar Company	-	-	-	-	-	-	-	-	40.3	-	-	40.3
Butali Sugar Mills	-	-	-	-	-	-	-	-	33.6	-	-	33.6
Commercial	11.5	-	-	-	-	-	-	-	-	-	1.5	13.0
Small Scale Businesses	1.8	-	-	-	-	-	-	-	-	-	-	1.8
Medium or Large businesses	9.7	-	-	-	-	-	-	-	-	-	1.5	11.2
Agriculture	6.2	-	-	15.5	-	-	-	-	-	-	9.3	31.0
Agro_demand	6.2	-	-	15.5	-	-	-	-	-	-	9.3	31.0
Community	4.5	-	-	-	-	-	-	-	-	0.1	-	4.6
Education	0.9	-	-	-	-	-	-	-	-	-	-	0.9
Health	0.5	-	-	-	-	-	-	-	-	-	-	0.5
Community build.	1.8	-	-	-	-	-	-	-	-	-	-	1.8
Street L	1.3	-	-	-	-	-	-	-	-	0.1	-	1.4
Transportation	-	1,124.0	-	520.3	-	-	-	-	-	-	-	1,644.3
Passenger Transportation	-	610.5	-	135.4	-	-	-	-	-	-	-	745.9
Freight Transportation	-	513.5	-	384.9	-	-	-	-	-	-	-	898.5
Total Demand	35.8	1,124.0	434.8	535.8	3.3	-	230.7	4.2	221.3	1.2	10.8	5,755.6

2.3 Climatic conditions and resource assessment in Kakamega

The average temperature in Kakamega is 20.8 °C with the monthly average been shown in Graph 4. The average hours of sunlight are 12.1 per day and the average precipitation is 2.923 mm/day (monthly average shown in Graph 5) (NASA - Atmospheric Science Date Center n.d.). This is a good indicator for rain-fed crops described later on the report.



Graph 4. Average daily temperature range in Kakamega (°C) (NASA - Atmospheric Science Date Center n.d.).



Graph 5. Average monthly precipitation in Kakamega (°C) (NASA - Atmospheric Science Date Center n.d.).

2.3.1 Biomass and Bioenergy crops

Kakamega has a total area of 3,020 km² from which around 30% is devoted to cash crops (mainly sugar cane (60%) in the south region and partially sunflower, coffee and tea in the rest of the county). According to the KETS report in 2013 the average yield of sugar cane in the region of Kakamega was 55 tons/hectare or 5,500 tons/km². Based on the same report, the estimated land devoted in the cultivation of sugar cane in Kakamega was 540 km² (KETS 2013). Cultivation of maize is also very important in the county and it occupies almost 70% of the land of north Kakamega (Lugari and North Kakamega districts).

The gazette forest land is 282 km² mainly comprising by five areas namely Viz, Lugari, Turbo, Nzoia, Malava and Kakamega forest) (Ngetich 2013). The forest area accounts for 9.3% of the county's territory, number which is slightly below the marginal value of 10% of tree coverage, set by the National Forest Policy in 2014 (Ministry of Environment, Water and Natural Resources 2014). That is, the aforementioned forest areas should not be accounted as potential biomass source in the County.

Additionally, by taking into consideration the population and the area of the biggest towns in Kakamega it was calculated that on average every urban citizen occupies 5×10^{-4} km². That is, with the urban population being 15.2%, the total urban area of Kakamega is roughly 128 km². Finally, it was estimated that the total area not occupied by human activities showing the potential for natural biomass is about 936 km².

Table 13. Estimated land use in the county of Kakamega.

Land Use	Area (sq. km)	% of total
Cash crops	906	30.0
Food crops	768	25.4
Gazetted forests	282	9.3
Urban	128	4.2
Other	936	31.0
Kakamega	3020	100

Woody biomass

According to FAO, the area of Kakamega has a woody biomass availability to a rate of 12,500 tons/km² (Drigo 2005). That means that the available woody biomass in the county (based on the 963 km² value) is estimated around 11.7 million tons or approximately 93.6 million GJ¹⁰.

Bagasse and Vegetal Wastes (Molasses)

The calculation of bagasse availability in the county was based on the sugar cane production on the base year. Therefore, in 2013 around 2.9 million tons of sugar cane were crushed yielding 1,105,382 tons of bagasse (KETS 2013). The calorific value of bagasse depends on the moisture content and lies between 8 – 14 MJ/kg (The Engineering Toolbox 2015). Hence, the available bagasse in Kakamega was estimated to be approximately 12.5 million GJ.

Following the same procedure based on the sugar cane crushed in Kakamega on the base year the availability of vegetal waste (specifically molasses) was estimated. Every ton of sugar cane crushed yields 3.5% of molasses (Francis X. Johnson 2012) which have a calorific value of 12.13 MJ/kg (The Engineering Toolbox 2015). In 2013, approximately 102.2 thousand tons of molasses were produced implying an available energy equivalent of 1.24 million GJ.

Sustainability index for various energy crops

As described above, Kakamega has a very good potential for rain-fed crops. Table 14 presents which energy crops could be cultivated in the County. Figures 20 – 30 (APPENDIX E) illustrate the geographic area where each energy crop yields a high sustainability index for high input, rain-fed conditions.

¹⁰ Using an average heating value of 8MJ/kg.

Table 14. Energy crops with high sustainability index (IRENA Global Atlas 2015).

Crop	Sustainability Index	Crop	Sustainability Index
Barley	High sustainability index	Reed canary grass	No
Cassava	High sustainability index	Sorghum	High sustainability index
Coconut	No	Soybean	High sustainability index
Jatropha	High sustainability index	Sugar beet	No
Maize	High sustainability index	Sugarcane	High sustainability index
Miscanthus	High sustainability index	Sunflower	High sustainability index
Palm oil	No	Switch grass	No
Rape	High sustainability index	Wheat	High sustainability index

2.3.2 Solar Resource

Global Horizontal Insolation

According to GIS data retrieved from IRENA Global Atlas, the average annual global horizontal irradiance in Kakamega is approximately 237 W/m² as shown in Figure 7. This provides a potential of about 5.7 kWh/m²/day of global horizontal insolation (GHI) as shown in Figure 8.

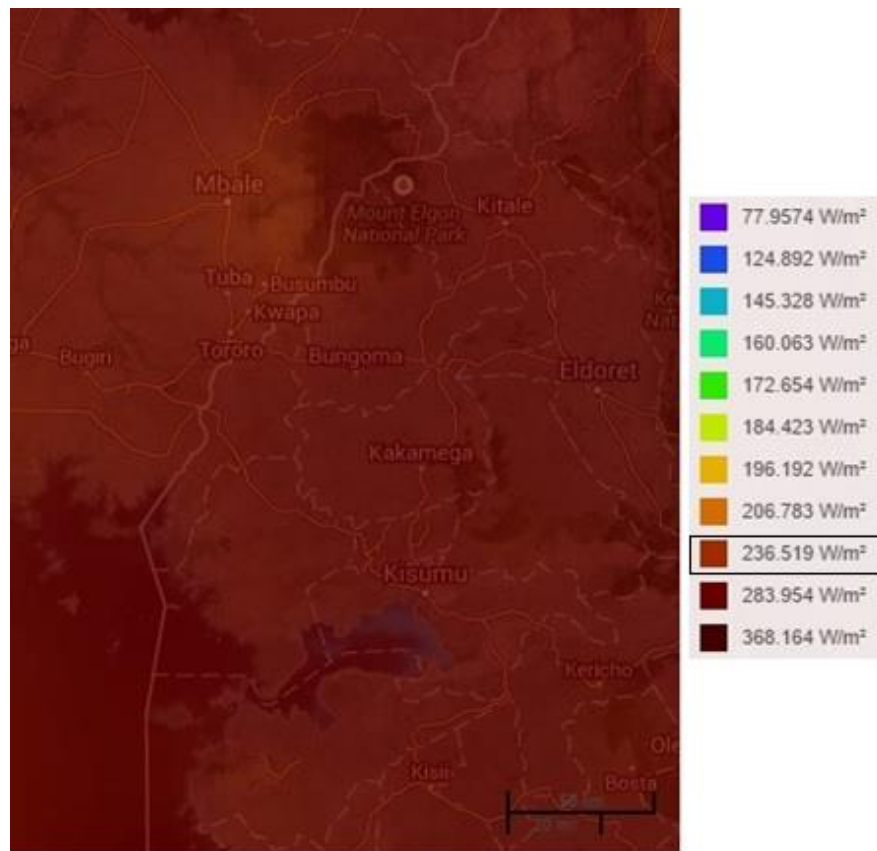


Figure 7. Global Horizontal Irradiance annual average for Kakamega County (IRENA Global Atlas 2015).

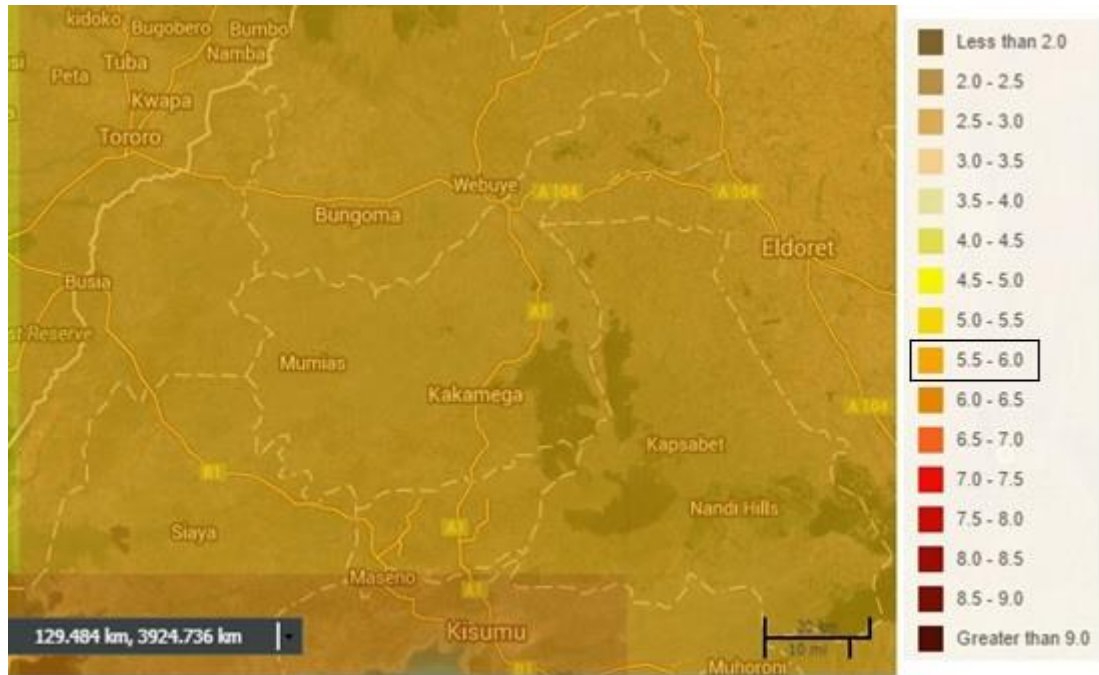
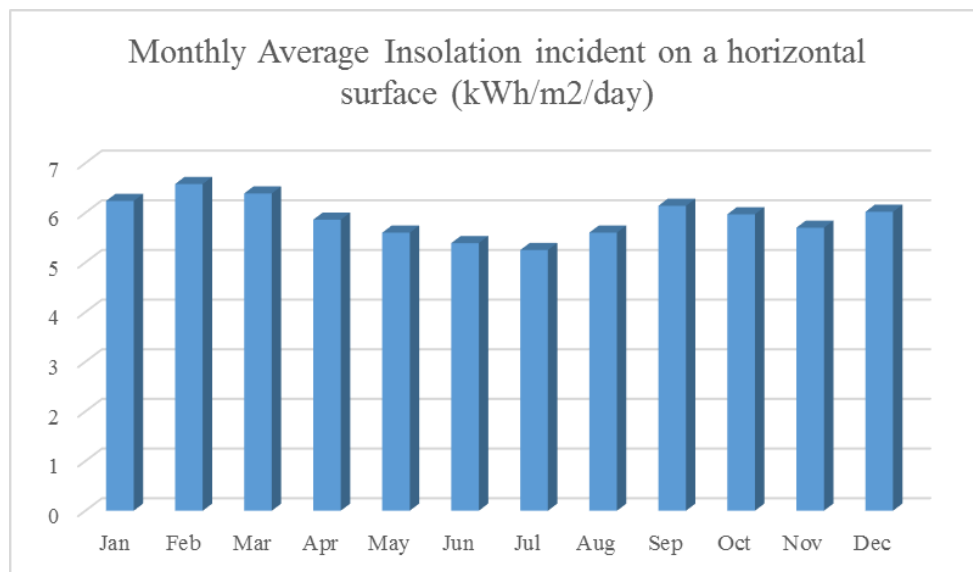


Figure 8. Global Horizontal Insolation in kWh/m²/day for Kakamega County (IRENA Global Atlas 2015). Furthermore, NASA Laboratory, providing 22 year average data (1982 – 2004) gives a monthly average global horizontal Insolation for Kakamega which averages at 5.9 kWh/m²/day (Graph 6).



Graph 6. Monthly Average Insolation kWh/m²/day (NASA - Atmospheric Science Date Center n.d.).

Direct Normal Insolation

According to GIS data retrieved from IRENA Global Atlas, the average annual Direct Normal Insolation (DNI) in Kakamega is between 4 – 4.5 kWh/m²/ day as shown in Figure 9.



Figure 9. Direct Normal Insolation in kWh/m²/day for Kakamega County (IRENA Global Atlas 2015).

SolarGIS ranges the DNI potential between 1700 – 1900 kWh/m² per year, which implies an average value of about 4.9 kWh/m²/day as shown in Figure 10.

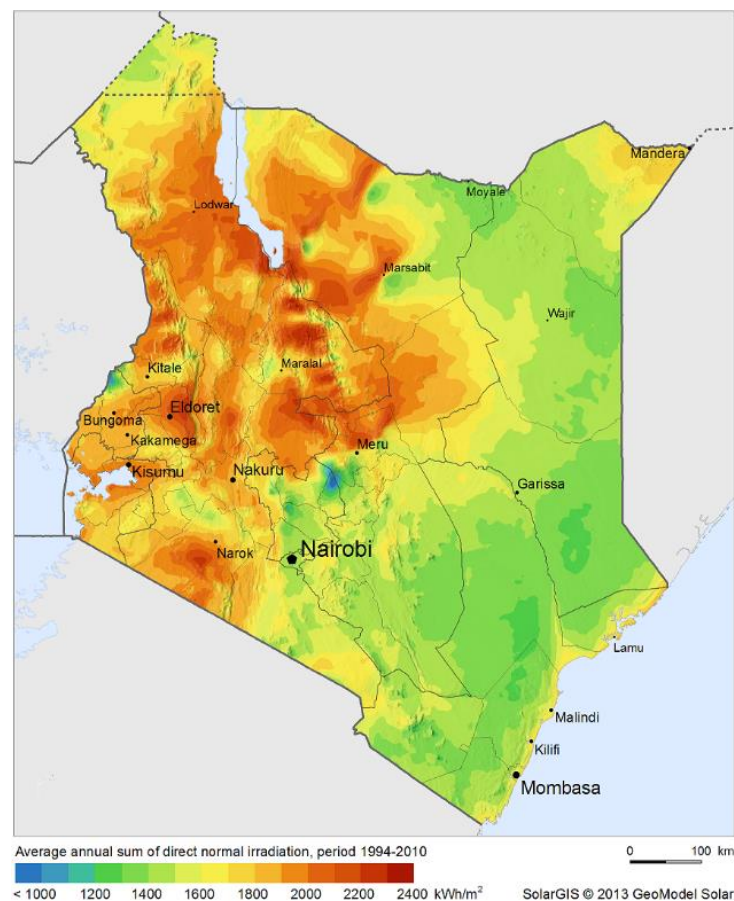


Figure 10. Direct Normal Insolation on annual basis for Kenya in kWh/m² (SolarGIS, 2015).

Therefore, the average global horizontal insolation for the region of Kakamega was estimated to be 5.8 kWh/m²/day while the direct normal insolation 4.6 kWh/m²/day (value correlated with concentrated solar

power technologies). In order to estimate the solar potential in the county the area of 187¹¹ km² was assumed to be suitable for solar technologies deployment. Hence, the available solar energy in the county was estimated at 1.43 billion GJ per year.

2.3.3 Hydro Resource

River Nzoia and River Yala are the major water resources in the County. However, there are about fifteen tributaries to Nzoia River (Ikhamala, Isuikhu, Lusumu, Kipkareen, Sasala, Laini Luandati, Sergoit, Lumakanda, Vivatsi and many other smaller). It is estimated the monthly average of the water sources ranges between 2.8 – 10.8 m³/s. River Nzoia has an estimated annual average discharge between 52 – 70 m³/s (Ngetich 2013).

In order to quantify the hydro potential in the county a number of assumptions were made. Kakamega lies within a latitude of 250 – 2,000 meters (SoftKenya n.d.). Therefore a net head of 250 meters for the prospective power plants can be a plausible scenario. By accumulating the monthly average flows of the aforementioned rivers and assuming an efficiency of 0.85, the hydropower equation¹² yields a power potential of around 280 MW in the county. It was further assumed that 1 MW yields approximately 5.22 GWh¹³ per year (Renetech 2015). Hence, under these assumptions the energy potential from small hydro in Kakamega was found to be 5.3 million GJ per year.

2.3.4 Wind Resource

The wind potential is not high in Kakamega and it ranges between 3 - 4.2 m/s throughout the year. Figure 11 illustrates the annual average wind speed in the County measured at 10 m height while Graph 7 the wind speed frequency throughout the year. Nasa Laboratory etc. provides a monthly average of the wind speeds in the area measured at 50 m height (Graph 8). The average wind speed is estimated at 4.07 m/s which implies a low potential for commercial exploitation of the wind resource in the area.

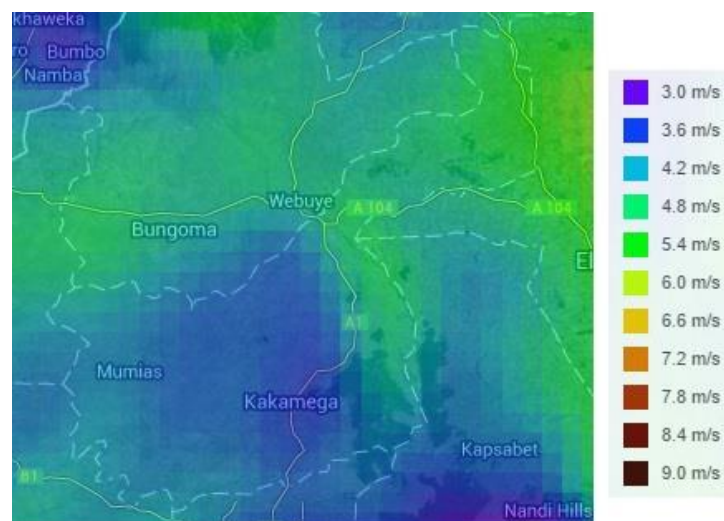
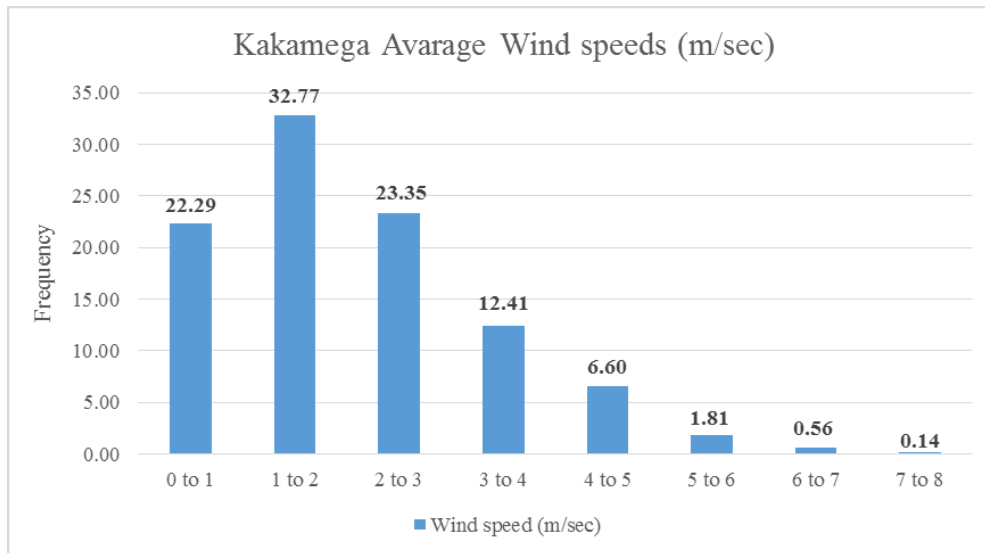


Figure 11. Wind speeds in m/s throughout Kakamega County (measurement at 10 m) (IRENA Global Atlas 2015).

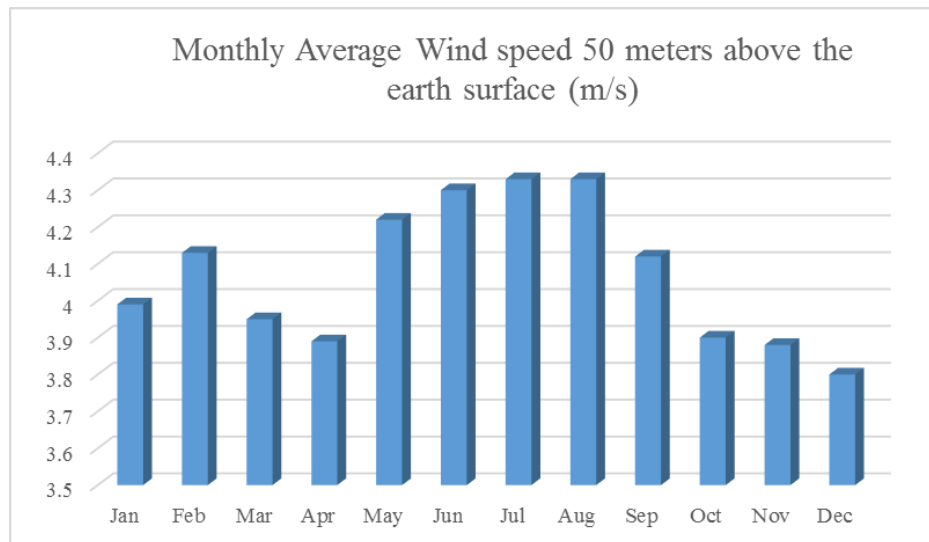
¹¹ This value represents the 20% of the land area in Kakamega that is not directly occupied by human activities according to table 13.

¹² $P=Q \cdot \rho \cdot g \cdot h \cdot \eta$ where Q: water flow (m³/s), ρ : density of water (kg/m³), g: acceleration of gravity (m/s), h: net head (m), η : efficiency.

¹³ Based on Renetech AB calculations on a similar project.



Graph 7. Wind speed frequency on annual basis for Kakamega region in m/s (measurement at 10 m) (NASA - Atmospheric Science Date Center n.d.).



Graph 8. Monthly Average wind speed for Kakamega region in m/s (NASA - Atmospheric Science Date Center n.d.).

2.3.5 Geothermal Resource

There are no finding of geothermal sources available to a commercial level in Kakamega. Therefore, such a solution will not be further researched under the framework of this thesis.

Conclusions

The preliminary assessment of the renewable energy resources in Kakamega showed that solar, small hydro and biomass have high potential. The forms under which these sources of energy could be leveraged in the future, are explicitly described in *CHAPTER 3. ENERGY MODELLING*.

CHAPTER 3. ENERGY MODELLING

This chapter elaborates on the energy modelling process followed in this thesis. It specifically describes the LEAP model structure developed for the representation of the energy sector in Kakamega and the insertion of data under current, reference and alternative scenarios.

3.1 Energy modelling & methodology

Energy models are a useful tool in order to analyze, plan and manage the energy transition which is evident around the world nowadays. There is a wide range of energy models, employing different techniques and serving various purposes. However, according to Frauke Urban, there are roughly twelve models suitable for the evaluation of energy systems behavior in developing countries (AIM, ASF, IMAGE-TIMER, MARIA, MARKAL, MiniCAM, LEAP, PowerPlan, RETScreen, MESSAGE, SGM, WEM) (Urban 2009). Long-range Energy Alternative Planning system (or LEAP) is one of them and was selected for the purposes of this thesis.

LEAP is a scenario based energy/environment accounting modelling tool that can follow a top-down, a bottom-up or a mixed/hybrid approach of energy systems representation. Through the collection of disaggregated data on different energy related activities it allows the analysis of explicitly set technological specification and detailed end-user choices. That implies that the model is independent from market behavior and production frontiers. Furthermore, its ability to incorporate electrification rates, urban-rural classification, power sector performance and traditional biofuels in the analysis, makes it suitable for regional energy modelling in developing countries (Urban 2009) (Stockholm Environment Institute 2015).

However, there are some characteristics of the energy sector in developing countries which are not taken into consideration such as the impact of the informal economy on GDP, supply shortages, structural economic changes and investment decisions. In order to cope with these issues, a different approach was selected during the model formation, inspired by the multi-tier methodology of energy access measurement, proposed by ESMAP.

This methodology proposes an alternative, more comprehensive approach of assessing the energy access of the subject areas. Instead of the traditional binary metrics (Access or No Access), eight attributes and five tiers are used in order to provide a more disaggregated and detailed analysis of energy access. It is believed that information exchange between this methodology and the LEAP model, can provide a broad understanding of how energy interventions can be translated into development goals, and thus was highly encouraged in this thesis.

Therefore, under the scope of this thesis, LEAP was used in order to create a benchmark of energy uses and greenhouse gas emissions over six sectors in the selected area (Kakamega County, Kenya), make projections of the energy supply and demand over a fifteen year planning horizon and identify the cost requirements for the suggested renewable energy integration. The following paragraphs provide a detailed description of the modelling process.

3.2 Data structure

As mentioned previously, six main sectors were selected at the initial stage of data collection. The modelling process required however a more disaggregated approach, which was inspired by the multi-tier methodology of energy access measurement. Figure 12 illustrates the level of disaggregation followed in the model formation. This structure allows each sector to be processed and analyzed individually but also offers aggregated results if necessary. In order to get a better understanding of the modelling process a more detailed description of each sector characteristics is presented below.

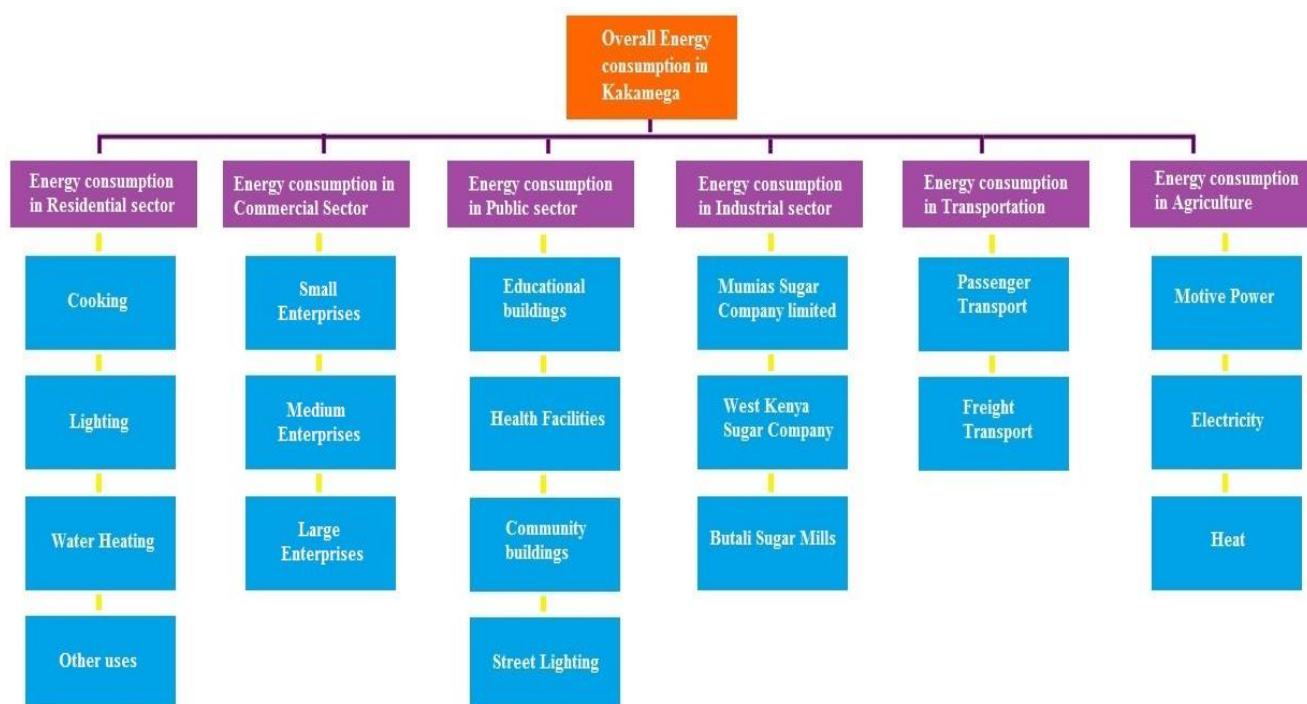


Figure 12. Developed based on the Comprehensive Measurement of Energy Access for productive uses using the multi-tier approach, Adaptation from the World Bank Group (2014)

General basic parameters

The initial stage of the model formation involved the regulation of the model's basic parameters. Thus, the energy units were set in GJ, the currency in United States Dollars (USD), the base year at 2013, the first scenario year at 2015 and time series years at 2017, 2022 and 2030. Furthermore, a sub national scale was selected involving energy and non-energy sector environmental loadings.

Key assumptions

The following step involved the insertion of macroeconomic and demographic characteristics of the selected area, in the form of key assumptions. The selection of these characteristics is very important for the development of the model, as they are the base on which the demand tree is structured. Table 15 presents the key assumptions that were taken into consideration in this model. The data were collected through literature review and assumptions as described in *CHAPTER 2. KAKAMEGA COUNTY CASE STUDY*.

Table 15. Key assumptions for Kakamega’s energy model.

Population	1644328	people
Population growth	2.5	%
GDP per capita	1245.5	USD
County's GDP	2.048	Billion USD
GDP growth	5.2	%
Households	349857	HH
Household's size	4.7	people
Household growth	2.54	%
Urbanization	15.2	%
Urbanization growth	4.1	%
Roads in County	600	km
Vehicles	74443	Veh
Vehicle_km	2604212719	Veh_km
Passenger_km	2435848474	Pass_km

3.3 Demand modelling

For each sector a current and a reference scenario were created based on business as usual patterns and expected development of the aforementioned indicators. Here is how every sector in Kakamega is expected to develop till 2030.

3.3.1 Residential sector

The residential sector model structure was based on the number of households existing in Kakamega. The households were then divided into electrified and un-electrified. This division was selected as the two categories show different energy consumption patterns and different fuel mix. Table 16, shows the differences between these two categories and how that was imported in to the model. It should be mentioned at this point that no further urban/rural division was introduced, as it would add complexity but not added value to the model, due to the small size of the selected area.

Reference scenario

The reference scenario describes the energy consumption projections in the sector under business as usual activities. In this scenario the population growth (2.5% per year) is the main driver of the energy consumption increase in the residential sector as it is directly connected to the number of households. In addition to that, the electrification rate is expected to increase to 9.5% by 2017, 19% by 2022 and 38% by 2030. The increasing electrification rate is expected to affect the fuel mix, basically in the prospective electrified households.

Lighting seems to be the main end use of electricity in the region implying a decrease of kerosene consumption in the following years. Furthermore, a turn towards cleaner and more efficient ways of cooking is expected in the County. That, because of the government’s acts against deforestation, the increase in popularity of efficient cook stoves and the individual income growth. All these factors indicate a decrease in firewood consumption, while the adoption of other fuels is expected to expand (mainly charcoal and electricity (grid and or off-grid solar). Finally, an increase in water heating capacity is expected in the County mainly due to governmental plan.

It should be mentioned here that fuel stacking is a significant characteristic to consider in energy modelling, especially in developing countries. Predicting the fuel transition is a quite challenging task especially on a local level. Therefore, due to the lack of explicit data for Kakamega, the fuel transition was estimated based on the aforementioned expected changes in the county. These estimations were applied to the end year of the simulation (2030), leaving the interpolation method on LEAP to change them gradually over the years. The reference scenario in numbers is shown in Table 17.

Table 16. Residential sector segmentation based on energy consuming activities and fuel mix.

Current scenario							
Residential 355679	Electrified 5.60%	Cooking 100%	Charcoal	44	750	kg/year	HouseHold
			Firewood	39.7	2448	kg/year	HouseHold
			Electricity	8.9	986	kwh/year	HouseHold
			Kerozene	5.4	200	l/year	HouseHold
			LPG	1.5	128.6	l/year	HouseHold
			Biogas	0.5	321.8	m ³ /year	HouseHold
		Lighting 100%	Electricity	100	209	kwh/year	HouseHold
		Water Heating 45%	Electricity	50	117	kwh/year	HouseHold
			Solar	15	749	MJ/year	HouseHold
			Firewood	35	165	kg/year	HouseHold
		Other 100%	Electricity	100	336	kwh/year	HouseHold
	Unelectrified 94.40%	Cooking 100%	Firewood	86.9	2448	kg/year	HouseHold
			Charcoal	9	750	kg/year	HouseHold
			kerozene	2.5	200	l/year	HouseHold
			LPG	1	128.6	l/year	HouseHold
			Biogas	0.6	321.8	m ³ /year	HouseHold
		Lighting 100%	Kerozene	99.3	126	l/year	HouseHold
			Solar	0.7	24.64	kwh/year	HouseHold
		Water Heating 20%	Electricity	0	117	kwh/year	HouseHold
			Solar	5	749	MJ/year	HouseHold
			Firewood	95	165	kg/year	HouseHold

Table 17. Residential sector expected changes by the end year (2030) under the suggested reference scenario.

Reference scenario							
Residential growing	Electrified growing			Share	Final Consumption	Units	Base (per)
		Cooking 100%	Charcoal	49	750	kg/year	HouseHold
			Firewood	10	2448	kg/year	HouseHold
			Electricity	30	986	kwh/year	HouseHold
			Kerosene	7	200	l/year	HouseHold
			LPG	2.5	128.6	l/year	HouseHold
			Biogas	1.5	321.8	m ³ /year	HouseHold
		Lighting 100%	Electricity	100	209	kwh/year	HouseHold
		Water Heating 75%	Electricity	60	117	kwh/year	HouseHold
			Solar	10	749	MJ/year	HouseHold
			Firewood	30	165	kg/year	HouseHold
		Other 100%	Electricity	100	336	kwh/year	HouseHold
	Unelectrified decreasing	Cooking 100%	Firewood	75	2448	kg/year	HouseHold
			Charcoal	18	750	kg/year	HouseHold
			kerosene	3.5	200	l/year	HouseHold
			LPG	2	128.6	l/year	HouseHold
			Biogas	1.5	321.8	m ³ /year	HouseHold
			Lighting 100%	Kerosene	70	l/year	HouseHold
			Solar	30	24.64	kwh/year	HouseHold
		Water Heating 50%	Electricity	0	117	kwh/year	HouseHold
			Solar	10	749	MJ/year	HouseHold
			Firewood	90	165	kg/year	HouseHold

3.3.2 Industrial sector

In the industrial sector the model structure was based on the activity level and final energy intensity. For the three sugar companies identified in the area, the annual production of sugar (tons/year) was used in order to estimate the energy consumption in terms of heat and electricity throughout the year.

Reference scenario

The reference scenario describes the energy consumption projections in the sector under business as usual activities. In this scenario the sugar production growth is the main driver of the energy consumption in the industrial sector. The main expected changes are presented below:

Mumias

- Sugar production increase at an annual rate of 2.7%
- An energy intensity drop is expected to 2.9 GJ/metric ton by 2022 and 2.88 GJ/metric ton by 2030 due to few efficiency measures taken by the company.

West Kenya Sugar Company (WKSC)

- Sugar production increase at an annual rate of 3.2%
- The energy intensity is expected to grow on an annual basis at 0.7% due to aging.

Butali Sugar Mills

- Sugar production increase at an annual rate of 2.45%
- The energy intensity is expected to be stable till 2030.

3.3.3 Transportation sector

As described in chapter 2 the transportation sector model was structured based on two indicators, passenger transportation and vehicle transportation. Further segmentation was introduced based on the type of vehicle and the respective fuel used (gasoline – diesel).

Reference scenario

The reference scenario describes the energy consumption projections in the sector under business as usual activities. In this scenario there are different factors affecting the energy consumption in the transportation sector. The main expected changes are presented below.

Passenger transportation

The number of vehicles responsible for passenger mobility in the county is highly related with the population and the income level. In order to depict this change an elasticity of 1.15 was set between income growth rate and passenger-km (Olof Johansson 1997). That is, 1% of income growth will impose a 15% increase in the passenger-km value, implying an increase of vehicles used. In parallel the vehicle fleet is further affected by the population growth.

Freight transportation

The activity of freight transport is related with the economic activity of the county in terms of GDP. As the GDP grows (incorporating also a population growth) the energy intensity in terms of ton-km/USD is also growing. Energy intensity is correlated with the energy consumption which also shows a growing projection.

3.3.4 Commercial sector

In the commercial sector the base on which the model structure was developed is the number of buildings devoted to commercial activity. The segregation was also based on the enterprise size. It was assumed that small enterprises require a small amount of electricity in order to cover their energy needs, while medium and large enterprises require both electricity and heat. The energy intensities were set based on the estimations explained in chapter 2.

Reference scenario

The reference scenario describes the energy consumption projections in the sector under business as usual activities. In this scenario the growing number of enterprises is the main driver of the energy consumption in the commercial sector. More specifically, it is expected that the number of small stores will be 1200 by 2022 and 1500 by 2030. Similarly, the number of medium/large commercial stores is expected to reach 750 by 2022 and 800 by 2030. Not any significant changes in energy consumption are expected in this sector.

3.3.5 Community sector & Public services

The segregation in the public sector involved the following areas: education, health, community buildings and street lighting. Here the base of the model was the number of building devoted for the relevant activities. Street lighting energy consumption was based on the electrification coverage of the streets in Kakamega. Table 18 shows in detail the energy intensities for the selected categories.

Table 18. Structure of the public (community) sector activities as introduced into the LEAP model.

Community	Activity (Buildings)		Intensity	Units		Units	Base (per)
	Hospitals/Health Clinics						
	Hospitals						
	17	Electricity	25	kwh/day	9125	kwh/year	building
	Health Clinics						
	32	Electricity	15	kwh/day	5475	kwh/year	building
	Health Centers/Dispensaries						
	70	Electricity	7.5	kwh/day	2738	kwh/year	building
	Education Buildings						
	Primary/Secondary						
	1214	Electricity	1.515	kwh/day	553	kwh/year	building
	Higher Education						
(one univ. of 5000m ²)	5000	Electricity	0.12	kwh/day/m2	43.8	kwh/year/m2	building
Commuity Buildings				Share			
100		Electricity		100	18250	kwh/year	building
		Lighting		25	4562.5	kwh/year	building
		Office equipment		75	13687.5	kwh/year	building
Street Lighting							
300	(km of coverage)	Electricity	9333	kWh/km/year	2799900	kWh/year	

Reference scenario

The reference scenario describes the energy consumption projections in the sector under business as usual activities. In the education subsector, an elasticity of 1 was set between the population growth and the number of buildings. Similarly, in the health sector an elasticity factor was set to 1 as very small improvements are expected in the health provision over the next years. The number of community buildings is not expected to significantly increase, thus a small 1% annual growth was induced. Finally, street lighting projections involved the installation of 5,000 new lamps by 2030, while by that year it is expected that 20% of the municipality lamps will be powered by solar.

3.3.6 Agriculture, Aquaculture & Livestock

A bottom-up modelling of the agricultural sector is very challenging as it incorporates factors that are volatile and highly dependent on individual activities. Therefore, in this case a top-down approach was followed by taking into consideration the total energy consumption in the sector. The consumption was then shared by three forms of energy: Mechanical (motive power), thermal and electrical. The base on which the model was built was the energy intensity in terms of kWh per USD yielded.

Reference scenario

The reference scenario describes the energy consumption projections in the sector under business as usual activities. Agriculture is related with the GDP through its added value. Under this scenario it is assumed that by 2030 agriculture will remain contributing at about 10% to the county's GDP. It is also assumed

that the level of mechanization will show an increase, implying a respective increase in diesel consumption in the sector. Thus, the useful energy intensity is expected to grow with a rate of 1.5% per year while no changes are expected in the equipment efficiency.

3.4 Supply modelling

The modelling of energy generation in various forms is structured in LEAP under the transformation section. In Kakamega, three main energy conversion activities were identified: Electricity generation, charcoal production and ethanol production. Each of them is explicitly described in the following paragraphs.

3.4.1 Electricity generation

Mumias Sugar Company cogeneration plant is the solely source of utility electricity generation in Kakamega. Its generation capacity is 26 MW with an availability of 7,200 hours/year. According to the MoEP annual report 2015, the electricity generation of Mumias plant was 57 GWh in the period 2013-2014 implying a capacity factor of 0.3 on a yearlong basis (MoEP 2015). Bagasse, deriving from sugar production, is used as a fuel for the plant. As its availability is based on the harvesting seasons, storage is required in order to maintain the capacity factor at a constant level. No additional capacity is expected in the following years and the dispatch rule was defined as proportional to the plant's capacity. It should be mentioned at this point that the electricity generated in the plant is exported to the national grid. Therefore, transmission and distribution losses needed to be included in the analysis. More specifically, the losses due to that factor were set at 18% at the base year (EAPIC 2014) with an estimated reduction to 12% by 2030.

3.4.2 Charcoal production

Charcoal supply is acquiring significant economic importance in Kakamega due to the increasing urbanization, which implicitly imposes a fuel transition from firewood towards alternative, more accessible fuels. However, even though its efficiency and environmental performance is better than firewood's, its growing adoption rate has caused concerns regarding the sustainability of its production.

According to the Kenyan Ministry of Environment, Water and Natural resources, 156,000 bags of charcoal were licensed through the movement permit system in Kakamega in 2013 (MoEWNR 2013). That is, 5,460¹⁴ tons of charcoal produced in the county, which accounted for 18.7% of the indigenous charcoal consumption in that year. The remaining demand is covered by imports mainly from Elgeyo Marakwet, Turkana and neighbor counties (Eldoret, Busia and Narok) (MoEWNR 2013) (Wambua, Household energy consumption and dependency on common pool forest resources: The case of Kakamega forest, Western Kenya 2011). It should be mentioned at this point that charcoal distribution channels in Kakamega involve wholesalers, retailers and street vendors (hawkers), with the latter being the main source of indigenous charcoal production. That consequently implies that charcoal production in Kakamega follows an informal norm.

Even though there are several technologies for charcoal production, the traditional earth mound kilns are predominant in Kakamega. It was assumed that 95% of the production in 2013 was conducted using this technology with an estimated efficiency of 17.5% (UNDP - KFS 2011). The rest 5% of the production

¹⁴ A bag of charcoal weights approximately 35 kg.

was assumed to come from improved kilns with a respective efficiency of 28% (UNDP - KFS 2011). Finally, Mekko kiln (Biochar) alternative technology was included in the model, even though there was no actual utilization of it in Kakamega in 2013. However, its impressive efficiency of 62.5% (UNDP - KFS 2011) and its side-effect advantages (mobility, gasses recycling, heat recovery) are expected to facilitate its market adoption in the following years, both for small and large scale production. Ultimately, at the reference scenario, the import ratio of charcoal is expected to remain near 20% in Kakamega by 2030, while new technology adoption is expected to develop as shown in Table 19.

Table 19. Charcoal production technology adoption by 2030 in Kakamega County, Kenya.

Technology	2013	2017	2022	2030
Traditional earth mounds kilns (%)	95	90	83	77
Improved kilns (%)	5	8	12	15
Mekko kilns (%)	0	2	5	8

3.4.3 Ethanol production

One of the promising energy resources in Kakamega is ethanol. Ethanol is solely produced by Mumias Sugar Company based on molasses deriving as by-product of the sugar production process. It is estimated that the final molasses weight is 4% of the sugar cane (3.0 - 4.5) and that one tone of molasses yields approximately 220 liters of ethanol (Francis X. Johnson 2012) (BUTUNYI 2009). In 2013, Mumias Sugar Company crushed 1,964,063 tons of sugar cane producing 68,742 tons of molasses and therefore approximately 15.1 million liters of ethanol (KETS 2013). It should be mentioned at this point that the full capacity of the company is 22 million liters of ethanol per year (BiofuelsDigest 2010) (Andae 2015). The conversion efficiency¹⁵ on an energy base was set at 38.9% for this process.

As there is no current domestic use, all the ethanol produced is primarily exported either for alcohol production in the beverage industry either as resource in various industries in the country. However, ethanol could be introduced either in transportation sector (ethanol – petrol blends) or in residential – commercial sector in order to cover cooking and lighting fuel needs. Further investigation of these scenarios can be found in the following chapters.

3.5 Alternative scenarios

The formation of the new policy scenarios for Kakamega was basically inspired by the SE4ALL fundamental goals regarding universal access to modern energy services, improved efficiency and increase share of renewables in the energy balance. In addition, priority was given to the productive use of energy, which is regarded as a significant driver of economic growth and social progress in the area. Therefore, under this framework two main scenarios were created:

- Productive use of energy expansion scenario
- Sustainable energy for all scenario
 - Demand Side Management scenario
 - Renewable energy generation scenario

A more detailed description of their basic parameters is presented in the following paragraphs.

¹⁵ Based on the calorific value of molasses as 12 MJ/kg and ethanol energy density as 21.2 MJ/liter.

3.5.1 Productive use of energy expansion scenario (PUoE)

PUOE in a nutshell

- *25% contribution of Agriculture to GDP by 2030.*
- *1,000 new small enterprises by 2030 with increased access to energy (from 5 to 10 kWh/day/enterprise).*
- *Increase energy intensity by 15% in medium and large scale enterprises.*
- *Increase the inpatient availability service at the level of 15 beds per 10,000 people by 2030.*
- *Increase the energy intensity per institution by 30% till 2030.*
- *Double and triple the energy intensity in primary and secondary educational buildings respectively by 2030.*
- *Increase by 50% the energy intensity in the office equipment of public institutions by 2030.*
- *100% lighting coverage of the main roads identified in Kakamega by 2030.*

Energy has a significant role in achieving goals related to both human and economic development. In the context of providing access to modern energy services in rural areas of developing countries, a productive use of energy involves the application of energy aiming to create goods and/or services which directly or indirectly lead to value creation (R. Anil Cabraal 2005). In economic terms that means that such activities add value which is taxable in form of VAT if they are part of the formal economy (Anna Brüderle 2011). That mainly concerns agricultural, commercial and industrial activities. In social terms, productive use of energy means that energy is used in order to enhance human welfare resulting people being more productive in the activities they are involved (better health services, improved education, gender equality, women empowerment etc.) (R. Anil Cabraal 2005).

The productive use of energy expansion scenario was used in order to project the energy demand in Kakamega if the energy intensity and utilization in productive activities increase with higher rate than in the business as usual scenario. More specifically, under PUoE scenario the following changes are expected in each sector respectively.

Agriculture

- The contribution of agriculture in the County's GDP increases from 10% to 25% by 2030.

The introduction of modern forms of energy can underpin the creation and upgrading of agricultural value chains, facilitate the diversification of economic structures and reduce vulnerability to multiple stresses and external shocks. That is expected to increase the productivity in the sector which however comes along with an increase in the useful energy intensity (kWh/USD produced). An elasticity factor of 1.35 was selected in order to define the correlation between the useful energy intensity and the GDP growth. That is, more energy is required per USD produced in the sector. It should be mentioned at this point that this value does not reflect real data but it was used here in order to facilitate the model development.

Commercial

The development of the commercial sector is critical for the economic reformation aimed for Kakamega. Under the productive use of energy expansion scenario two trends are assessed regarding both the quantity and the quality of energy demand projections. Therefore the following goals were set.

Small scale enterprises

- Promote entrepreneurship and micro/small business creation in the County. 500 new small enterprises by 2022 and 1000 by 2030.
- Increase the energy intensity from 5 kWh per day to 10 kWh per day.

Medium/large scale enterprises

- Increase the energy intensity in all the respective activities in this subsector by 15% till 2030.

Community

As aforementioned, the productive use of energy under a social framework indicates that energy is utilized in order to enhance the welfare in a society. Under these terms the following goals were set for the public sector.

Hospitals & health clinics

- Increase the inpatient availability service at the level of 15 beds per 10,000 people by 2030.
- Increase the energy consumption per institution by 30% till 2030.

According to the current situation in Kakamega the availability of inpatient services is estimated at 3 beds per 10,000 people. According to the World Health Organization that index lies far below the necessary requirements for adequate healthcare, which depending of the country is about 20-30 beds per 10,000 people (WHO 2009). Under the PUoE scenario the goal set at 15 beds per 10,000 people by 2030. That means that the number and the inpatient availability¹⁶ (capacity) of the institutions will grow at the level shown in Table 20. Furthermore, a 30% increase in energy consumption was set in order to represent the acquisition of additional equipment. It should be also mentioned that under this scenario the number of health institutions was found to grow elastically with the population at a range of 2.23.

Table 20. Productive use of energy goals in health sector of Kakamega by 2030.

Type of building	Current situation			PUoE scenario by 2030		
	Number	Number of beds/building	Energy intensity/building	Number	Number of beds/building	Energy intensity/building
Hospitals	17	20	25 kWh/day	34	70	114 kWh/day
Health Centers	32	10	15 kWh/day	50	15	29.3 kWh/day
Dispensaries	72	2	5 kWh/day	203	4	13 kWh/day
Inpatient availability	5 beds/10000 people			15 beds/ 10000 people		

Educational buildings

- Increase in primary and secondary educational buildings which follows an elastic correlation (value: 1.7) with the population.
- Double and triple the energy consumption in primary and secondary educational buildings respectively by 2030.

The increased number of buildings will provide better educational environment as schools will be less crowded. Special focus should be given on the secondary education by increasing the attendance from 55% to 80% by 2030. The expected number of schools and their relative intensities under the PUoE scenario are presented in Table 21.

¹⁶ The increase in capacity means that hospitals will increase their daily consumption by 3.5 times, health clinics by 1.5 time and health centers and dispensaries by 2 times.

Table 21. Productive use of energy goals in education in Kakamega by 2030.

Type of building	Current situation			PUoE scenario by 2030		
	Number	Number of students	Energy intensity/building	Number	Number of students	Energy intensity/building
Primary	938	420	1.52 kWh/day	1664	350	3 kWh/day
Secondary	276	360	1.52 kWh/day	715	290	5 kWh/day
Higher	1	-	600 kWh/day	1	-	600 kWh/day

Community buildings

- Increase by 50% the energy intensity in the office equipment by 2030.

Street Lighting

- 100% lighting coverage of the main roads identified in Kakamega by 2030.

The electrification of the roads will increase security in the county and will allow to the extension of the commercial activities beyond dusk.

3.5.2 Sustainable energy for all (SE4ALL) scenario

Demand Side Management (DSM) scenario

DSM in a nutshell

- Increase the electrification rate in households to 15% by 2017, 30% by 2022 and 70% by 2030.

Already electrified households

- Promote the use of efficient firewood and charcoal based cook stoves (min. efficiency of 25%).
- Facilitate the adoption of new fuels in cooking mainly electricity (60%), ethanol (15%) and charcoal (10%) by 2030.

Un-electrified households

- Reduce of the use of firewood as fuel for cooking by 50% and promote efficient firewood and charcoal fired cook stoves (min. efficiency of 25%)
- Facilitate the adoption of new fuels in cooking, charcoal (35%), ethanol (10%) and LPG (8%).
- Facilitate the adoption of solar technologies for lighting to an extent of 50% and for water heating to an extent of 60% by 2030.

Commercial sector

- Reduce the amount of firewood used for heating purposes by 30% by 2030.
- Promote alternative fuels for heat generation like electricity and/or bio-charcoal from agricultural waste in order to cover 50% of the sector's demand by 2030.

Transportation

- Promote public transportation by increasing the number of busses to a level of serving 20% of the total local passenger transport by 2030.

Agriculture

- Increase electricity penetration to a level of 50% of the total useful energy demand in the sector.

Kakamega is at the forefront of an ambitious socioeconomic development. That consequently implies that the energy consumption in the county is expected to increase significantly over the next fifteen years. In order this transition to be stable and effective, proper management of the increasing demand is necessary. It terms of scenario formation that is translated into three fundamental goals:

- 1) The promotion of new, more efficient technologies
- 2) The introduction and expansion of new fuels in the market
- 3) Proper management of natural resources

It is evident that in Kakamega the energy demand is mainly derived by the residential sector. Therefore, primary goal in the development plan should be to propose alternative consumption patterns that will improve the activities' performance in terms of energy consumption and relieve the environmental stress caused by the local resource depletion (mainly deforestation). Hence, under the DSM scenario, the following trends are expected.

Residential sector

- Increase the electrification rate in households to 15% by 2017, 30% by 2022 and 70% by 2030.

The higher electrification rate is expected to lower the consumption of firewood and kerosene, as it will replace their utilization in cooking and lighting activities respectively. Electricity based residential equipment shows higher efficiency while it will eliminate to a great extent the health problems related with the currently used fuels. Furthermore the following goals are suggested.

Electrified households

- Promote the use of efficient firewood and charcoal based cook stoves (min. efficiency of 25%).
- Facilitate the adoption of new fuels in cooking mainly electricity (60%), ethanol¹⁷ (15%) and charcoal (10%) by 2030.

Un-electrified households

- Reduce of the use of firewood as fuel for cooking by 50% by 2030.
- Promote efficient firewood and charcoal fired cook stoves (min. efficiency of 25%)
- Facilitate the adoption of new fuels in cooking, charcoal (35%), ethanol (10%) and LPG (8%).
- Facilitate the adoption of solar technologies for lighting to an extent of 50% by 2030.
- Facilitate the adoption of solar technologies for water heating to an extent of 60% by 2030.

Commercial sector

- Reduce the amount of firewood used for heating purposes by 30% by 2030.
- Promote alternative fuels for heat generation like electricity and/or bio-charcoal from agricultural waste in order to cover 50% of the sector's demand by 2030.

Agriculture

- Increase electricity penetration to a level of 50% of the total useful energy demand in the sector.

Transportation

- Promote public transportation by increasing the number of busses to a level of serving 20% of the total local passenger transport by 2030.

Charcoal Production

- Promote the adoption of improved kilns in charcoal production by 10% by 2017, 20% by 2022 and 30% by 2030.
- Promote the adoption of Mekko kilns in charcoal production by 5% by 2017, 10% by 2022 and 20% by 2030.

¹⁷ Ethanol energy density was assumed at 21.2 MJ/liter while the stove efficiency at 0.6 based on the performance of KIKE cook stove. Η πηγή που καθορίστηκε δεν είναι έγκυρη..

Costing data

The following paragraph elaborates on the cost factors taken into consideration in order to have an initial cost estimation of the transformation proposed in Kakamega based on the DSM scenario. This scenario involves radical changes mainly in the energy consumption pattern of the residential sector. These changes are associated with a cost deriving by both the capital (acquisition) cost of new equipment and also operational cost based on the fuel used. These costs are presented in Table 22 and Table 23 respectively.

Table 22. Acquisition cost of residential equipment.

Costing Data - Residential sector			
		Cost per unit	Lifetime (years)
Cooking			
(stoves)	Firewood	300 Ksh	1
	Charcoal	800 Ksh	2
	Ethanol	2500 Ksh	6
	Kerosene	3500 Ksh	6
	LPG	6000 Ksh	6
	Electricity	10000 Ksh	15
	Biogas	15000 Ksh	15
Lighting			
(lanterns)	Kerosene	290 Ksh	8 months
	Electricity	35 Ksh	1
	Solar	3500 Ksh	5
Water Heating			
(complete systems)	Electricity	9000 Ksh/unit	15
	Solar	15000 Ksh/unit	15
	Firewood	-	-

It is estimated that in order to increase the efficiency of cook stoves based on firewood and charcoal to a minimum 25%, an incremental cost of about 600 Ksh per stove is required (Boulkaid 2015). In lighting, the costs were estimated by taking into account the average number of units per household and their respective operative duration. A typical household in Kakamega possesses 1.8 kerosene lamps (779 Ksh/household/year) or when electrified uses on average 4.5 light bulbs (156 Ksh/household/year). On the other hand, solar lamps show high initial cost however they do not have any operational cost and their lifespan is significantly higher (Sklivaniotis, Integrated Techno-Economic Comparative and Socio-Economic Impact Study for Increasing Energy Access in Rural Kenya 2014). In water heating, a high initial cost (acquisition and installation) is evident especially in solar based systems which however are free of operational costs (Waruru 2014) (Climate Innovation Center 2012).

An equally important parameter that should be taken into consideration in the cost evaluation is the current and projected price of the main fuels involved in the energy sector in Kakamega. It should be mentioned at this point that the projection of fuel prices in a span of fifteen years is a quite challenging task as there are numerous factors associated with their fluctuation. Table 23, presents the authors assumptions based on lasting literature review and the experience acquired during the duration of the project.

Table 23. Cost of the main fuels involved in the energy sector of Kakamega.

Fuel Price		
Type	Current (2013-2014)	Estimated 2030
Firewood (purchased)	9 Ksh/kg	15 Ksh/kg
Charcoal	14 Ksh/kg	16 Ksh/kg
Kerosene	84.7 Ksh/liter	55.4 Ksh/liter
LPG	256.2 Ksh/liter	213.5 Ksh/liter
Electricity	12.6 Ksh/kWh	9.6 Ksh/kWh
Gasoline	107.4 Ksh/liter	107.4 Ksh/liter
Diesel	101.2 Ksh/liter	101.2 Ksh/liter
Molasses	1.2 Ksh/kg	1.2 Ksh/kg
Bio-charcoal	23 Ksh/kg	18 Ksh/kg
Ethanol	200 Ksh/liter	133 Ksh/liter
Bagasse	-	-
Biomass residues	-	-
Biogas	-	-

The extensive use of firewood in Kakamega has raised serious concerns regarding its origin, which is associated with the high deforestation rate in the area. Restriction measures are expected from the National Forest Policy (Ministry of Environment, Water and Natural Resources 2014) and will likely impose an increase in the price of the purchased firewood over the next years. This trend will internally affect the charcoal production as well. Therefore, its price is also expected to slightly increase. On the other hand, the adoption of new production methods base of biomass residues and the increase in awareness of bio-charcoal products (briquettes) is expected to reduce their price in the next fifteen years (GVEP International 2015), (International Biochar Initiative 2014).

Kerosene shows a continuous decrease in price with the current level being 63.7 Ksh/liter (as in June 2015, in Kakamega) (Energy Regulatory Commission (ERC) 2015). Kerosene price is expected to further decrease by 15% by 2030. Similar trend is expected for the price of LPG as well, which according to Dalberg the end user cost of LPG has the potential to decrease by 20% by 2020 (Dalberg 2013). Gasoline price shows a general stabilized level in the last three years at around 107 Ksh/liter. On the other hand, diesel price has decreased by 18% over the last year with the current level being 85.5 Ksh/liter (Energy Regulatory Commission (ERC) 2015). Due to their erratic price fluctuation, both fuels were selected to follow projections equal to their current cost.

Ethanol is a critical fuel in the proposed DSM scenario as it will replace to a significant percent the traditional fuels. That strategy was selected due to the indigenous ethanol production in the County. The current price is quite high at around 200 Ksh/liter (Nderitu 2015) however with governmental support (through policies, regulations and subsidies) it is expected that the end user price will be reduced by 50% until 2030. This statement is further enhanced by the close proximity of the producer to the new market (reduced logistic costs) and the increasing purchasing power of the local population.

The price of electricity is also a very important parameter for the estimation of the intervention cost necessary for the successful implementation of the DSM scenario. The Energy Regulatory Commission of Kenya has established several tariff levels for different customer segments. The tariff for domestic users varies from 2.5 – 19.6 Ksh/kWh while the average price for commercial and industrial users is 14.2

Ksh/kWh. For simplicity, an average price of 12.6 Ksh/kWh was assumed as a representative tariff in this project (ERC 2013). According to the 5000+ plan, proposed by the Kenyan government in 2013, the electricity prices are expected to be reduced by 92% and 57% in the two customer segments respectively, by 2017 (Ministry of Energy and Petroleum MoEP n.d.). Therefore, the expected average electricity price is about 9.6 Ksh/kWh, value that was assumed as a plausible goal for Kakamega County by 2030.

Finally, bagasse and biomass residue may have a minor cost associated with their collection and transportation but such costs were assumed as negligible in this project.

Renewable energy generation (RENE) scenario

RENE in a nutshell

- **29 MW of additional capacity from Mini-Hydro power plants**
 - *3 mini hydro power plants with capacity of 2 MW each constructed by 2022.*
 - *An 8 MW mini hydro power plant constructed by 2025.*
 - *A 15 MW mini hydro power plant constructed by 2028.*
- **31 MW of additional capacity from Solar PV systems**
 - *A 15 MW solar power plants constructed by 2020 in the north of the County.*
 - *A 15 MW solar power plants constructed by 2020 in the north of the Count.*
 - *A 5 MW solar power plant constructed by 2027 northwest of Kakamega town.*
 - *1 MW additional capacity from roof top PV installations in community institutions.*

This scenario refers to the exploitation of the county's renewable energy sources in order to generate electricity in a local, decentralized and commercially viable manner that will enhance the self-dependency of the county and will support the aforementioned development. As described in Chapter 2, Kakamega shows high solar and hydro potential. Therefore, this scenario elaborates on the deployment of photovoltaic technology and small hydro projects in the county.

An initial assessment of the location, the capacity and the economic viability of such projects was performed based on GIS maps provided by IRENA Global Atlas. Figure 13, illustrates a map showing the most economic rural electrification activity for the region of Western Kenya. The comparative analysis was performed based on the levelized cost of electricity expected by four selected electrification options: Grid extension, off grid photovoltaic systems, diesel generators and small hydro. A more detailed description of the modelling process can be found in the suggested literature (S. Szabo 2011) (Lily Parshall 2009) (IRENA Global Atlas 2015).

Utilizing the map as a base, the following assumptions were made. The grid extension seems to be the most economical solution in the south part of the county, mainly due to the existence of the cogeneration plant in Mumias which is connected to the national grid, as well as in the central-north part of the county due to the passing high voltage transmission line connecting Eldoret and Tororo, Uganda (Figure 14). Furthermore, photovoltaic systems were identified as most profitable solution in three main regions of the County while mini hydro projects could be commercially deployable in five identified areas. A description of the proposed projects follows in the next paragraphs. It should be mentioned at this point that the aforementioned assumptions are based on the author's preliminary assessment. Further analysis should necessarily be contacted in order to validate the suitability of these locations.

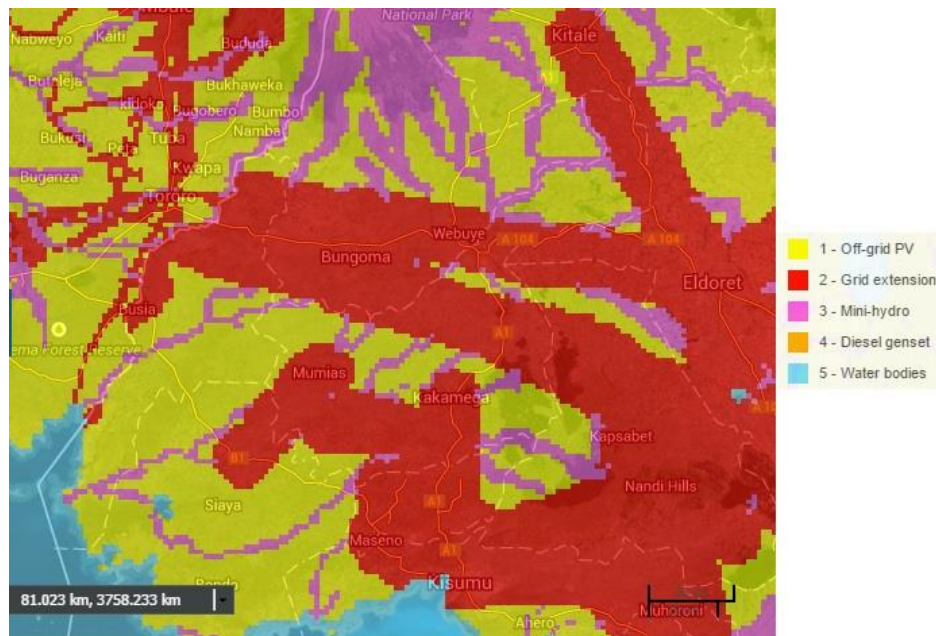


Figure 13. Most economic rural electrification activity in Kakamega County, acquired from IRENA Global Atlas (Masdar Institute of Science and Technology n.d.)

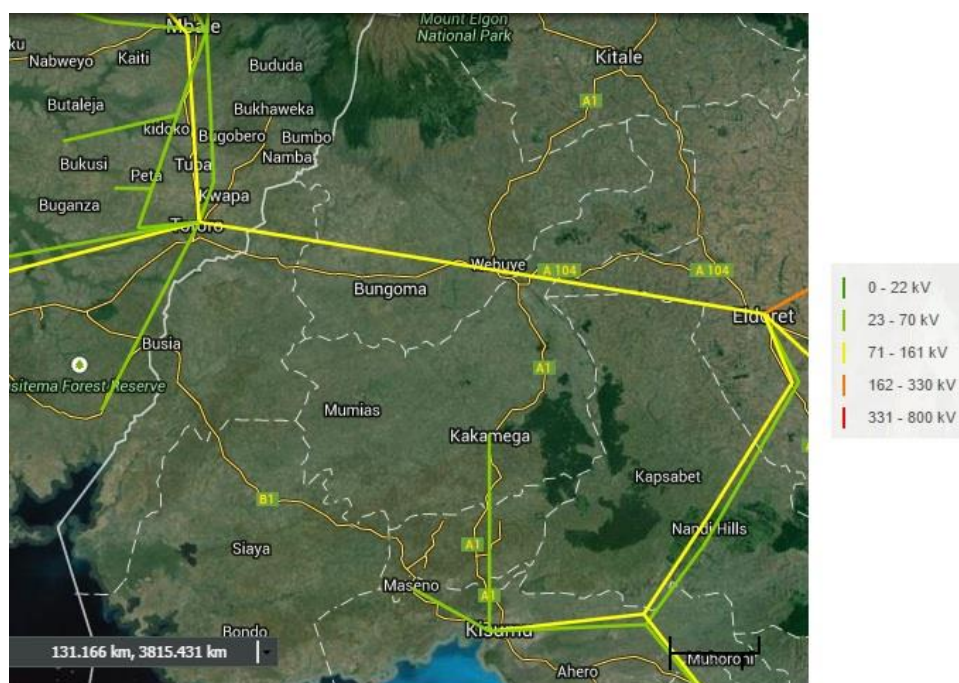


Figure 14. Transmission lines in the area of Kakamega, Kenya, acquired from IRENA Global Atlas (Masdar Institute of Science and Technology n.d.).

Small hydro exploitation

In chapter 2 it was calculated that Kakamega has potentially 280 MW of available capacity deriving from small hydro projects deployment. Five main areas were identified, where small hydro power plants could be economically sustainable according to IRENA (Figure 15).



Figure 15. Identified areas for the deployment of small hydro power plants in Kakamega, Kenya, base map acquired from IRENA Global Atlas database.

Under the renewable energy generation scenario a capacity of 29 MW is suggested to be deployed by 2030. The scenario involves the commissioning of 3 small hydro power plants with individual capacity of 2 MW constructed by 2022 in the central part of the county (points A, B and C in Figure 15). Additional capacity is suggested with the construction of an 8 MW hydro power plant by 2025 (point D) and a 15 MW power plant by 2028 in the north of the county (point F).

Solar energy exploitation

Grid-connected power plants

Based on the most economically effective solution for electrification discussed in the previous paragraph, three main areas were identified for the deployment of photovoltaic technology in Kakamega presented in Figure 16.

More specifically under the renewable energy generation scenario a capacity of 30 MW is suggested to be deployed by 2030. The scenario involves the commissioning of a 15 MW solar power plant by 2020 constructed in the north part of the county. It is assumed that the proximity to the transmission lines is an additional advantage for this location and will facilitate the interconnection with the national grid. Secondly, a 10 MW solar power plant by 2024 constructed northwest of Mumias town and a 5 MW plant by 2027 northwest of Kakamega town. It is estimated that such projects require approximately 20,000 m² per MW deployed (Mevin Chandel 2013), (Scatec Solar n.d.), (Tata Power Solar n.d.). Thus, the land use for the proposed projects is estimated at roughly 600,000 m² (60 ha).

Off-grid systems

In terms of off-grid solar systems, the renewable energy generation scenario involves the deployment of individual PV systems covering partially the needs of social institutions (educational and health buildings). The scenario suggests that the capacity of such systems will reach 1 MWp by 2030 setting the goal of 375 Wp of PV modules on every roof top.

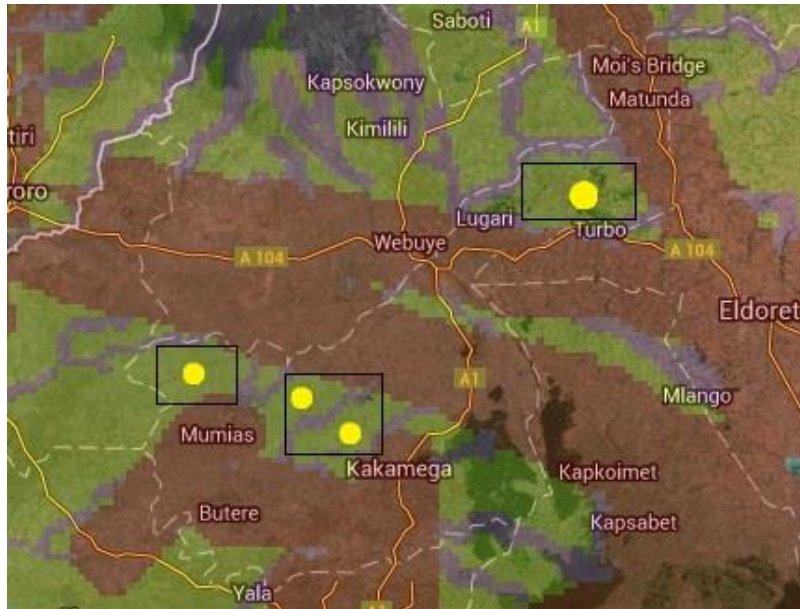


Figure 16. Identified areas for the deployment of PV solar parks in Kakamega, Kenya, base map acquired from IRENA Global Atlas database.

Costing data

The following paragraph elaborates on the cost factors taken into consideration in order to have an initial cost estimation of the transformation proposed in Kakamega based on the renewable energy scenario. This scenario involves changes in the energy generation pattern of the County. A preliminary assessment of the associated costs in LEAP, involves capital costs (feasibility studies, development of the site, construction of the generating facility and applicable taxes and waivers (VAT, import duties etc.) and operation & maintenance costs (fixed and variable). These costs are presented below for the respective technologies.

Photovoltaic systems

PV is one of the fastest growing renewable energy technologies at the moment, trend that is continuously reducing the cost of its implementation. It should be noticed that the cost of PV systems is certainly affected by numerous factors such as technology, location, financial environment etc. however an in depth analysis of the cost structure associated with PV system deployment is out of the scope of this thesis. Thus, at this stage the cost estimation will be principally based on capital (CAPEX) and operating & maintenance (OPEX) costs as they were retrieved for the Kenyan region.

According to IRENA estimations, the total installed PV system cost for utility scale systems in 2015 is around 2,700 USD/kW showing a declining trend, expecting to reach 1,800 USD/kW by 2020 (IRENA 2012). A study conducted in 2013, estimated that the investment cost for a utility scale 10 MW solar power plant in Mombasa, Kenya requires an investment cost of 2,566 USD/kW (Ondraczek, Are we there yet? Improving solar PV economics and power planning in developing countries: The case of Kenya 2013). According to MoEP, the initial cost PV systems in the range of 0-10 MW is estimated at 2,670 USD/kW (MoEP, Ramboll, ECA, ERC, The World Bank 2014). Therefore, an average cost of 2,650 USD/kW installed was considered representative of utility scale systems for the region of Kakamega and was used in the model.

Systems with much lower capacity ($\ll 1$ MW) for residential/commercial sectors, show higher costs mainly due to their storage requirements and no benefit from economies of scale. According to US

Department of Energy, it is estimated that such installations necessitate an initial cost of nearly 3,500 USD/kW (IRENA 2012).

In addition, the fixed operation & maintenance costs were set at 1.5% of the initial cost whereas there are not associated variable costs. The capacity factor of such systems was set as 0.25, the lifetime at 25 years and the discount rate at 8% (MoEP, Ramboll, ECA, ERC, The World Bank 2014) (Ondraczek, Are we there yet? Improving solar PV economics and power planning in developing countries: The case of Kenya 2013).

Small hydropower systems

Hydropower is a capital intensive technology showing long lead times of deployment due to significant feasibility assessments, design planning and construction work requirements. The cost breakdown includes two main categories regarding infrastructure development costs on the one hand and electro-mechanical equipment costs on the other. Additional costs may derive from necessary environmental impact analyses and mitigation measures (wildlife, water quality etc.), licensing and miscellaneous activities (IRENA 2015).

According to IRENA, small hydro power plants (< 10 MW) in Africa have an initial cost of 3,700 USD/kW installed (IRENA 2015). MoEP estimates that in the Kenyan context, the capital cost for hydro power plants in the range 1-10 MW is roughly 2,500 USD/kW (MoEP, Ramboll, ECA, ERC, The World Bank 2014). According to Renetech AB, the estimated cost for a hydro plant in the same capacity range is 3,200 USD/kW (Renetech 2015). The average value of 3,130 USD/kW was assumed as representative and was further used in the LEAP model. The fixed operation & maintenance costs was set at 2.7%, the capacity factor at 0.6, the lifetime at 20 years and the discount rate at 8% (IRENA 2015) (MoEP, Ramboll, ECA, ERC, The World Bank 2014).

CHAPTER 4. RESULTS

This chapter presents the results of the simulation process as yielded by the LEAP model developed for Kakamega.

Kakamega is at the forefront of a socio-economic transformation spurred by the national and regional development plans. While the scope of this thesis is to translate the consequences of this transformation in terms of energy requirements, it is also worthy to review how the macroeconomic and demographic characteristics of the County are expected to evolve. Table 24 presents the projection of the main parameters based on which the model was structured.

Table 24. Projection of the population and GDP for the region of Kakamega at 2030.

	2015	2030	
Population	1,685,400	2,441,000	people
Households	358,700	522,600	HH
GDP per capita	1274.1	1792.1	USD
County's GDP	2.147	4.375	Billion USD

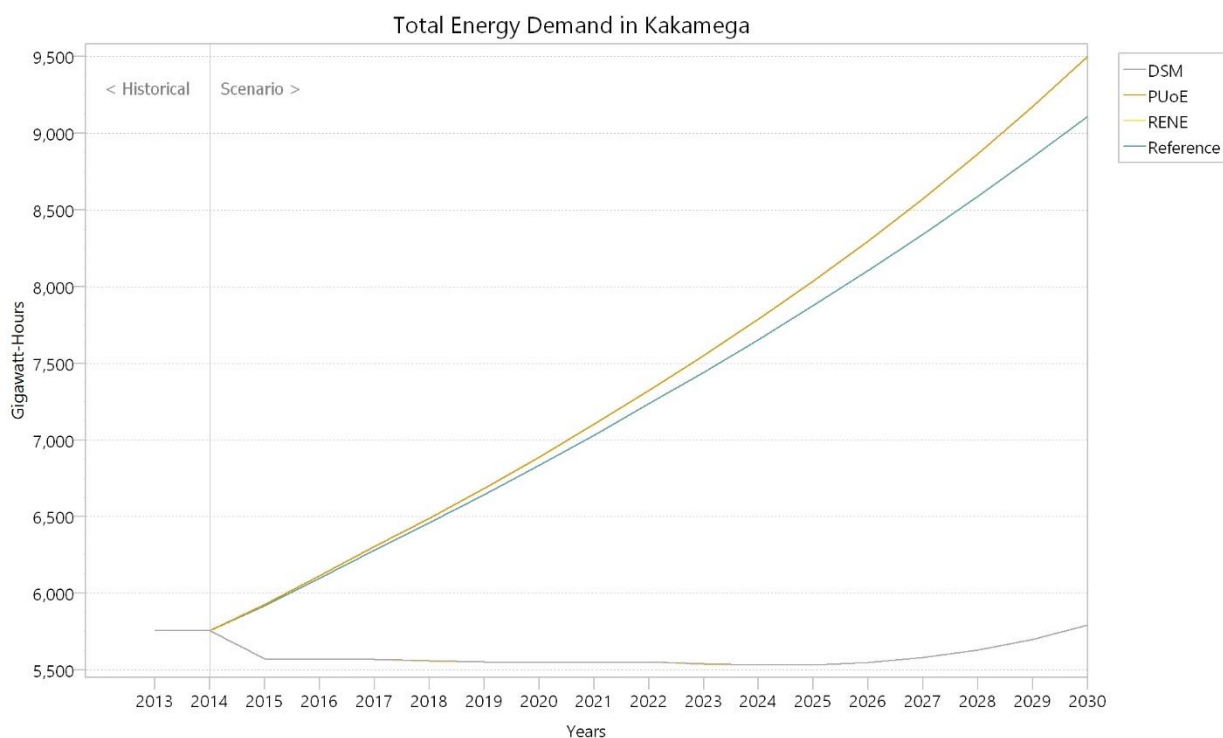
4.1 Energy demand

The energy demand in Kakamega is mainly characterized by the huge consumption of the residential and transportation sector. That is a pattern usually evident in developing regions especially in sub-Saharan Africa, and is predominantly caused by the consumption of low energy content fuels and/or the use of inefficient technologies. As expected, the continuous growing population and the prospective economic development in such regions tend to increase the energy demand, however usually under an unsustainable and questionable environment. While access to energy services should not be contestable, alternative pathways should be identified and amplified in order to achieve a sustainable and prosperous future in these regions.

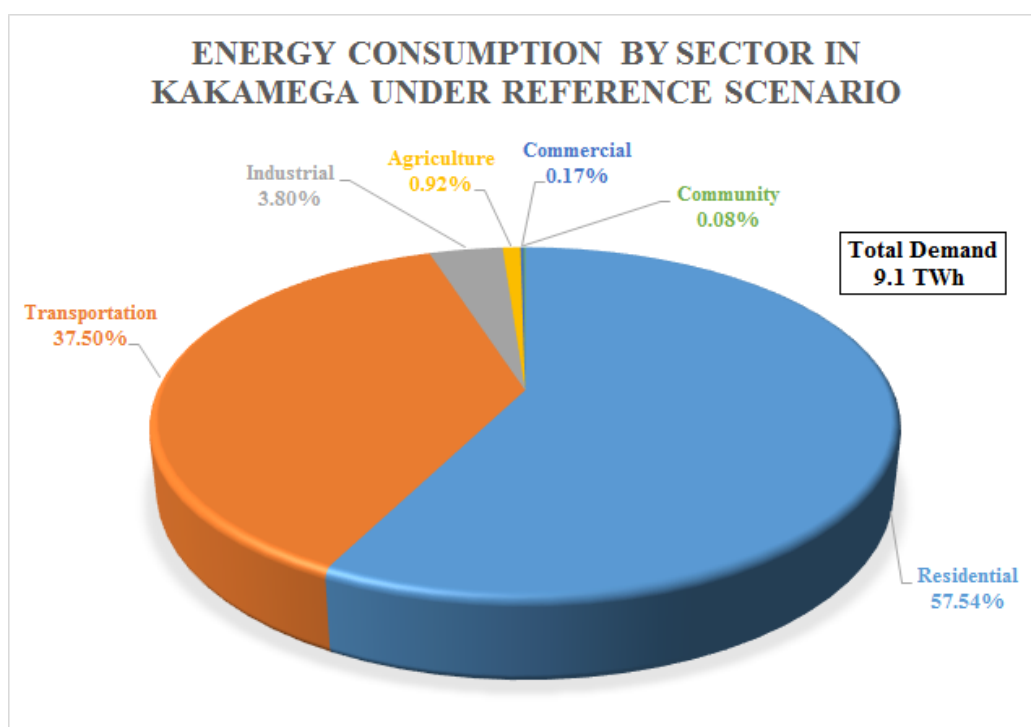
Under this perspective, the energy demand for Kakamega County was subjected to different scenarios, whose projections are illustrated in Graph 9. Under the reference scenario (BAU) the demand shows an increase from 5.9 TWh in 2015 to 9.1 TWh in 2030. The demand is further increased by 0.39 TWh under the PUoE scenario, which aims to induce however important improvements in the social and economic content of the County. This transformation does not necessarily mean that the energy consumption need to be skyrocketed. In fact, with the adoption of the SE4ALL scenario (combination of DSM and RENE) the energy demand can be stabilized at around 5.8 TWh by 2030 without restricting the aforementioned development goals. That is mainly due to the reduced energy demand in the residential sector caused by the fuel switching strategy proposed and the promotion of more efficient technologies.

The energy consumption by sector is also expected to differentiate under the alternative scenarios. Graphs 10, 11 & 12 show the share of every sector in the total energy demand of Kakamega by the year 2030. As expected the transportation sector will predominantly account for more than 50% of the energy demand in the County by the year 2030 under the combination of all the scenarios. That is due to the difficulty

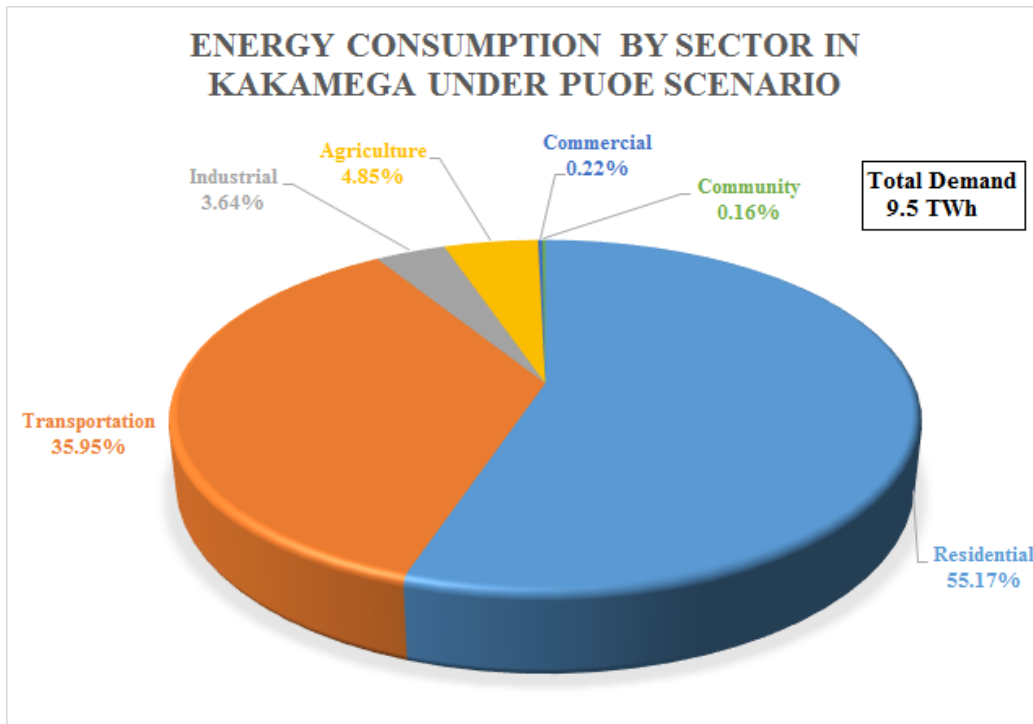
encountered in the formation of policy and the implementation of measures towards the reduction of energy consumption in this sector.



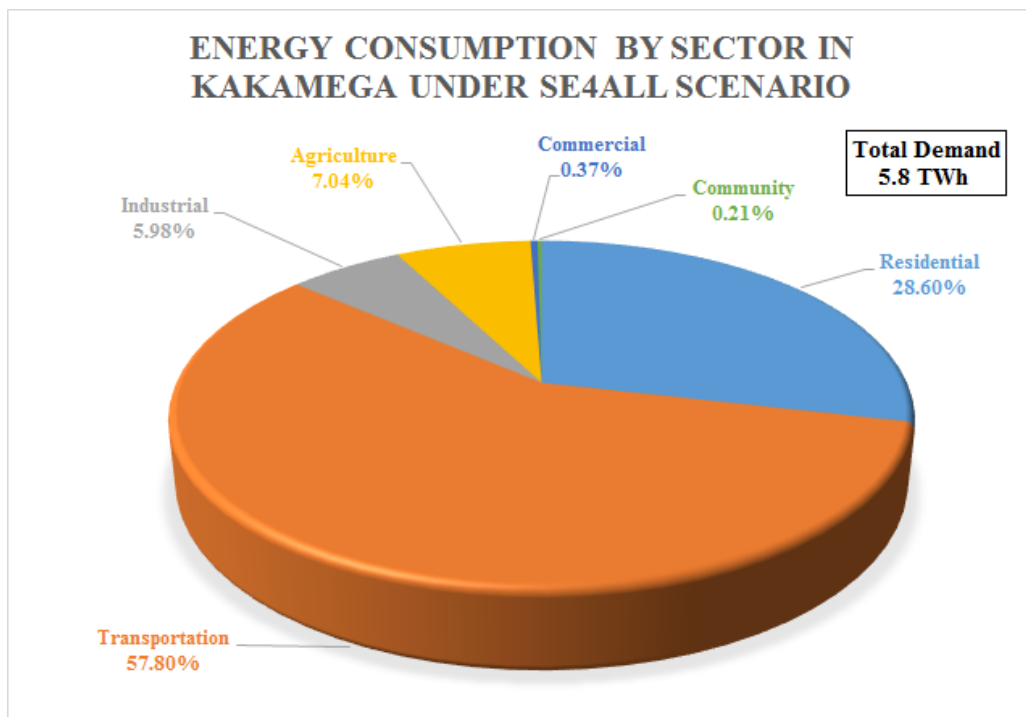
Graph 9. Total energy demand scenarios projection in Kakamega till 2030.



Graph 10. Total energy demand by sector in Kakamega till 2030 under reference scenario.



Graph 11. Total energy demand by sector in Kakamega till 2030 under PUoE scenario.



Graph 12. Total energy demand by sector in Kakamega till 2030 under SE4ALL scenario.

Tables 25, 26 and 27 provide a disaggregated description of the energy consumption in Kakamega per fuel and sector, as projected in 2030 under the reference and the two proposed alternative scenarios (PUoE and SE4ALL).

Table 25. Estimated fuel consumption in GWh per sector in Kakamega in 2030 under the reference (BAU) scenario.

Sectors	Electricity (grid)	Gasoline	Kerosene	Diesel	LPG	Wood	Charcoal	Biogas	Bagasse	Electricity (solar)	Heat	Total
Residential	142.5	-	446.1	-	5.7	3,934.1	701.2	5.9	-	6.5	-	5,242.0
Electrified	142.5	-	21.6	-	2.7	862.7	525.7	2.0	-	3.1	-	1,560.4
Unelectrified	-	-	424.5	-	3.0	3,071.4	175.5	3.9	-	3.4	-	3,681.7
Industry	-	-	-	-	-	-	-	-	346.3	-	-	346.3
Mumias Sugar Company Limited	-	-	-	-	-	-	-	-	222.2	-	-	222.2
West Kenya Sugar Company	-	-	-	-	-	-	-	-	74.7	-	-	74.7
Butali Sugar Mills	-	-	-	-	-	-	-	-	49.4	-	-	49.4
Commercial	13.5	-	-	-	-	-	-	-	-	-	1.6	15.1
Small Scale Businesses	2.7	-	-	-	-	-	-	-	-	-	-	2.7
Medium or Large businesses	10.7	-	-	-	-	-	-	-	-	-	1.6	12.4
Agriculture	16.8	-	-	42.0	-	-	-	-	-	-	25.2	84.0
Agro_demand	16.8	-	-	42.0	-	-	-	-	-	-	25.2	84.0
Community	6.5	-	-	-	-	-	-	-	-	0.5	-	7.0
Education	1.3	-	-	-	-	-	-	-	-	-	-	1.3
Health	0.8	-	-	-	-	-	-	-	-	-	-	0.8
Community build.	2.1	-	-	-	-	-	-	-	-	-	-	2.1
Street L	2.2	-	-	-	-	-	-	-	-	0.5	-	2.8
Transportation	-	2,091.4	-	1,324.9	-	-	-	-	-	-	-	3,416.3
Passenger Transportation	-	814.8	-	368.0	-	-	-	-	-	-	-	1,182.8
Freight Transportation	-	1,276.7	-	956.9	-	-	-	-	-	-	-	2,233.6
Total Demand	179.2	2,091.4	446.1	1,366.9	5.7	3,934.1	701.2	5.9	346.3	7.1	26.8	9,110.8

Table 26. Estimated fuel consumption in GWh per sector in Kakamega in 2030 under the productive use of energy (PUoE) scenario.

Sectors	Electricity					Electricity						Total
	(grid)	Gasoline	Kerosene	Diesel	LPG	Wood	Charcoal	Biogas	Bagasse	(solar)	Heat	
Residential	142.5	-	446.1	-	5.7	3,934.1	701.2	5.9	-	6.5	-	5,242.0
Electrified	142.5	-	21.6	-	2.7	862.7	525.7	2.0	-	3.1	-	1,560.4
Unelectrified	-	-	424.5	-	3.0	3,071.4	175.5	3.9	-	3.4	-	3,681.7
Industry	-	-	-	-	-	-	-	-	346.3	-	-	346.3
Mumias Sugar Company Limited	-	-	-	-	-	-	-	-	222.2	-	-	222.2
West Kenya Sugar Company	-	-	-	-	-	-	-	-	74.7	-	-	74.7
Butali Sugar Mills	-	-	-	-	-	-	-	-	49.4	-	-	49.4
Commercial	19.3	-	-	-	-	-	-	-	-	-	1.9	21.2
Small Scale Businesses	7.3	-	-	-	-	-	-	-	-	-	-	7.3
Medium or Large businesses	12.0	-	-	-	-	-	-	-	-	-	1.9	13.9
Agriculture	92.2	-	-	230.6	-	-	-	-	-	-	138.3	461.2
Agro_demand	92.2	-	-	230.6	-	-	-	-	-	-	138.3	461.2
Community	13.7	-	-	-	-	-	-	-	-	1.1	-	14.8
Education	3.4	-	-	-	-	-	-	-	-	-	-	3.4
Health	2.9	-	-	-	-	-	-	-	-	-	-	2.9
Community build.	2.9	-	-	-	-	-	-	-	-	-	-	2.9
Street L	4.5	-	-	-	-	-	-	-	-	1.1	-	5.6
Transportation	-	2,091.4	-	1,324.9	-	-	-	-	-	-	-	3,416.3
Passenger Transportation	-	814.8	-	368.0	-	-	-	-	-	-	-	1,182.8
Freight Transportation	-	1,276.7	-	956.9	-	-	-	-	-	-	-	2,233.6
Total Demand	267.7	2,091.4	446.1	1,555.5	5.7	3,934.1	701.2	5.9	346.3	7.6	140.2	9,501.8

Table 27. Estimated fuel consumption in GWh per sector in Kakamega in 2030 under the sustainable energy for all (SE4ALL) scenario.

Sectors	Electricity (grid)	Gasoline	Kerosene	Diesel	LPG	Wood	Charcoal	Biogas	Ethanol	Bagasse	Electricity (solar)	Heat	Total
Residential	446.8	-	147.1	-	16.3	640.1	290.5	5.6	92.9	-	17.4	-	1,656.7
Electrified	446.8	-	39.8	-	5.0	194.8	123.3	3.7	72.3	-	5.7	-	891.3
Unelectrified	-	-	107.3	-	11.3	445.2	167.3	1.9	20.6	-	11.7	-	765.4
Industry	-	-	-	-	-	-	-	-	-	346.3	-	-	346.3
Mumias Sugar Company Limited	-	-	-	-	-	-	-	-	-	222.2	-	-	222.2
West Kenya Sugar Company	-	-	-	-	-	-	-	-	-	74.7	-	-	74.7
Butali Sugar Mills	-	-	-	-	-	-	-	-	-	49.4	-	-	49.4
Commercial	19.3	-	-	-	-	-	-	-	-	-	-	1.9	21.2
Small Scale Businesses	7.3	-	-	-	-	-	-	-	-	-	-	-	7.3
Medium or Large businesses	12.0	-	-	-	-	-	-	-	-	-	-	1.9	13.9
Agriculture	136.0	-	-	163.2	-	-	-	-	-	-	-	108.8	408.1
Agro_demand	136.0	-	-	163.2	-	-	-	-	-	-	-	108.8	408.1
Community	5.8	-	-	-	-	-	-	-	-	-	8.8	-	14.6
Education	1.8	-	-	-	-	-	-	-	-	-	1.6	-	3.4
Health	1.5	-	-	-	-	-	-	-	-	-	1.5	-	2.9
Community build.	1.5	-	-	-	-	-	-	-	-	-	1.5	-	2.9
Street L	1.1	-	-	-	-	-	-	-	-	-	4.3	-	5.4
Transportation	-	1,971.2	-	1,377.0	-	-	-	-	-	-	-	-	3,348.2
Passenger Transportation	-	694.5	-	420.1	-	-	-	-	-	-	-	-	1,114.6
Freight Transportation	-	1,276.7	-	956.9	-	-	-	-	-	-	-	-	2,233.6
Total Demand	608.0	1,971.2	147.1	1,540.2	16.3	640.1	290.5	5.6	92.9	346.3	26.2	110.7	5,795.2

Residential sector

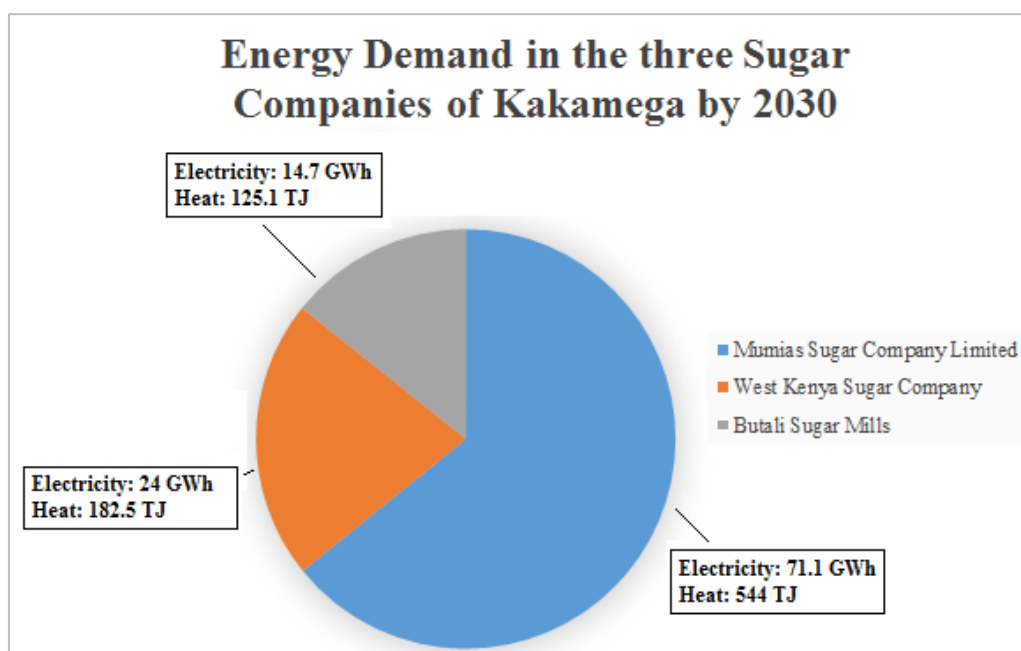
The implementation of the proposed scenarios is expected to transform the energy consumption profile in the residential sector. More specifically, the electrification rate is expected to reach 70% by 2030 which means that 365,820 households are expected to be given access to electricity by that year. In addition, the promotion of efficient technologies that was proposed by the DSM scenario, is expected to lower the energy demand in the sector by almost 53.7%. The cumulative energy demand including all fuels, is expected to reach 891.3 GWh for electrified households and 765.4 GWh for un-electrified households. Table 28, presents the energy demand by activity for both categories under the reference and SE4ALL scenario respectively.

Table 28. Energy demand (in GWh) per household activity under reference & SE4ALL scenario by 2030.

	Electrified Households				Un-electrified households			
	Reference		SE4ALL		Reference		SE4ALL	
	2015	2030	2015	2030	2015	2030	2015	2030
Cooking	191	1400.5	225.3	596.8	3241.1	3166.4	2889.9	632
Lighting	5.7	41.5	7	76.5	416.1	408.3	382.2	101.4
Water heating	4.4	45.3	6	83.4	45.3	107	49.4	32.1
Other	10	73.1	13.6	134.6	-	-	-	-

Industrial sector and Transportation

The energy demand in both sectors is expected to increase over the next fifteen years however no specific goals were set for these sectors. In the industrial sector the demand will increase in proportion with the sugar production in the County, which is expected to reach 422.7 ktons of sugar by 2030, an increase of 54.5% over the current stage (273.6 ktons). Graph 13 illustrates the energy demand in final units by 2030, for the three sugar companies operating in Kakamega.



Graph 13. Final energy demand by Sugar Company by 2030.

In the transportation sector the demand will increase in proportion with the population and the economic development in the County. More specifically, the energy correlated with the passenger transportation is

expected to increase by 53.7% under the reference scenario or by 45.8% under the SE4ALL scenario which implies intensification of the public transport (increased share of busses from 6% to 20% by 2030).

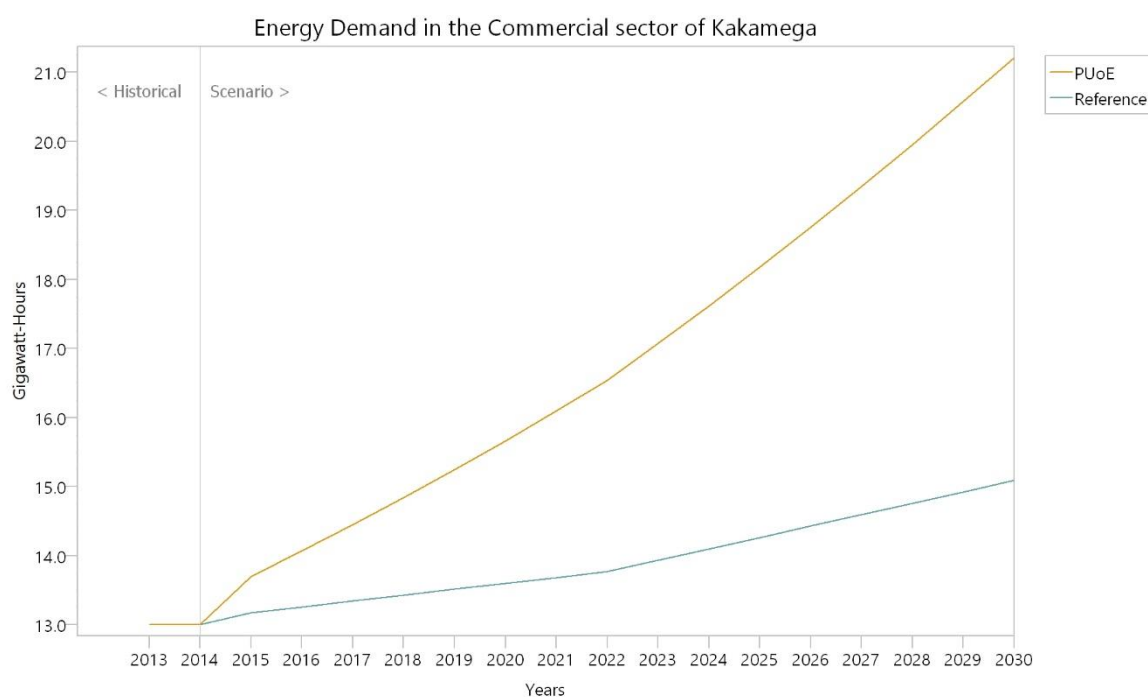
On the other hand, the prospective development in the commercial and agricultural sector is expected to impose an increase of 134.8% in the energy demand related to freight transportation. Table 29, presents the projected energy demand per vehicle category in Kakamega over the next fifteen years.

Table 29. Energy demand (in GWh) per vehicle category in Kakamega till 2030 (reference scenario).

	2015	2020	2025	2030
Passenger Transportation				
Cars	342.3	401.3	470.1	550.3
Motor Cycles	397.9	453.5	516.8	589.0
Busses/Mini Busses	29.4	33.5	38.2	43.5
Walking/Cycling	-	-	-	-
Freight Transportation				
Utilities/Parcel vans/Pick ups	750.1	997.0	1,325.2	1,761.5
Lorries/Heavy vans	139.6	185.6	246.7	327.9
Trailers	24.7	32.8	43.7	58.0
Other Vehicles	36.7	48.8	64.8	86.2
Total	1720.6	2152.4	2705.4	3416.3

Commercial sector

As described in *CHAPTER 3. ENERGY MODELLING*, the productive use of energy expansion, is a necessary prerequisite for the economic development of the County. The quantitative and qualitative increase of energy in the commercial sector was one of the main goals of the PUoE scenario. Graph 14, compares the energy demand in the sector under the reference and the PUoE scenario while Table 30 provides a more detailed description of the energy demand projection over the sub categories of the sector.



Graph 14. Energy demand projections for the Commercial sector in Kakamega by 2030.

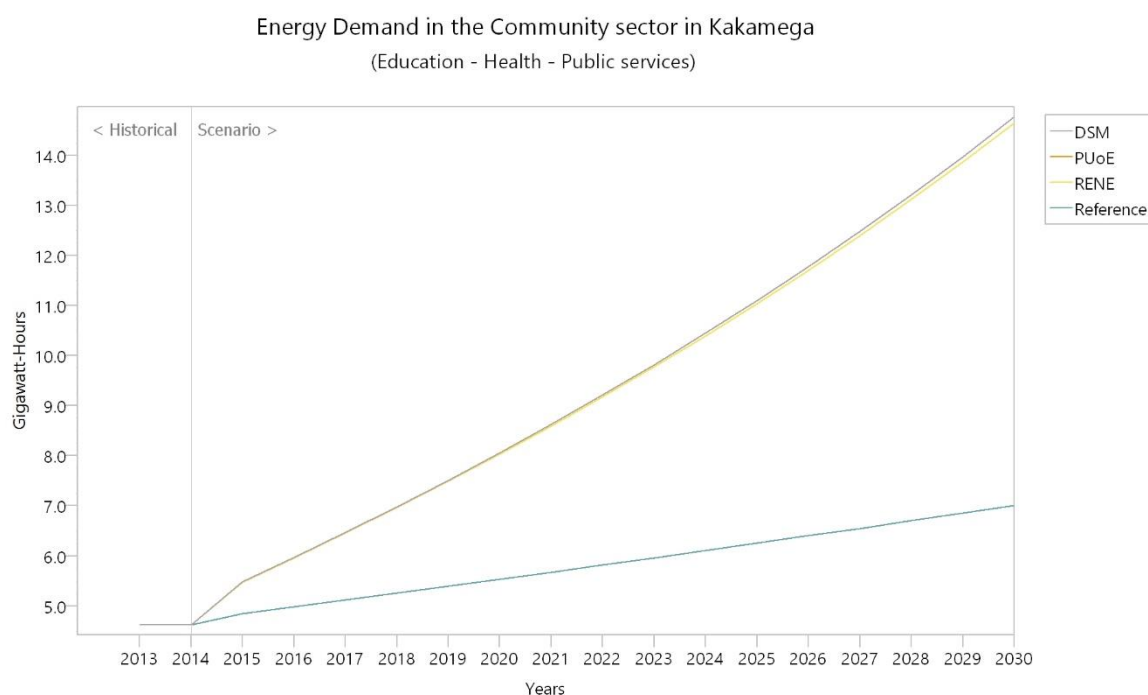
It is observed that the PUoE scenario implementation will impose an increase of about 55% in the energy demand of the sector. The increase is more evident in small scale businesses which intend to constitute the backbone of Kakamega's economy by 2030 with 1,000 new small enterprises established within the next fifteen years. In addition, the increased energy intensity proposed for the medium and large businesses is expected to improve the quality of services, assist in the expansion of operations and ultimately lead to revenue increase.

Table 30. Energy demand (in GWh) per vehicle category in Kakamega

	Reference		PUoE	
	2015	2030	2015	2030
Medium or Large businesses	11.3	12.4	11.4	13.9
Hotels_Guest houses_Restaurants	2.6	2.9	2.7	3.3
Super Market_Open Markets_Shopping Malls	6.4	7	6.5	7.7
Service Businesses	2.3	2.5	2.3	2.9
Small Scale Businesses	1.9	2.7	2.3	7.3
Total	13.2	15.1	13.7	21.2

Community Sector & Public services

The access to energy services is of vital importance for the community sector as it fundamentally enhances the welfare of the society. Substantial improvements were proposed by the alternative scenarios for this sector with aim to increase the quality in education, health and public services. The energy demand projections under different scenarios are presented in Graph 15.



Graph 15. Energy demand projections for the Community sector in Kakamega by 2030.

Under the reference scenario the energy demand is expected to grow with a rate of 45.8%, proportionally with the population increase projected for the County. However, the implementation of the goals proposed by the PUoE scenario require an increase of 169% in the energy demand. More specifically, the education sector is expected to require 3.4 GWh by 2030, the health sector 2.1 GWh, the community

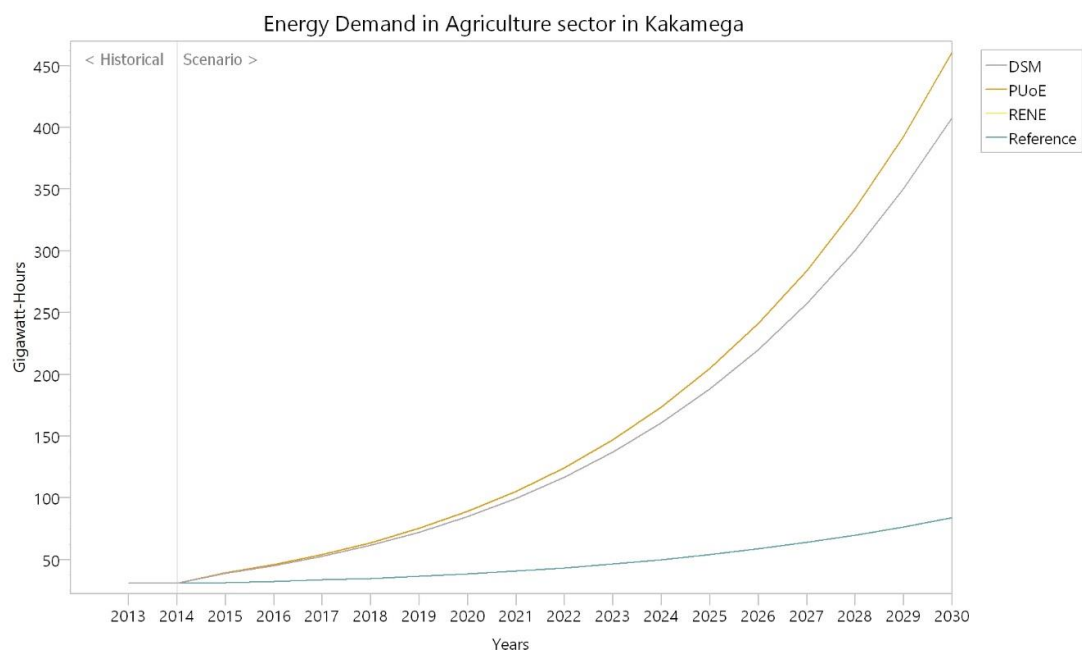
buildings 2.9 GWh and street lighting services 5.6 GWh. It should be mentioned in this point that the same demand pattern was followed in the formation of the RENE scenario as well. While the two scenarios (PUoE and RENE) show similar demand increase, their transformation profile changes significantly due to the increased adoption of solar powered technologies in the latter one. Table 31, provides a detailed description of the energy demand projections of various activities taken into consideration if the community sector.

Table 31. Energy demand projections (in GWh) per sub-category of the Community sector in Kakamega.

	Reference		PUoE	
	2015	2030	2015	2030
Education	0.9	1.3	1.0	3.4
Primary	0.5	0.8	0.6	1.8
Secondary	0.15	0.23	0.2	1.3
Higher	0.2	0.3	0.2	0.3
Health	0.6	0.8	0.6	2.9
Health centers/Dispensaries	0.2	0.3	0.2	1
Health Clinics	0.2	0.3	0.2	0.5
Hospitals	0.2	0.2	0.2	1.4
Community buildings	1.8	2.1	1.9	2.9
Street lighting	1.6	2.8	1.9	5.6
Total	4.9	7.0	5.4	14.8

Agriculture, Aquaculture & Livestock

The adoption and expansion of modern forms of energy in agriculture, is essential for Kakamega, whose economic welfare and development is highly interconnected with this sector. Therefore, special focus was given in the quantitative and qualitative increase of energy in the sector, aiming to improve productivity and upgrade of agricultural value chains. In order to make this possible, the PUoE scenario proposed that agriculture should contribute by 25% in the County's GDP by 2030. This goal was also escorted by an increase in energy intensity of the main agricultural activities, which basically implied the utilization of more energy per activity.



Graph 16. Energy demand projections for Agriculture sector in Kakamega by 2030.

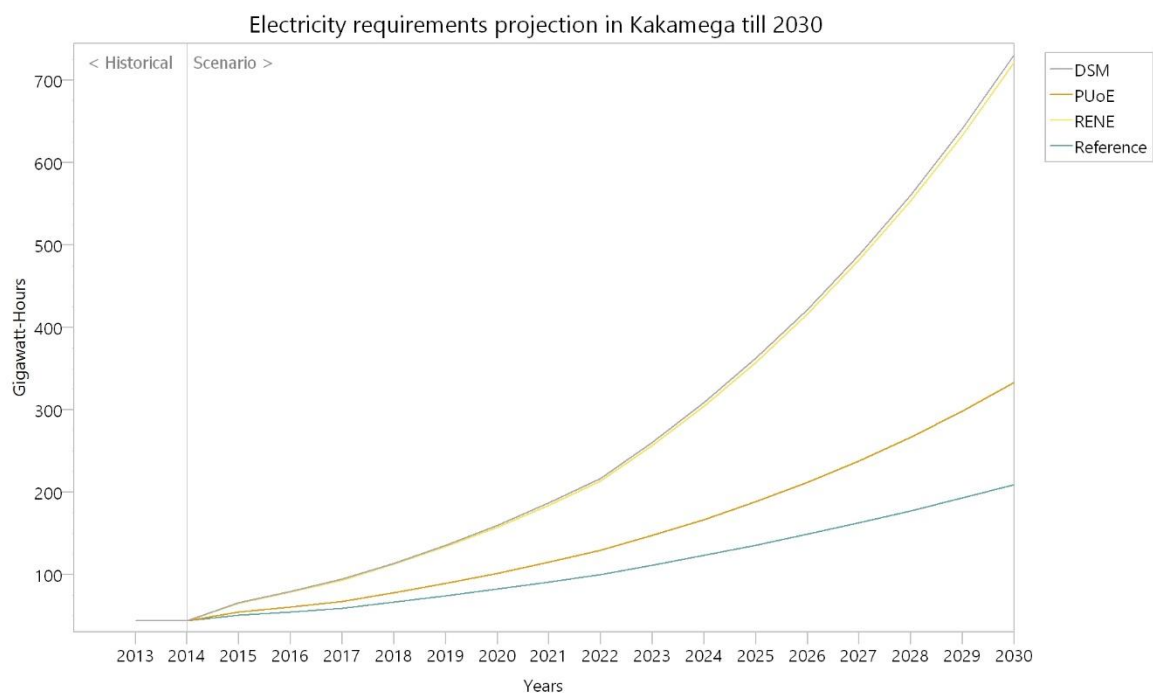
The modernization and energy intensification of the agriculture sector is expected to increase the demand to a significant extent. More specifically, the total energy demand in the sector is expected to increase from 39.4 GWh in 2015 to 461.2 GWh in 2030 under the PUoE scenario. The demand is slightly lower under the DSM scenario, which implies the adoption of electricity (more efficient form of energy) to a greater extent. Graph 16, illustrates the different demand projections under the four alternative scenarios.

4.2 Energy supply

The transformation of the energy sector in Kakamega cannot be integrated without improvements being implemented in the supply side as well. This chapter presents the energy generation projections under different scenarios over the next fifteen years.

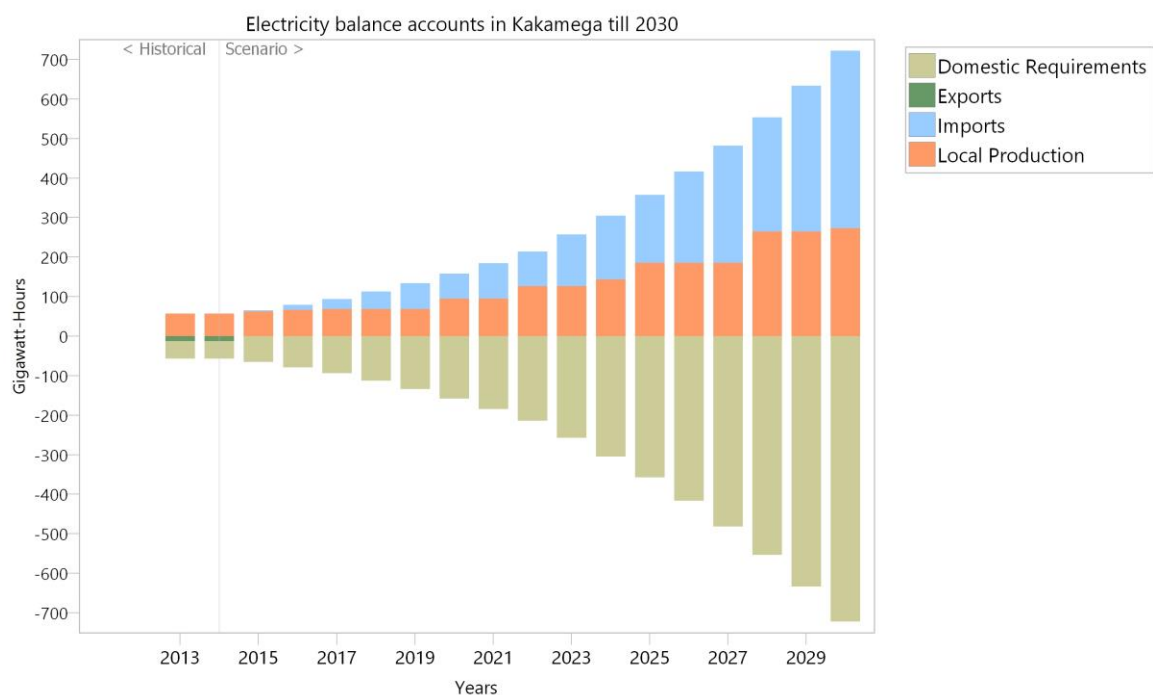
Electricity

Adequate electricity generation is fundamental for the successful implementation of the proposed scenarios, however the requirements are expected to vary. More specifically, as in 2014 the electricity requirements in Kakamega were estimated at 44.4 GWh. The reference scenario indicates that the electricity demand will be 4.7 times higher by 2030 reaching 209.3 GWh. The requirements are even higher under the PUoE scenario reaching 333.6 GWh by 2030. Finally, the intensive electrification plan proposed by the SE4ALL (and more specifically DSM) scenario will increase electricity requirements of the County to 730.8 GWh by 2030. Graph 17, shows the aforementioned projections over the next fifteen years.



Graph 17. Electricity generation requirements in Kakamega projected till 2030.

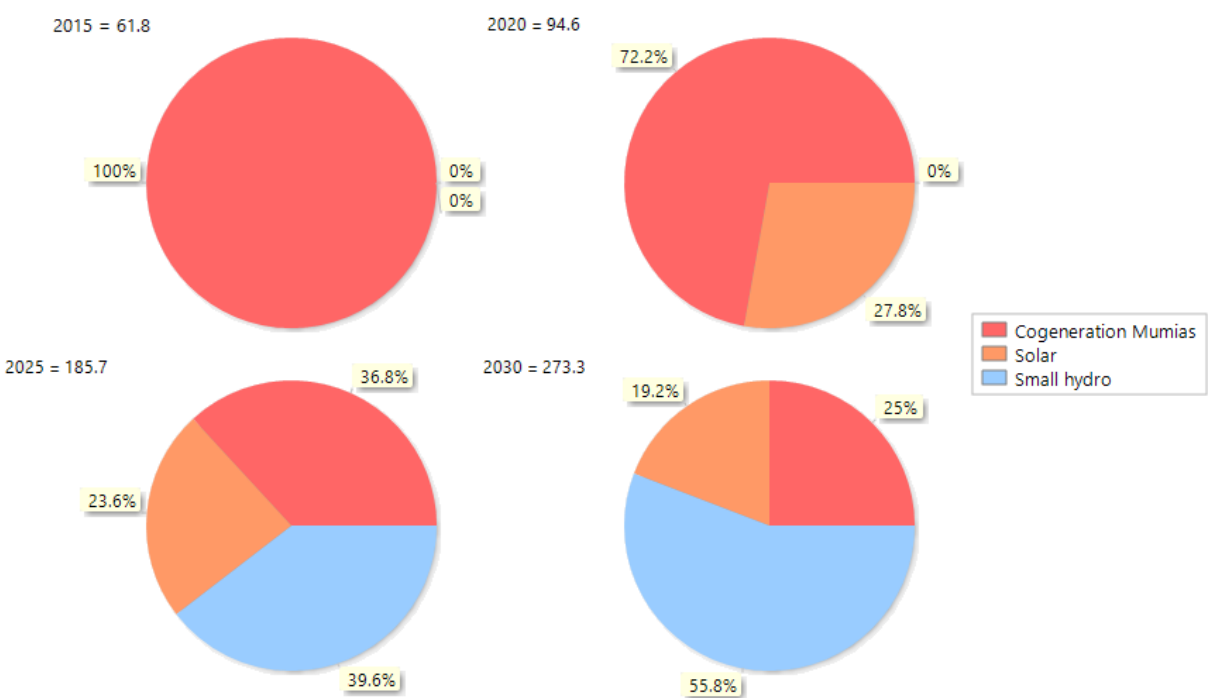
In order to cope with the increasing electricity demand, 59 MW of additional capacity were proposed with the RENE scenario. More specifically, a capacity of 30 MW of solar PV power plants is expected to be constructed in the County and be gradually commissioned by 2030.



Graph 18. Electricity balance in Kakamega projected till 2030.

The estimated electricity output is 52.6 GWh per year by 2030. Small and mini hydro power plants are expected to reach a capacity of 29MW by 2028 generating 152.4 GWh per year (by 2028). The generation capacity in the County is expected to cover 37.9% of the indigenous electricity requirements in 2030,

with the imports increasing significantly every year. Graph 18, illustrates the balances between indigenous production and imports of electricity in Kakamega over the next fifteen years under the RENE scenario. Graph 19, depicts the expected electricity generation mix in the County under the same scenario.

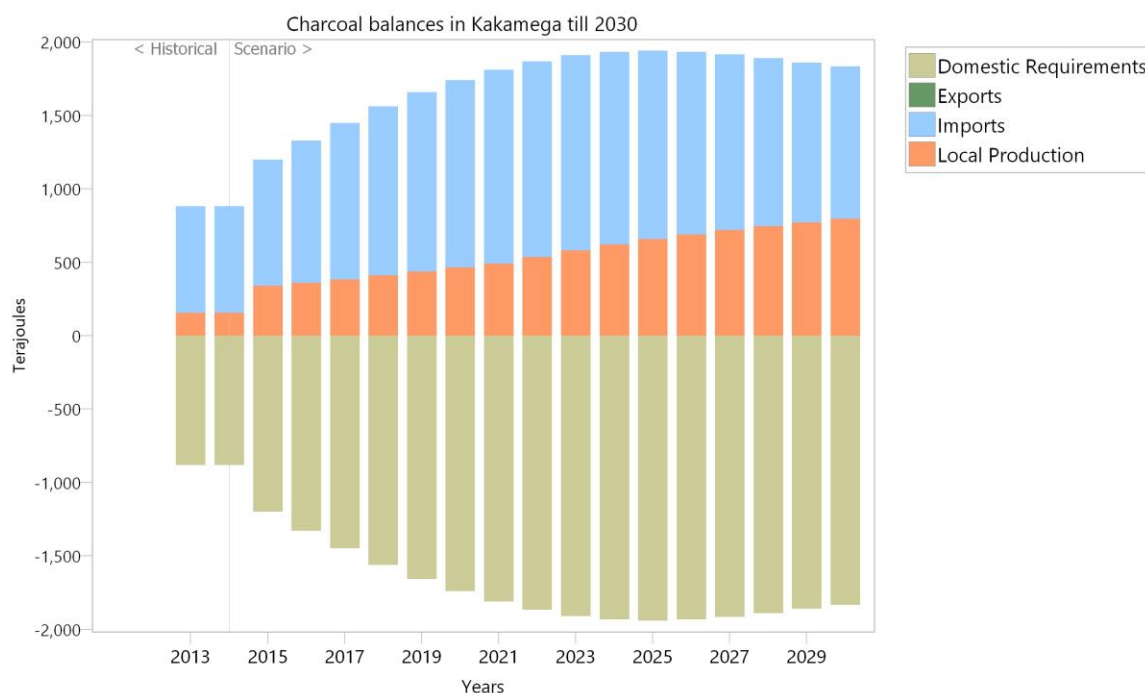


Graph 19. Electricity generation mix in Kakamega by 2030.

Charcoal

Charcoal is a significant fuel for Kakamega, as it is expected to assist in the transition from traditional to modern fuels especially in the residential sector. Its availability is therefore necessary for this transition to be smooth and effective. As in 2014, the requirements for charcoal in the County accounted for 882.4 TJ. Under the reference scenario the expected demand for charcoal by 2030 is 2,640 TJ. The demand is significantly higher (3,129 TJ) under the PUoE expansion scenario which suggests increased use of charcoal as fuel in the commercial sector. However, the implementation of the DSM scenario will induce the charcoal demand to stabilize and create a plateau after 2026 over 1,850 TJ/year. Graph 20, illustrates the charcoal balance accounts till 2030 under the DSM scenario.

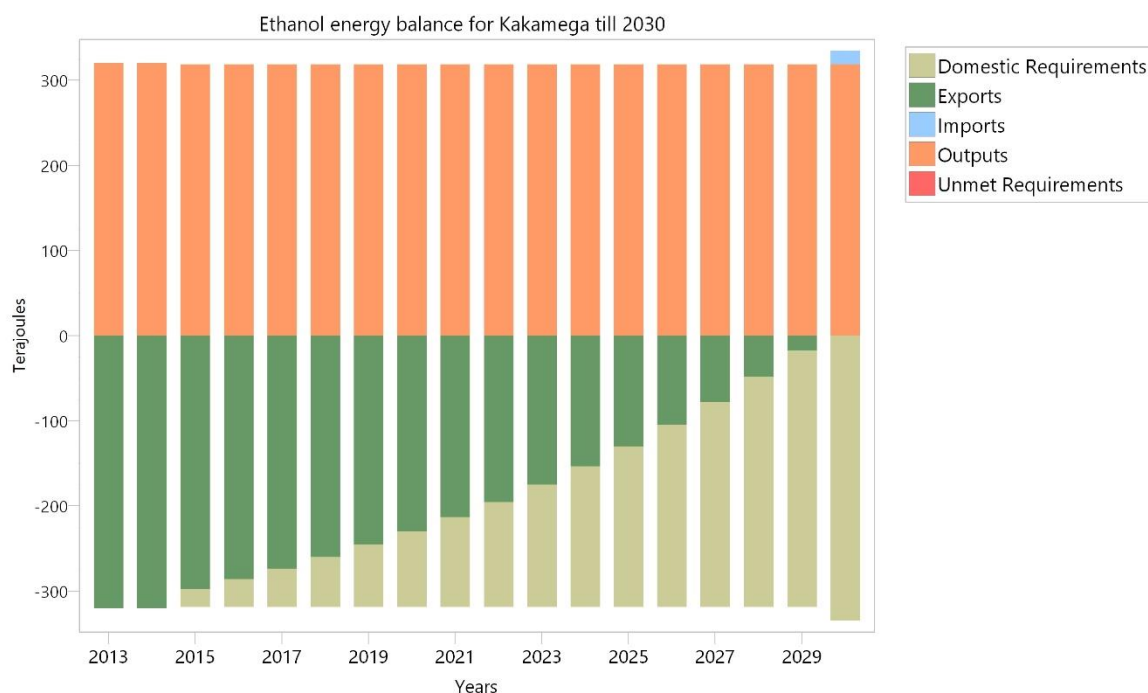
It should me mentioned at this point that the charcoal production capacity of Kakamega in not unlimited. Deforestation issues in the County set restrictions and require that the increasing demand (and the proportionally increasing production capacity) should be covered by the adoption of modern, more efficient technologies in the sector (improved kilns, Mekko kilns etc.). Under the DSM scenario it is expected that the indigenous charcoal production will cover 43.5% of the total domestic demand in 2030, almost double the share at the current stage, without compromising the sustainability of the gazette forests in the region.



Graph 20. Charcoal balances in Kakamega till 2030 under the DSM scenario.

Ethanol

As described in *CHAPTER 3. ENERGY MODELLING*, Ethanol is solely produced by Mumias Sugar Company based on molasses deriving as by-product of the sugar production process. The full capacity of the company is 22 million liters of ethanol per year which is translated to 465.95 TJ per year. However, the plant's production availability was set and kept at 68.4% in order to consort with the real data found for the base year (2013). Therefore the ethanol production capacity is constant at 318.7 TJ (around 15 million liters) per year till 2030. Under the reference scenario all the ethanol produced in the County is exported. On the other hand, under the DSM scenario part of the produced ethanol is circulated within the borders of the County in order to cover the domestic requirements, deriving by the increasing use of ethanol stoves in the residential sector. Graph 21 illustrates the ethanol balances till 2030 as projected under the DSM scenario for Kakamega.



Graph 21. Ethanol energy balance in Kakamega till 2030 under the DSM scenario.

4.3 Resources

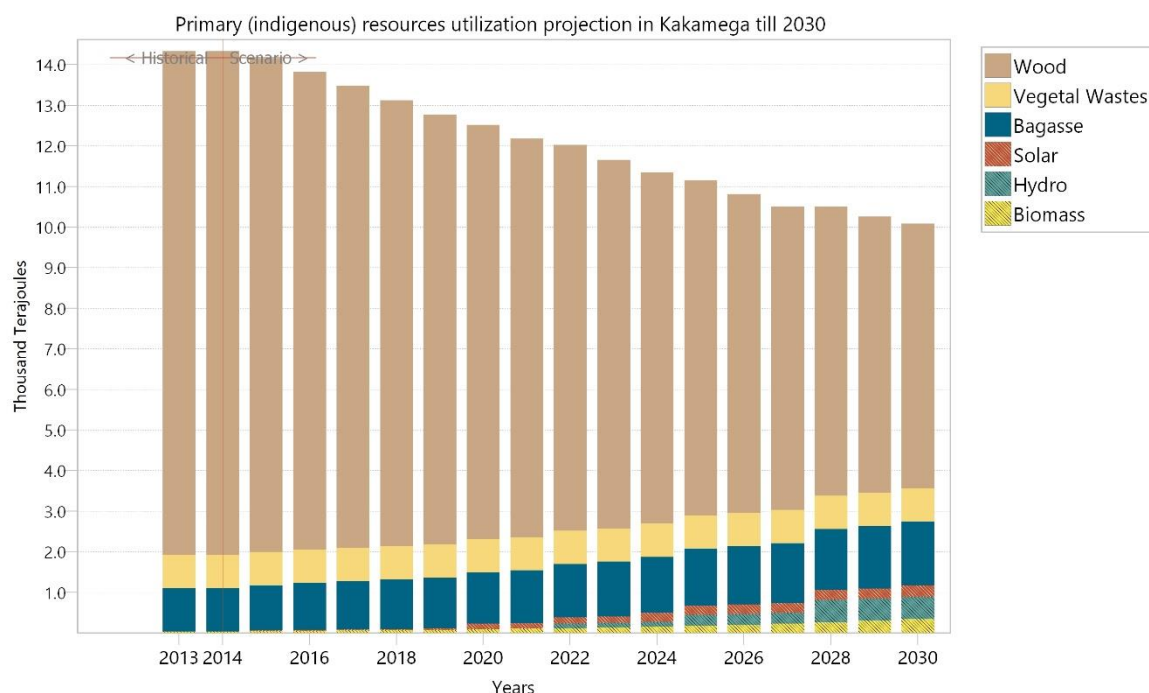
The prospective energy transformation in Kakamega is highly interconnected with the ability of the County to have access to essential resources. As described in *CHAPTER 2. KAKAMEGA COUNTY CASE STUDY*, the main primary energy resources in Kakamega are solar, hydro and different sources of biomass (wood, bagasse, agroforestry residues). The secondary resources are mainly comprising by charcoal, kerosene, gasoline, diesel, LPG, electricity and biogas. While access to secondary resources should not be contestable, it is necessary for the County do strengthen its self-sustainability by following pathways that will effectively utilize available resources and minimize the dependency on imported fuels.

As in 2014, the total resource requirements in Kakamega were estimated at 22,300 TJ with the primary resources accounting 64% and secondary the rest 36%. As an initial remark, it should be mentioned that the total energy resource requirements are expected to grow under any scenario. More specifically, under the reference scenario, it is expected that they will reach 35,900 TJ by 2030 with primary and secondary resources accounting for 55% and 45% respectively. As expected the PUoE scenario will impose an increase in the resource requirements in the County, which however could be mitigated by the implementation of the SE4ALL scenario. In fact, the combination of these scenarios is expected to reduce the total resource requirements in Kakamega by 27.6% (in respect with the reference scenario) by 2030 to 26,000 TJ. Table 32, provides a comprehensive description of the resources projections over the next fifteen year in Kakamega.

Table 32. Energy demand projections (in TJ) per sub-category of the Community sector in Kakamega.

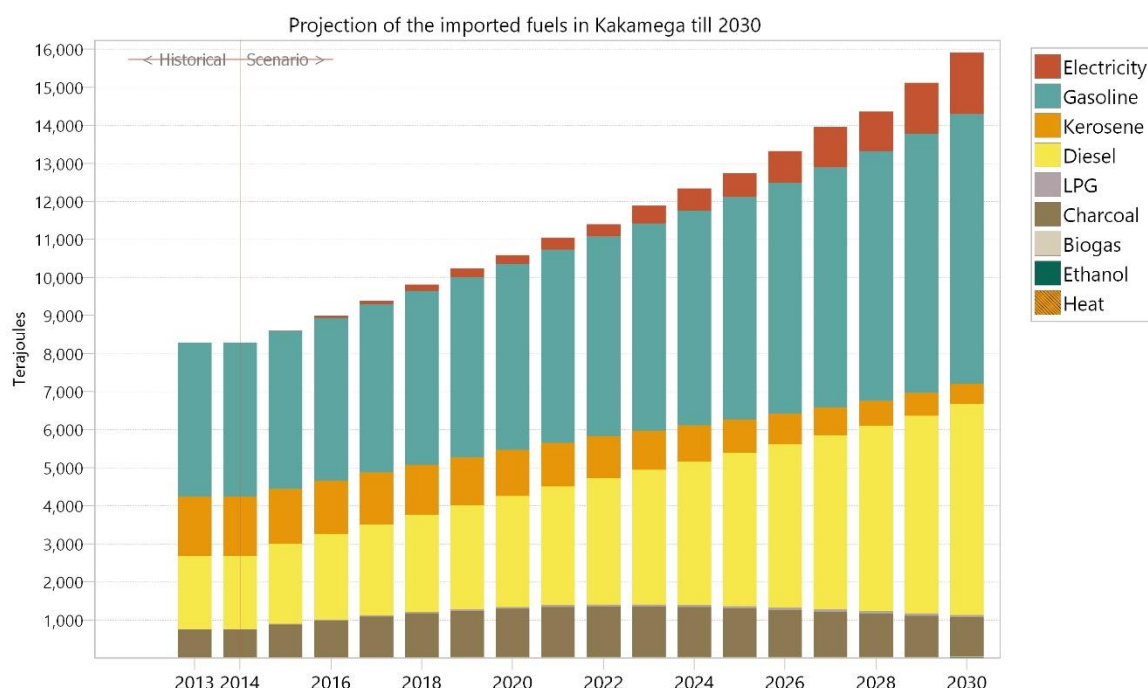
	Reference Scenario			PUoE scenario			SE4ALL scenario		
	2015	2022	2030	2015	2022	2030	2015	2022	2030
Primary Resources	14,931	17,388	19,596	15,051	17,889	21,877	14,148	12,025	10,087
Wood	13,016	15,199	17,118	16,117	15,644	19,153	12,155	9,506	6,524
Bagasse	1,063	1,323	1,575	1,077	1,323	1,575	1,115	1,323	1,575
Molasses	819.3	819.3	819.3	819.3	819.3	819.3	819.3	819.3	819.3
Solar	5.5	13	25.4	5.6	13.8	27.4	12.2	147.5	284
Hydro	0	0	0	0	0	0	0	113.5	549
Biomass	26.1	33.1	57.8	31.7	88.7	302.2	46.2	115.1	337
Secondary Resources	8,260	11,184	16,337	8,275	11,528	17,839	8,302	11,201	15,915
Kerosene	1,575	1,665	1,606	1,575	1,665	1,606	1,447	1,105	529.7
Gasoline	4,181	5,450	7,529	4,180	5,450	7,529	4,142	5,251	7,096
Diesel	2,071	3,115	4,921	2,084	3,261	5,600	2,103	3,324	5,545
LPG	12.2	15.4	20.4	12.2	15.4	20.4	21	48	58.7
Charcoal	723.9	1,124	2,050	723.9	1,216	2,427	857	1,330	1,035
Ethanol	318.7	318.7	318.7	318.7	318.7	318.7	297.7	195.3	318.7
Biogas	15.4	18	21.3	15.4	18	21.3	15.3	17.6	15.7
Electricity	0.7	115.3	507.4	2.4	221.5	955.1	14.1	316	1,615
Total	23,190	28,571	35,932	23,326	29,417	39,716	22,450	23,226	26,002

Graph 22, shows the projected utilization of the indigenous energy resources under the SE4ALL scenario till 2030. A reduction is evident mainly due to the replacement of wood as fuel in the residential sector and the adoption of more efficient transformation technologies (cook stoves, charcoal kilns etc.) while the majority of the rest primary resources seem to follow an increasing trend.

**Graph 22.** Primary energy resources utilization in Kakamega till 2030 under the SE4ALL scenario.

Graph 23, shows the projection of imported fuels in Kakamega under the SE4ALL scenario till 2030. As in 2015, the total imported fuels in Kakamega accounted for 8,599 TJ. Gasoline and diesel dominate the imports as they are correlated with a sector (transportation), whose energy demand is difficult to mitigate. Hence, an increase in necessary imports in these two fuels is necessary in order to meet the demand in Kakamega. On the other hand, kerosene and (imported) charcoal are expected to decrease over the next fifteen years. Finally, electricity imports are expected to reach 449 GWh by 2030 accounting for the

62.1% of the total electricity requirements that year. The total imported fuels are expected to account for 15,915 TJ by 2030.



Graph 23. Projection of imported fuels in Kakamega till 2030 under the SE4ALL scenario.

4.4 Environmental assessment

The sustainable energy transition described in the previous paragraphs will create new possibilities for Kakamega in terms of growth while at the same time it will secure the precious natural resources, the environment and human welfare. The Global Warming Potential (GWP) index (100 years horizon) was selected in order to present the environmental assessment of the energy sector in Kakamega (EPA n.d.), (IPCC n.d.).

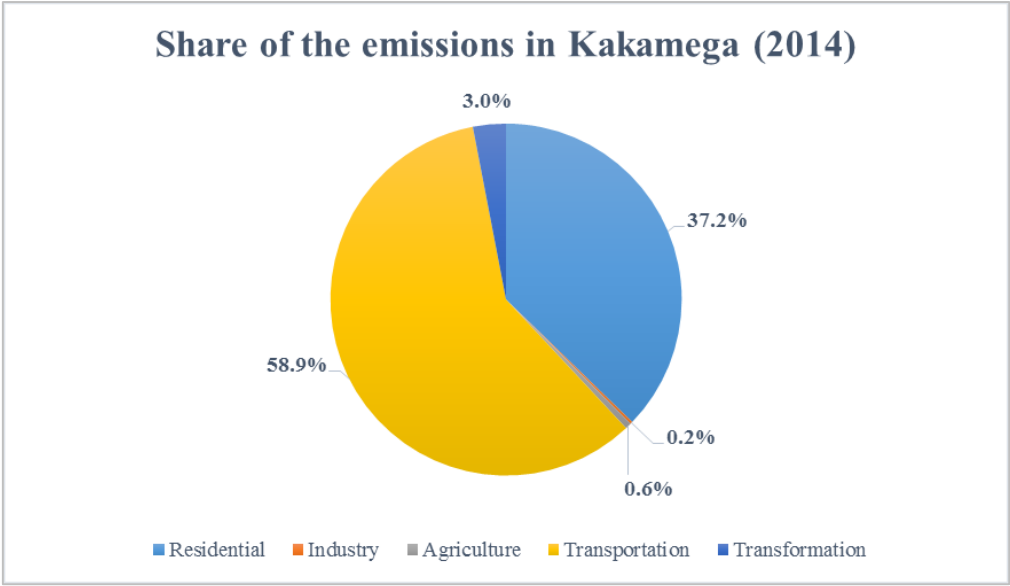
As in 2014, the total emissions were estimated at 708.7 ktons of CO_{2e}, with 58.9% coming from the transportation sector, 37.2% from the residential sector, 3% from the transformation sector (energy production), 0.6% from agricultural activities and 0.2% from the industrial sector. Graph 24, illustrates the share of estimated emissions in Kakamega in 2014.

More specifically, in the residential sector, the un-electrified households are responsible for the 92% of the emissions due to the use of traditional and severely polluting fuels such as firewood, charcoal, kerosene etc. Cooking is the most polluting activity followed by lighting. In transportation, 55% of the emissions are deriving from freight transport (Lorries, heavy vans, utility cars) while the rest 45% is associated with passenger transport (cars and motorcycles). In the transformation sector 51.4% of the emissions are correlated with heat generation, 46.2% with charcoal production and only 2.4% with electricity generation.

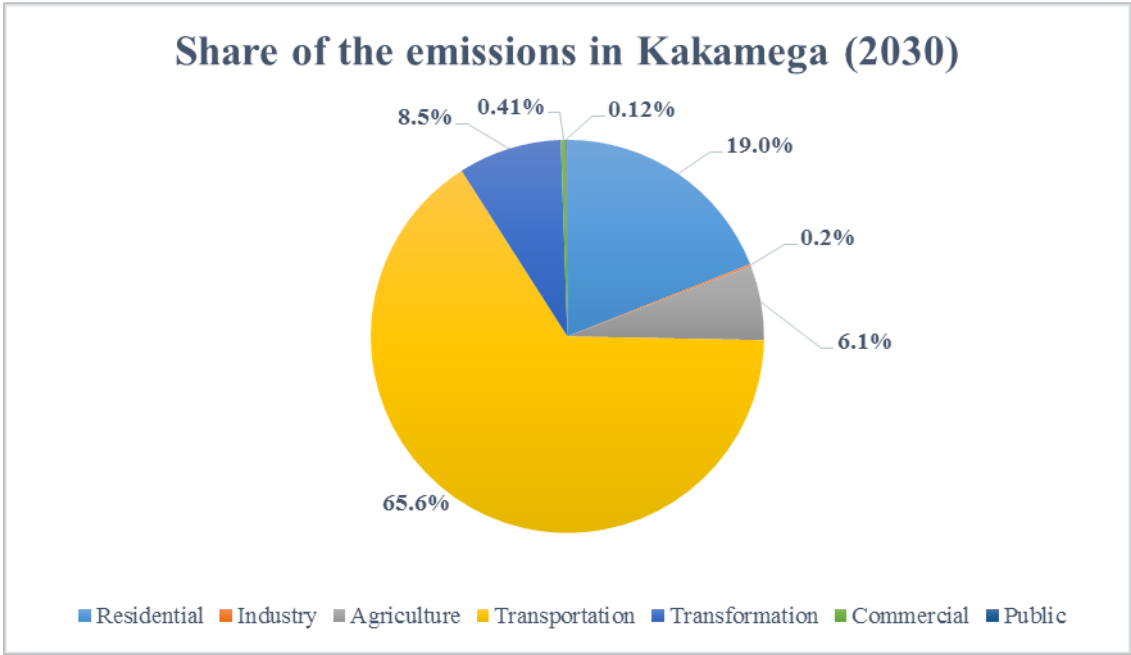
It should be mentioned at this point that emissions associated with electricity are a critical point in this analysis. LEAP provides the possibility of assigning environmental loadings to various generation technologies. In Kakamega, the existing and proposed electricity generation is deriving from renewable

energy sources implying very low to zero emission levels. According to the proposed alternative scenarios, electricity is expected to show increasing penetration rates in the following years. That is, the imported from the grid electricity is expected to increase. However, environmental loadings for imported electricity is not an option in LEAP. Therefore, the emissions associated with the imported electricity were calculated manually according to the emission factors associated with the national grid in Kenya.

The methodology used, associated the emission factors of carbon dioxide (CO₂), methane (CH₄) and nitrous oxide (N₂O) to the electricity consumed (in this case imported). In particular, one kWh of electricity imported was associated with 0.332 kgCO₂, 13.07x10⁻⁶ kgCH₄ and 2.62x10⁻⁶ kg N₂O (Matthew Brander 2011), (Mott MacDonald 2010), (IEA 2014). The aggregated GWP index was then estimated for the different scenarios.



Graph 24. Share of estimated emissions by sector in Kakamega (2014).

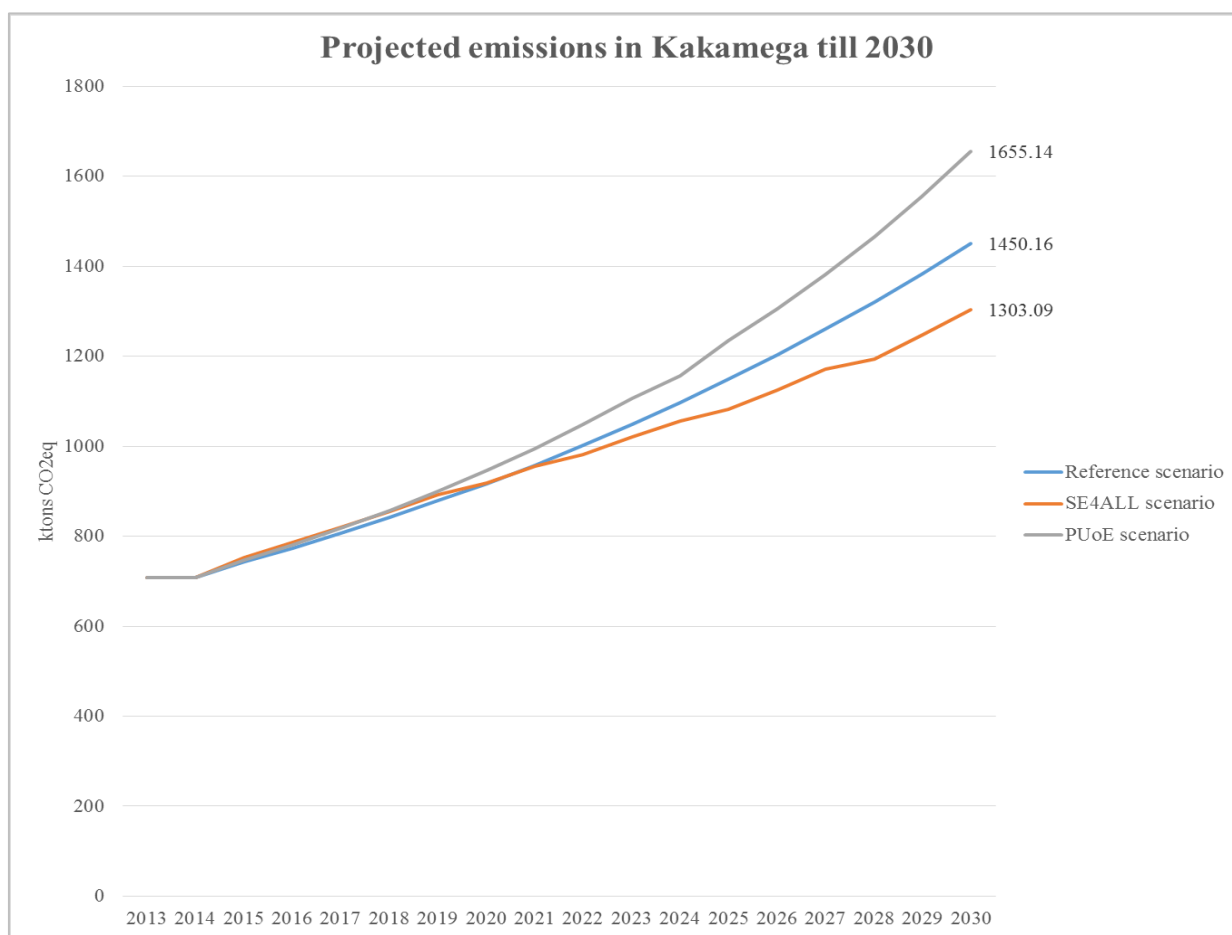


Graph 25. Share of estimated emissions by sector in Kakameg under SE4ALL scenario.

According to the reference scenario, GWP index is expected to increase significantly reaching 1,450 ktons CO_{2e} while the increase is even higher under the PUoE scenario, reaching 1,655 ktons CO_{2e} by 2030. However, the adoption and implementation of the SE4ALL scenario, will lead to lower greenhouse gas emissions over the next fifteen years for the County, resulting to 1,303 ktons CO_{2e} in 2030.

According to the SE4ALL scenario it is expected that 65.6% of the emissions will be associated with the transportation sector, 19% with the residential sector, 8.5% with the transformation sector (energy production), 6.1% with agricultural activities and 0.41% with the commercial sector, 0.2% with the industrial sector and 0.12% with the public sector. Graph 25, illustrates the share of estimated emissions in Kakamega in 2030 under the SE4ALL scenario.

Comparing the SE4ALL and the reference scenarios, the accumulated GHG emission savings were estimated at 297 ktons CO_{2e} by 2030. **Σφάλμα! Το αρχείο προέλευσης της αναφοράς δεν βρέθηκε.**, compares the Global Warming Potential (GWP) index (100 years horizon) between the proposed scenarios for Kakamega till 2030.



Graph 26. Global Warming Potential (GWP) among different scenarios for Kakamega till 2030.

4.5 Cost assessment

The transformation proposed for Kakamega with the PUoE and SE4ALL scenarios, involve radical changes both in the energy consumption and generation pattern of the County. These changes are associated with a various social costs deriving by both the capital (acquisition) costs of new equipment

and operating costs. A preliminary assessment was performed in order to roughly estimate these costs and provide a monetary perspective to the aforementioned transformation. The values presented below were calculated under a discount rate of 5%.

Thus, it was estimated that the changes in the residential sector proposed by the DSM scenario will impose an accumulative cost of 11.6 million USD. Furthermore, the RENE scenario proposal for 1MW additional PV capacity in the community sector is expected to have an additional cost of 0.5 million USD. Therefore, the total social costs related with the demand management in Kakamega require a cost of 12.1 million USD.

The construction of the proposed 59 MW of small hydro and photovoltaic capacity by 2030 is expected to have an additional cost of 104.6 million USD.

On the other hand, it is estimated that the changes in the indigenously produced resources, the decreasing imports of certain fuels and additional exports will lead to a reduction in the resource expenses by 243.4 million USD in regards with the reference scenario.

Consequently, the proposed transformations in Kakamega under the combination of the scenarios are expected to have a positive balance characterized by a net present value of 126.7 million USD.

DISCUSSION AND CONCLUSIONS

A discussion of the results and the conclusions that the author has drawn during the Master of Science thesis are presented in this chapter. The conclusions are based from the analysis with the intention to answer the formulation of questions that is presented in Introduction.

In this study, a comprehensive analysis of the energy sector for Kakamega County in Western Kenya was performed. The study initiated with a base line survey recording and estimating the current energy demand level in the county for six selected sectors: Residential, Industrial, Transportation, Commercial, Public and Agricultural. As an outcome, a benchmark of energy uses and their respective technological solutions was created. Additionally, the renewable energy resources potential was assessed at local level using GIS and other available data.

At a secondary level, the study proceeded with the County's energy supply and demand modelling in LEAP. The model was created based on the data acquired from the previous stage of the study while three 15-year projection scenarios were developed. The first scenario (Reference) was used to describe the energy projections in the County under Business As Usual activities. The second scenario (Productive Use of Energy Expansion) incorporated energy related enhancements in various activities, aiming to increase their associated productivity. Finally, the third scenario (Sustainable Energy for All) induced changes targeting at universal access to modern energy services, increased energy efficiency and maximized share of renewables in the County's energy mix.

The implementation of the scenarios followed the same hierarchy, being inserted into the model in the form of twenty five energy related interventions with specified target goals and timelines. The disaggregated nature of the model, mandates that the results should be analyzed according to the respective question needed to be answered. However, here are some of the key elements outlining the pathway suggested by the aforementioned scenarios.

The electrification rate of households in Kakamega is estimated to reach 70% by 2030. Significant increase in the intensity, quality and productive use of energy is expected in agriculture, commercial and public sector aiming to amplify the socioeconomic conditions in the County. However, despite the increased energy requirements, the total demand in 2030 is expected to be retained at current levels (approx. 5.8 TWh/year) due to demand side management interventions.

In addition, the primary requirements of kerosene are expected to be reduced by 67%, of wood by 62% and of charcoal by 49.5% by 2030. Furthermore, the indigenous electricity generation is expected to cover 37.9% of the County's requirements, all deriving from renewable sources. While the GHG emissions will continue to show an increasing trend over the next years, it should be noticed that the suggested alternative scenarios will induce a 10.1% reduction of them by 2030 in comparison with the reference (BAU) scenario. Finally, a preliminary cost assessment showed that the proposed interventions will yield a beneficiary cost balance with net present value of 126.7 million USD.

To conclude, this thesis intended to create a framework aiming to facilitate sub-national energy planning in developing countries through a holistic approach securing the economic, social and environmental sustainability of the selected region. While further enhancements are necessary, it is expected that the findings will be complementary to already existing energy planning models but also the base for future research towards energy poverty elimination.

RECOMMENDATIONS AND FUTURE WORK

Paying respect to the global initiatives towards energy poverty elimination through sustainable development and promoting the continuous research on the field of renewable energy assessment and implementation, the following future work is suggested.

This thesis was elaborated on findings, estimations and assumptions derived mainly from literature review. While cross referencing was fundamentally followed, validation of the data through a field study is necessary and would add significant value to the work performed. Therefore is highly suggested as a next step.

Further improvement of the LEAP model is also recommended by increasing the disaggregation level of energy activities in all sectors. Another option would be the incorporation of the multi-tier methodology of energy access measurement, proposed by ESMAP, into the LEAP model. That would allow a more detailed and comprehensive representation of energy access in all sectors.

Finally, a more elaborated analysis based on GIS approach could be performed in order to accurately quantify the renewable resource potential and identify the optimal electrification option in the selected region.

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APPENDIX A

Existing isolated mini-grids in Kenya (MoEP 2015), (ECA & Trama TecnoAmbiental 2014).

Site & County	Year Commissioned	Number of customers	Installed Capacity (kW)	Proposed actions
Wajir (Wajir)	1988	3,360	1,746 - Diesel	800, 850 – Solar 300 - Wind
Merti (Isiolo)	2007	287	128 - Diesel 10 - Solar (2011)	100 – Solar 100 – Wind
Habaswein (Wajir)	2010	779	640 - Diesel 20 - Wind (2012) 30 - Solar (2012)	100 – Solar 100 – Wind
Lodwar (Turkana)	2007	1,610	1,440 - Diesel 60 – Solar (2012)	250, 500 – Solar Wind
Hola (Tana River)	2007	1,300	800 - Diesel 50 - Solar (2012)	100 – Solar
Marsabit (Marsabit)	2011	2,194	800 - Diesel 500 - Wind (2011)	
Mandera (Mandera)	2007	3,270	1,600 - Diesel 300 - Solar (2012)	200, 650 – Solar Wind
El Wak (Mandera)	2009	535	360 - Diesel 50 - Solar (2012)	100 – Solar Wind
Baragoi (Turkana)	2009	199	128 - Diesel	100 Solar, Wind
Mfangano (Homa Bay)	2011	101	584 - Diesel 11 - Solar (2013)	100 – Solar
Garissa (Garissa) (KenGen)	1996	NA	6,700 - Diesel	Connection to the national grid
Lamu (Lamu) (KenGen)	1989	NA	2,900 - Diesel	Connection to the national grid
Rhamu (Mandera)	2012	210 (1,770 potential)	184 - Diesel 50 - Solar	50, 150 - Solar
Eldas (Wajir)	2012	80 (1,370 potential)	184 - Diesel 30 - Solar	30, 150 – Solar
Takaba (Mandera)	2012	153 (2,526 potential)	184 - Diesel 50 - Solar	50, 150 – Solar
Lokichoggio (Turkana)	2012	10,980 potential	640 - Diesel	150 – Solar Wind
Lokori (Turkana)	2014	8,261 potential	184 - Diesel	150 – Solar
Faza (Lamu)	2014	1,681 potential	380 - Diesel	100 – Solar 100 – Wind
Laisamis (Marsabit)	2014	1,456 potential	184 - Diesel 80 - Solar	150 – Solar
North Horr (Marsabit)	2014	1,883 potential	184 - Diesel	100 – Solar 100 – Wind
Lokitaung (Turkana)	2014	7,239 potential	184 - Diesel	150 – Solar
Kiunga (Lamu)	Installation in progress	762 Households	184 - Diesel	150 – Solar
Hulugho (Garissa)	Installation in progress	759 Households	184 - Diesel	150 – Solar
Dadaab (Garissa)	Installation in progress	10,064 Households	640 - Diesel	100 – Solar 100 - Wind
Maikona (Marsabit)	In progress	1,208 Households	640 - Diesel	100 – Solar, Wind
Lokiriana	In progress	482	380 - Diesel	150 – Solar

(Turkana)		Households		
Banisa (Mandera)	In progress	3,217 Households	184 - Diesel	100 – Solar 100 – Wind
Kamoliriban (Mandera)	In progress	NA	NA	Not specific plans for Renewables
Kotulo (Mandera)	In progress	NA	NA	Not specific plans for Renewables
Khorondile (Wajir)	In progress	NA	NA	Not specific plans for Renewables
Kakuma (Turkana)	In progress	NA	NA	600 - Wind

APPENDIX B

Transmission network extension activity (Kenya Electricity Transmission Company Ltd. KETRACO n.d.).

Connected locations	Voltage	Status
National level		
Thika – Kiganjo (Nyaga)	32 km of 132 kV single circuit line Assosiated substations	In progress
Ishiara – Kieni – Embu	35 km of 132 kV line 23 MVA substation	In progress
Sultan Hamud – Wote – Kitui	37 km of 132 kV line 5 MVA substation	In progress
Bomet – Sotik	33 km of 132 kV line 23 MVA substation	In progress
Olkaria – Narok	68 km of 132 kV line 23 MVA substation	In progress
Lessos – Kabarnet	65 km of 132 kV 15 MVA substation	In progress
Nanyuki – Nyahururu	79 km of 132 kV line 23 MVA substation	In progress
Kisii – Awendo	44 km of 132 kV line 23 MVA substation	In progress
Eldoret – Kitale	60 km of 132 kV line 23 MVA substation	In progress
Kindaruma – Mwingi – Garissa	250 km of 132 kV line 23 MVA substation	In progress
Nairobi Metropolitan ring: Sub-stations	Part of 100 km of 400 kV line	In progress
Nairobi Metropolitan ring: Suswa – Isinya line	Part of 100 km of 400 kV line	In progress
Rabai – Malindi – Garsen – Lamu	320 km of 220 kV line Associated substations	In progress
Olkaria – Lessos – Kisumu	300 km of 220 kV double circuit line 90 MVA substation	In progress
Loiyangalani – Suswa	430 km of 400 kV line	In progress
Mombasa – Nairobi	475 km of 220/400 kV Associated substations	In progress
Kilimambogo – Thika – Githambo	17 km of double circuit 132 kV line 50 km 132 kV line 132 kV double substation	In progress
Kindaruma – Athi River	150 km of 220 kV line 75 MVA substation	Planned
Sondu – Kendu – Homa Bay	80 km 132 kV single circuit 2x23 MVA substation	Planned
Baringo – Rongai	80 km of 220 kV double circuit 2x75 MVA substation	Planned
Arusha – Nairobi	260 km of 400 kV line	Planned
Konza – Machakos	45 km of 132 kV line 23 MVA substation	Planned
Konza – Kajiado	45 km of 132 kV line 23 MVA substation	Planned
Meru – Isiolo – Nayuki	75 km of 132 kV line 23 MVA substation	Planned

Substation at Mariakani & Isinya	400/220 kV substations	Planned
Suaswa & Loiyangalani	400/220 kV substations	Planned
Nairobi Metropolitan ring: Suswa – Ngong line	45 km of double circuit line 2x23 MVA substation 220/66 kV substations	Planned
Regional level		
Lessos (Kenya) – Tororo (Uganda)	127 km of 400 kV line 2x75 MVA transformers 220 kV line upgrading	In progress
Eastern Africa Electricity Highway	700 km of 500 kV HVDC 400 kV substation	In progress
Kenya – Tanzania	500 km of 400 kV	Planned

APPENDIX C

Full list of stakeholders in the Kenyan Energy Sector.

1. Ministry of Energy & Petroleum (MoEP)
 2. Energy Regulatory Commission (ERC)
 3. Energy Tribunal
 4. National Electrification and Renewable Energy Authority
 5. The National Energy Institute
 6. Rural Electrification Authority (REA)
 7. County Energy Authorities
 8. Kenya Electricity Generating Company Limited (KenGen)
 9. Geothermal Development Company Limited (GDC)
 10. Independent Power Producers (IPPs) – 9 IPPs accounting for 24% of the installed capacity (as at December 2014):
 - Iberafrica Power (E.A.) Company Limited (thermal power plant) – 108.5 MW
 - Tsavo Power Company Limited (thermal power plant) – 74 MW
 - Mumias Sugar Company Limited (co-generation) – 26 MW
 - Orpower 4 Inc (geothermal power plant) – 110 MW
 - Rabai Power Company Limited (thermal power plant) – 90 MW
 - Imenti Tea Factory Company Limited (mini-hydro) – 0.3 MW
 - Gikira Hydro (mini-hydro) – 0.514 MW
 - Thika Power Limited (thermal power plant) – 87 MW
 - (Gulf Power Limited (thermal power plant) not in KPLC report)
 11. Kenya Electricity Transmission Company Limited (KETRACO)
 12. The Kenya Power & Lighting Company Limited (KPLC)
 13. Kenya Petroleum Refineries Limited (KPRL)
 14. Kenya Pipeline Company Limited (KPC)
 15. National Oil Corporation of Kenya Limited (NOCK)
 16. Oil Marketing Companies (OMCs)
 17. Petroleum Institute of East Africa (PIEA)
 18. Oil Exploration and Production Companies (OIEPs)
 19. Kenya Nuclear Electricity Board (Corporation) (KNEB)
 20. Centre of Energy Efficiency and Conservation (CEEC)
 21. Kenya Revenue Authority (KRA)
 22. National Environmental Management Authority (NEMA)
 23. Directorate of Occupational Safety and Health Services (DOSHS)
 24. Water Resources Management Authority (WRMA)
- Other players Kenya Railways (KR), Kenya Truckers Association (KTA), Kenya Association of Manufacturers (KAM), Kenya Bureau of Standards (KEBS), Kenya Maritime Authority (KMA) and of course consumers.

APPENDIX D

Health clinics categorization according to energy requirements.

		A	B	C = AxB	D	E = CxD	F = E/1000
Category I Health Clinic Power and Energy Consumption		Quantity	Power (Watts)	Total Watts	On-Time (watt hours/day)	Wattage (hours/ day)	kWh/day
Description of Devices	Vaccine Refrigerator/Freezer	1	60	60	6.0 - 12.0	360 - 720	0.36 - 0.72
	Small Refrigerator (non-medical use)	1	300	300	5.0	1,500	1.5
	Centrifuge	1	575	575	1.0 - 1.5	575 - 862.5	0.575 - 0.8625
	Hematology Mixer	1	28	28	1.0 - 1.5	28 - 42	0.028 - 0.042
	Microscope	1 - 2	15	15 - 30	3.0 - 4.0	45 - 120	0.045 - 0.12
	Lighting	2	10	20	2 - 10	40 - 200	0.040 - 0.2
	Incubator	1	400	400	2.0 - 12.0	800 - 4,800	0.8 - 4.8
	Water Bath	1	1,000	1,000	1.0 - 2.0	1,000 - 2,000	1 - 2
	Communication via VHF Radio	1					
	Stand-by		2	2	12.0	24	0.024
	Transmitting		30	30	1.0 - 2.0	30 - 60	0.030 - 0.060
	Total			2,430 - 2445		4,402 - 10,328.5	4.4 - 10.3

Figure 17. Category 1 health clinic energy consumption (USAID n.d.).

		A	B	C = AxB	D	E = CxD	F = E/1000
Category II Health Clinic Power and Energy Consumption		Quantity	Power (Watts)	Total Watts	On-Time (watt hours/day)	Wattage (hours/ day)	kWh/day
Description of Devices	Vaccine Refrigerator/Freezer	2	60	120	6.0 - 12.0	720 - 1,440	0.72 - 1.44
	Small Refrigerator (non-medical use)	1	300	300	5.0	1,500	1.5
	Centrifuge	1	575	575	2.0	1,150	1.15
	Hematology Mixer	1	28	28	2.0	56	0.056
	Microscope	2	15	30	5.0	150	0.15
	Lighting	2 - 3	15	30 - 45	10.0	300 - 450	0.3 - 0.45
	Sterilizer Oven (Laboratory Autoclave)	1	1,564	1,564	1.0	1,564	1.564
	Incubator	2	400	800	2.0 - 12.0	1,600 - 9,600	1.6 - 9.6
	Water Bath	1	1,000	1,000	2.0	2,000	2.0
	Communication via VHF Radio	1					
	Stand-by		2	2	12.0	24	0.024
	Transmitting		30	30	2.0 - 3.0	60 - 90	0.060 - 0.09
	Desktop Computer	1 - 3					
	Power Save Mode		50	50 - 150	7.0	350 - 1,050	0.35 - 1.05
	In Use		150	150 - 450	5.0	750 - 2,250	0.750 - 2.25
	Printer	1	65	65	3.0	195	0.195
	Total			4,744 - 5,159		10,569 - 21,519	10.6 - 21.5

Figure 18. Category 2 health clinic energy consumption (USAID n.d.).

		A	B	C = AxB	D	E = CxD	F = E/1000
Category III Health Clinic Power and Energy Consumption		Quantity	Power (Watts)	Total Watts	On-Time (watt hours/day)	Wattage (hours/ day)	kWh/day
Description of Devices	Vaccine Refrigerator/Freezer	3	60	180	6.0 - 12.0	1,080 - 2,160	1.08 - 2.16
	Small Refrigerator (non-medical use)	1	300	300	5.0	1,500	1.5
	Centrifuge	1	575	575	3.0	1,725	1.725
	Hematology Mixer	1	28	28	3.0	84	0.084
	Microscope	3	15	45	6.0	270	0.27
	Lighting	3 - 4	15	45 - 60	10.0	450 - 600	0.45 - 0.6
	Incubator	3	400	1,200	2.0 - 12.0	2,400 - 14,400	2.4 - 14.4
	Water Bath	1	1,000	1,000	2.0	2,000	2.0
	Communication via VHF Radio	1					
	Stand-by		2	2	12.0	24	0.024
	Transmitting		30	30	3.0 - 4.0	90 - 120	0.090 - 0.120
	Desktop Computer	3 - 6					
	Power Save Mode		50	150 - 300	7.0	1,050 - 2,100	1.05 - 2.1
	In Use		150	450 - 900	5.0	2,250 - 4,500	2.25 - 4.5
	Printer	1	65	65	4.0	260	0.26
	Portable X-Ray Machine	1	3,000	3,000	1.0	3,000	3.0
	Total			8,634 - 9,249		17,597 - 34,157	17.6 - 34.2

Figure 19. Category 3 health clinic energy consumption (USAID n.d.).

APPENDIX E

Sustainability index (on GIS maps) of various rain-fed energy crops in Kakamega County.

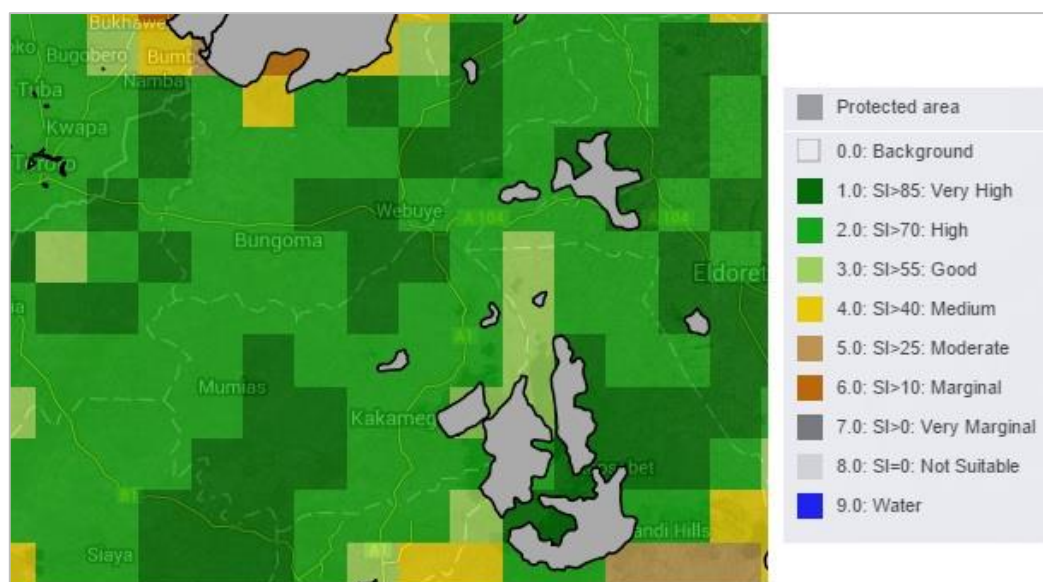


Figure 20. Barley sustainability index characterization for Kakamega County (IRENA Global Atlas 2015).

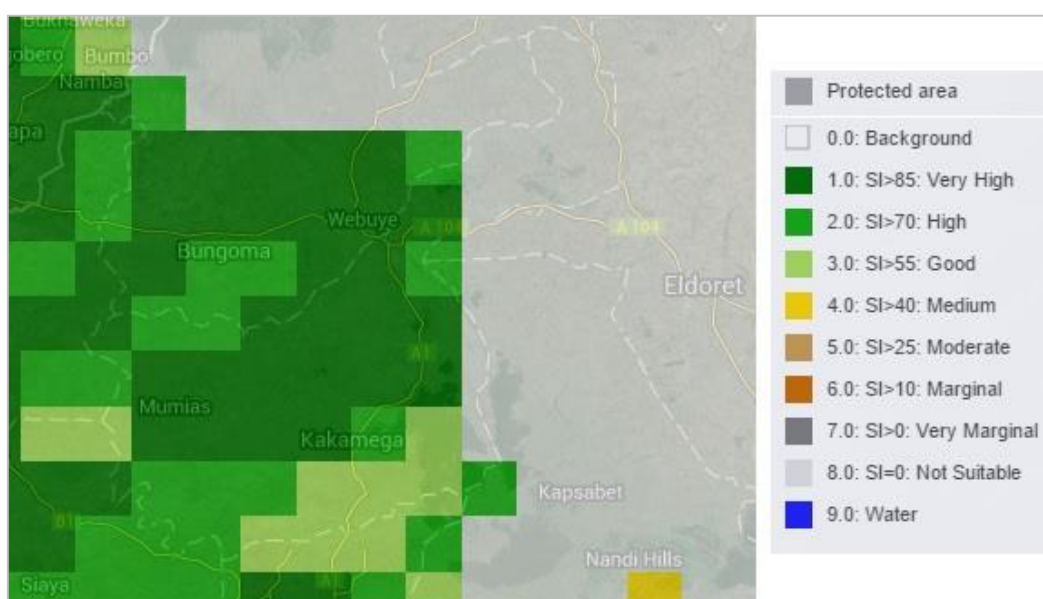


Figure 21. Cassava sustainability index characterization for Kakamega County (IRENA Global Atlas 2015).

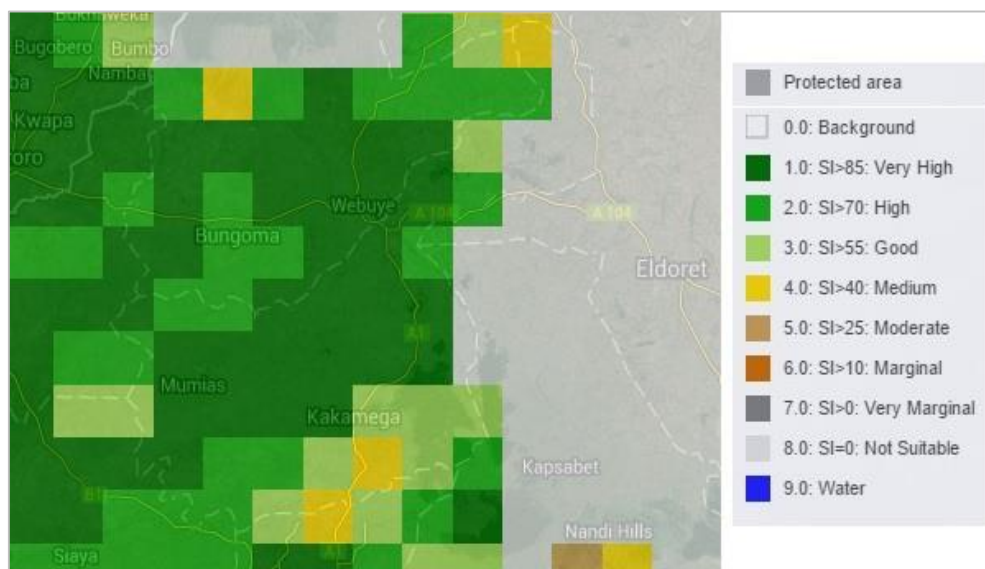


Figure 22. Jatropha sustainability index characterization for Kakamega County (IRENA Global Atlas 2015).

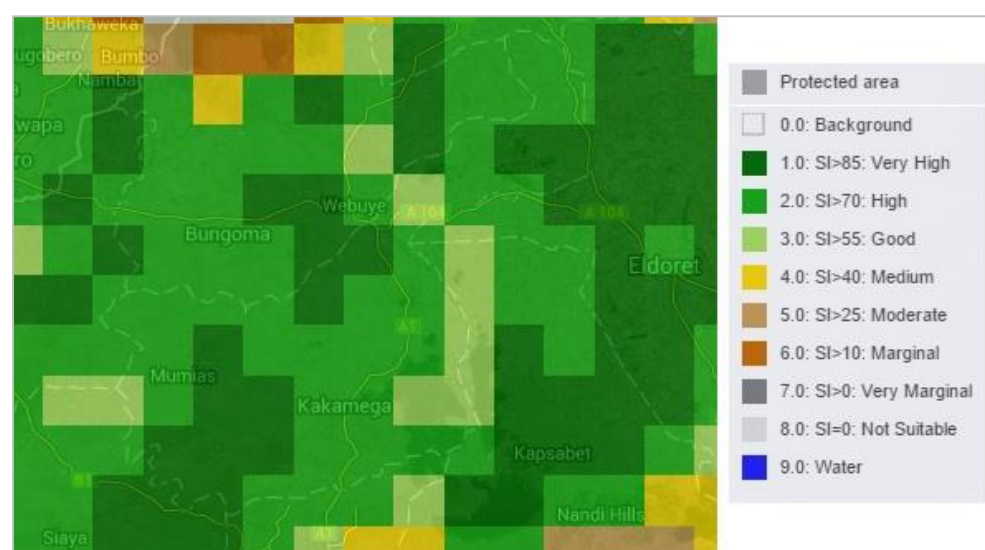


Figure 23. Maize sustainability index characterization for Kakamega County (IRENA Global Atlas 2015).

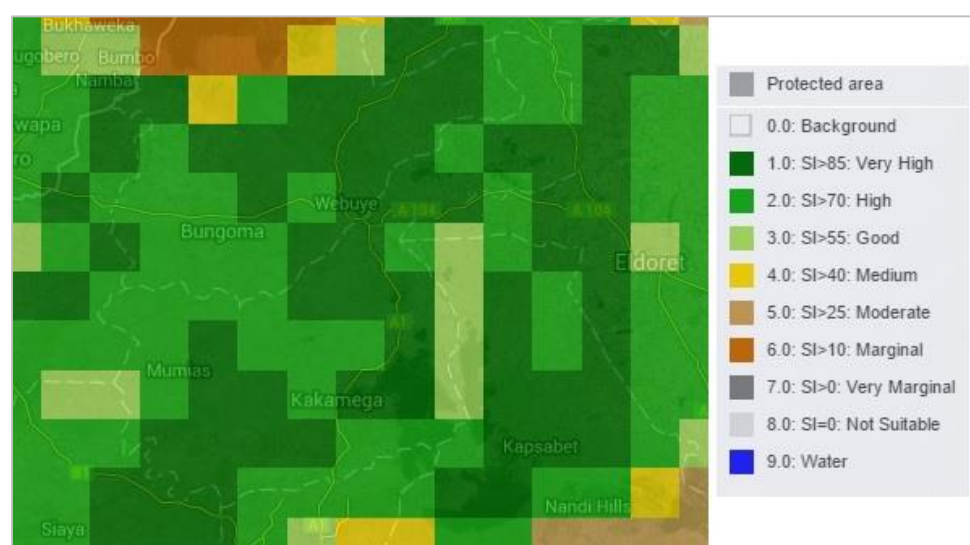


Figure 24. Miscanthus sustainability index characterization for Kakamega County (IRENA Global Atlas 2015).

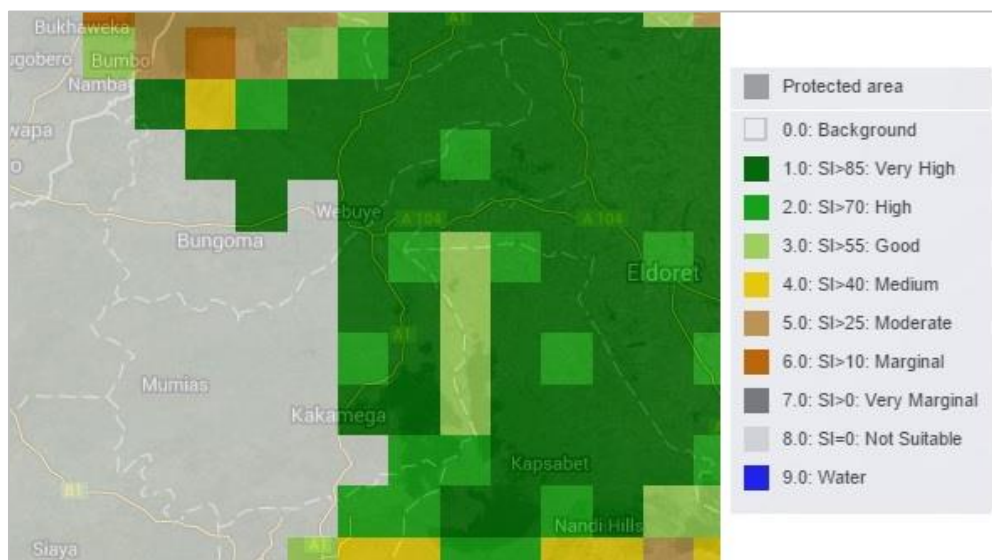


Figure 25. Rape sustainability index characterization for Kakamega County (IRENA Global Atlas 2015).

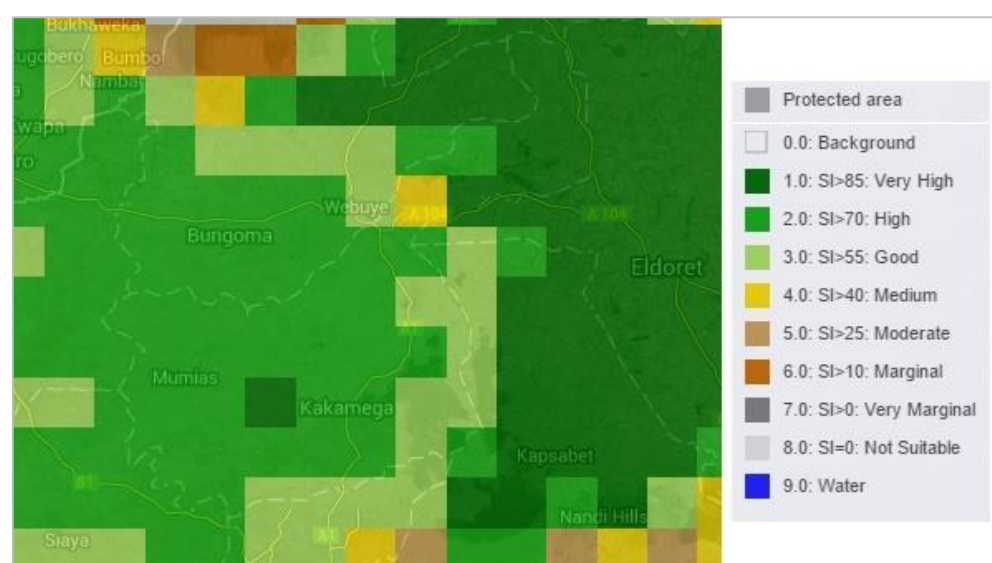


Figure 26. Sorghum sustainability index characterization for Kakamega County (IRENA Global Atlas 2015).

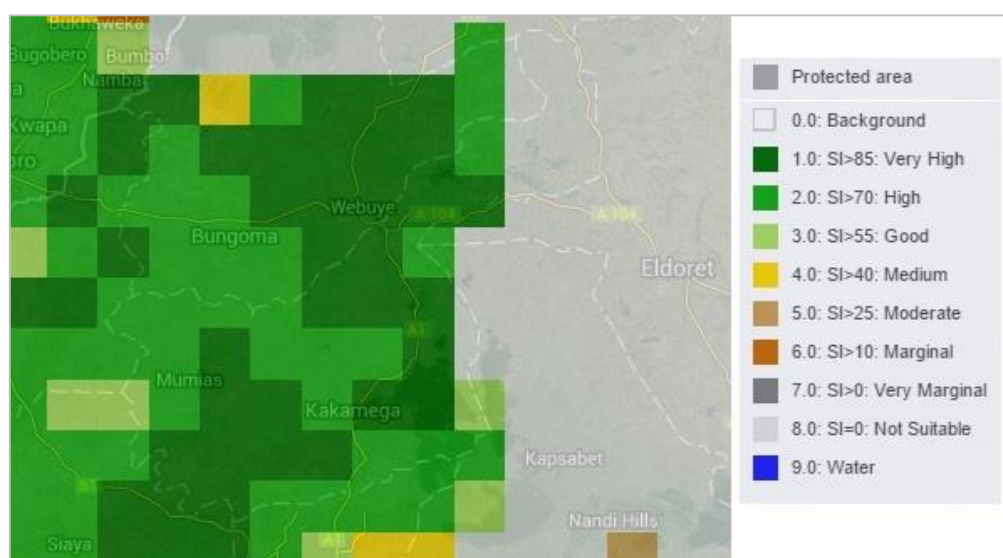


Figure 27. Soybean sustainability index characterization for Kakamega County (IRENA Global Atlas 2015).

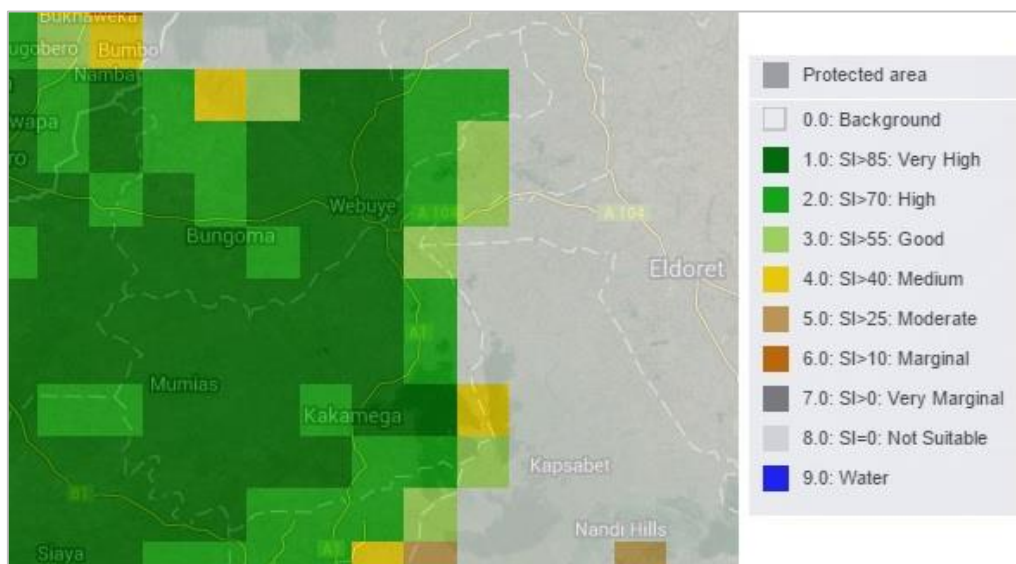


Figure 28. Sugarcane sustainability index characterization for Kakamega County (IRENA Global Atlas 2015).

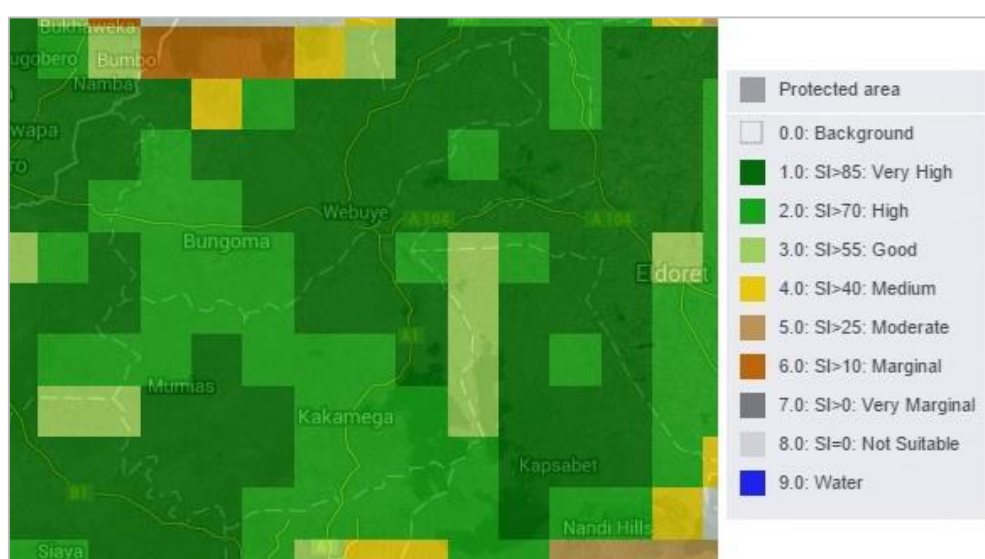


Figure 29. Sunflower sustainability index characterization for Kakamega County (IRENA Global Atlas 2015).

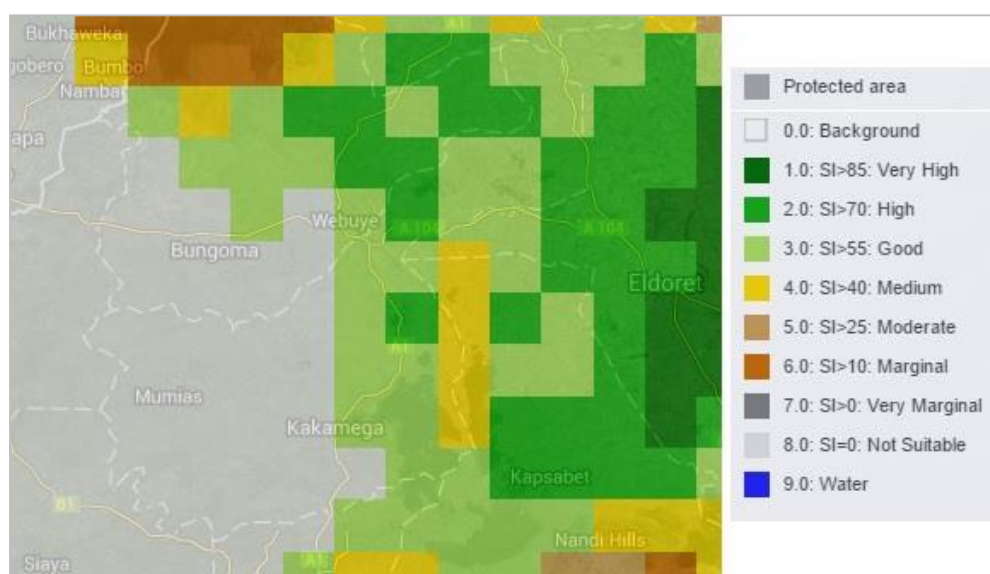


Figure 30. Wheat sustainability index characterization for Kakamega County (IRENA Global Atlas 2015).